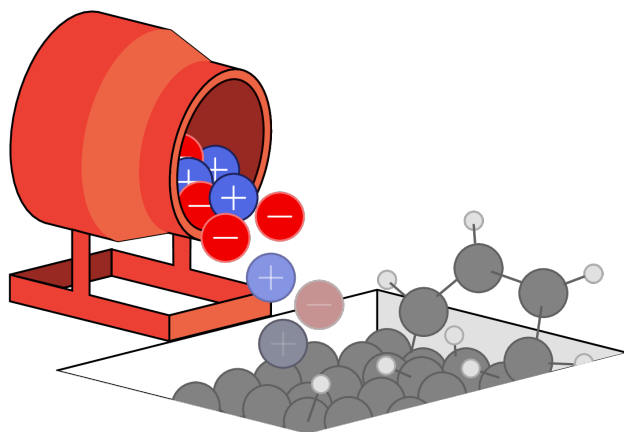


From Abstract to Concrete: Advancing the Globally Optimal Catalyst Method

Dissertation



in partial fulfillment of the requirements for the degree

Dr. rer. nat.

of the Faculty of Mathematics and Natural Sciences at

Kiel University

submitted by

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Kiel, 2024

From Abstract to Concrete:

Advancing the Globally Optimal Catalyst Method

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Kiel, June 24, 2024

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Date of oral examination: September 11, 2024

Approved for publication: September 11, 2024

ABSTRACT

This Thesis explores avenues of extension and improvement to the Globally Optimal Catalyst (GOCAT) approach. In this inverse design method, the goal is to find *in silico* an optimal catalytic embedding for arbitrary chemical reactions, in the shape of abstract interaction entities. These have, in the past, mostly taken the form of point charges with fixed spatial positioning. This general idea is extended here to include van-der-Waals interactions and even molecular fragments in a QM/MM setting. Additionally, new implementation work is done in the framework of OGOLEM, in which the GOCAT method is implemented. It allows a new way of optimizing a frame-wise embedding for sterically involved reactions, and the inclusion of interaction entities close to reaction centers in adaptive re-optimization of the reaction path, which was not possible before.

Furthermore, the complexity required in the interaction entities was previously unknown, i.e. how many point charges are needed for optimal catalysis. By employing a fitting scheme in OGOLEM using its evolutionary algorithm (EA), it could be shown that very complex GOCAT structures can be reduced to a minimum number of required interactions.

One of the most pressing issues of inverse design methods is, however, their transferability to actual chemical structures. With the new additions to the method, especially the ability to include molecular fragments and to reduce the complexity of GOCATs to their bare minimum, the approach is inching closer to a wider applicability, e.g. in biocatalyst design. Early results also show that method can be used successfully as a guidance in the design of catalytic surfaces and organic frameworks.

KURZZUSAMMENFASSUNG

Diese Dissertation befasst sich mit der Methode der Globally Optimal Catalysts (GOCATs). In diesem inversen Designansatz werden optimale, katalytische Einbettungen für arbiträre chemische Reaktionen, in der Form von abstrakten Wechselwirkungsentitäten, erzeugt. Diese abstrakten Entitäten wurden, in der Vergangenheit, hauptsächlich durch Punktladungen ausgedrückt, die um die Reaktion herum, im Raum fixiert, positioniert wurden. Dieser generelle Ansatz wurde hier erweitert, nämlich um van-der-Waals-Wechselwirkungsentitäten und um Molekülfragmente in einem QM/MM-Ansatz. Weiterhin wurde Implementationsarbeit am OGOLEM-Programmpaket vorgenommen, in dem die GOCAT-Methode entwickelt wurde. Dabei wurde eine neue bildweise Einbettung entwickelt, die es erlaubt, dass die abstrakten Entitäten mit dem Reaktionspfad zusammen optimiert werden und sich somit von ihren vormals fixierten Positionen bewegen können. Das erlaubt nicht nur die Anwendung auf sterisch anspruchsvolle Reaktionen, sondern erlaubt es auch, die abstrakten Entitäten näher an der Reaktion zu positionieren, ohne die Optimierung neuer Reaktionspfade zu kompromittieren.

Weiterhin bestand bei dem GOCAT-Ansatz das Problem, dass die nötige Komplexität für eine gegebene Reaktion nicht bekannt war, und somit z.B. keine abschließende Aussage über die benötigte Menge an Punktladungen gemacht werden konnte. Mithilfe einer Regressionsmethode, die den evolutionären Algorithmus (EA) von OGOLEM nutzt, konnte gezeigt werden, dass sich sehr komplexe GOCATs auf ihre wesentlichen Bestandteile reduzieren lassen, ohne ihre Optimalität einzubüßen.

Die größte offene Frage des GOCAT-Ansatzes ist jedoch eine generelle Frage inverser Designansätze. Es handelt sich dabei um die Umsetzbarkeit der abstrakten Einbettungen als tatsächliche Moleküle. Mit den neuen Möglichkeiten, die dem Ansatz nun zur Verfügung stehen, wie der Erzeugung von GOCATs aus Molekülfragmenten, sowie der Möglichkeit, die Komplexität von GOCATs auf ihren Kern zu reduzieren, steht der Ansatz nun noch näher an einer breiten Ansatzmöglichkeit, z.B. im Design von Biokatalysatoren. Erste Ergebnisse zeigen auch, dass GOCATs als Ausgangspunkt für das Design von katalytischen Oberflächen und Gerüstverbindungen dienen können.

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INTRODUCTION

Chemical reactions form the molecules that make up our world and are elementary for our lives and our society. Nature has, over billions of years, found wondrous ways to produce substances and structured materials that humankind, despite its vast technological advancements, struggles to reproduce synthetically^[1].

The most proficient machines that drive chemical reactions in nature are the *enzymes*. Many are highly specialized and can turn even the most unthinkable processes into manageable, bite-sized chunks that work at physiological conditions. One such example is the nitrogenase enzyme complex that can achieve the herculean cleavage of the triple bond in nitrogen at *ambient conditions*. This is a vital step in the production of nitrogen compounds indispensable for the existence of life^[2].

Enzymes such as the nitrogenase function as a *catalyst* to a chemical reaction. They lower the necessary energy barrier that is associated with the reaction, thus making it more efficient, or depending on the conditions, making it possible in the first place. During the reaction, they are generally required only in very low concentration and emerge, themselves, unchanged^[3].

Our society relies heavily on the industrial production of chemicals, such as pharmaceuticals and fertilizers, in large quantities. As such, the search for new and ever improved catalysts is a tremendously important research field in chemistry. Additionally, many established processes which produce basic compounds, such as the Haber-Bosch process for production of ammonia from nitrogen, are energy-hungry and have prominent carbon footprints. New, improved and altered catalysts could help transform the industry to use greener, more environmentally friendly alternatives^[4-7].

In addition to ever continuous research done in the lab, computational chemistry has been a growing and indispensable aid in this search for new catalysts. While theory has already helped chemists explain experimental findings for decades, it has increasingly also become a *predictive* tool, enabling the exploration and discovery of new mechanisms and materials *in silico*^[8].

In 2017, Houk described the ability to design an actual, functioning (bio)catalyst for any arbitrary reaction from scratch on the computer a “holy grail” of computational chemistry^[9]. Chemists have tried to chip away at this challenge from different angles and have made great progress in recent years. Houk himself and his coworkers have pursued this

goal with *theozymes*, which are manually built, ideal active sites for reactions using the building blocks of enzymes, amino acids^[10]. In an impressive collaborative effort, this approach has led to several successful proposals of newly designed enzymes^[11] which showed initial catalytic properties and could be further improved by directed evolution in the lab^[12,13]. Utilizing known catalytic or successful structures (from biology or elsewhere) in this way and modifying them is a practice that has proven both popular and immensely fruitful in catalyst and drug design^[14–16].

While undoubtedly successful, these approaches do not quite match the original holy grail idea. When a list of *predefined* candidates is sampled or chemical intuition is the driving force in design, there is already a bias towards very specific areas of the search space pertaining to the problem at hand. Thus, making fundamentally new discoveries is a fairly challenging prospect. An approach to catalyst design that seeks to avoid this inherent bias is the *inverse design*, in which the necessary ideal properties of the catalyst are elucidated and optimized first, without assuming any explicit molecular structure. There are many such approaches described in literature^[17,18], such as the gradient-driven molecule construction (GDMC)^[19]. The basic idea of this approach is to stabilize a target structure by introducing an external potential (termed “jacket potential”).

One big issue that inverse design methods have, in general, is that, while they are somewhat free of preconceptions, they are also unbounded by the reality of chemistry. An external potential, such as the one described above, must eventually be converted back into a molecule that can be synthesized in the lab and thus, conversion methods and rational constraints must be introduced to enable this step. One concept in this vein is that of the alchemical potential^[20]. By using a linear combination of atomic potentials, coefficients can be tuned to the desired effect and eventually rounded to obtain actual atoms from the periodic table. However, this approach has, up until now, mostly seen use in the modification of a select variety of known molecular systems^[21,22].

A particular mode of catalysis that is very interesting in the context of inverse design is the *electrostatic catalysis*. It has been shown to play a large role in many enzymes^[23] and has been researched extensively in the past decades^[24,25]. The electric field in the active site of an enzyme, ketosteroid isomerase (KSI), could also be determined experimentally^[26], showing clearly that electrostatics play a major role in its mechanism of catalysis. This spurred many investigations into the role of enzyme scaffolds and their electrostatic properties^[27–29].

Up until quite recently, the *tuning* of electrostatics specifically has played only a very limited role in synthetic chemistry. In the working group of COOTE, however, it was shown in experiment that electrostatics can not only catalyze reactions that are typically known

to have redox character, but also ones that involve strictly covalent species, such as Diels-Alder reactions^[30]. This was inspired by the work of SHAIK and co-workers, who previously had put forward the term oriented external electric field (OEEF) for an externally applied electric field that catalyzes reactions and can even control selectivity in theoretical studies^[31]. In the same vein, SOKALSKI and co-workers suggested the importance of electrostatic effects for the chorismate mutase^[32]. Building upon their model of differential transition state stabilization (DTTS) and catalytic fields, they also could show the electrostatic impact of directed evolution for a de novo designed enzyme and correlate the mutations directly with the lowering of the reaction energy barrier^[33].

Consequently, electrostatic catalysis is something that has not only become accessible, but highly promising for inverse design studies. With the ability to tune an external field in actual molecules (such as enzymes or other scaffolds), results from the inverse design can act as a guideline for improvement, or even the de novo design, of such molecules.

1.1 Globally Optimal Catalysts

The Globally Optimal Catalyst (GOCAT) concept is the inverse design method central to this work. It was first introduced by DITNER and HARTKE in 2018^[34] and much like the previously mentioned GDMC, it builds upon the fact that chemical reactions are always influenced by their immediate surrounding, even if it is not an active part of the reaction itself. Functionally, this influence is driven by interaction through Coulomb and dispersion forces: It is an inherent property of the given surrounding, regardless of whether it is a solvent, an enzyme or a metal surface. GOCATs seek the *maximum possible* influence such a surrounding can have on a chemical reaction, e.g., for the purpose of catalysis. In order to do so, the specific chemical structure of this surrounding is omitted and its explicit structure is replaced with abstract entities that represent only the Coulomb and/or dispersion interaction terms.

In practice, this is done by exploiting the very popular QM/MM approach, since it inherently lends itself to this purpose; in a QM/MM system, the surrounding is generally represented by parameterized force field atoms. By replacing these with abstract, unbound entities that are variable in their electrostatic and dispersion parameters, a setup emerges that makes it possible to tune the influence on a given reaction by optimizing both position and parameters of these entities.

Since the desired function of this influence is, in general, the catalysis of the embedded reaction, the goal with GOCATs is to find the best possible (spatial) arrangement and parameters such that the energy barrier of the embedded reaction is lowered maximally. This is achieved using a specifically tailored evolutionary algorithm (EA), implemented

in the program package OGOLEM^[35].

By including the information of an entire reaction path in the optimization, GOCATs target the influence on a reaction as a *whole*, and not only on a single structure, like the transition state. Additionally, by applying reasonable constraints for the abstract interaction particles, the approach also guarantees that an actual QM/MM embedding can be formulated which exhibits at least adequately close properties to that of the *globally optimal* one. While a study has yet to be done to quantify this reproducibility, first steps towards this have been taking with the establishment of an automated molecular construction scheme that can reproduce point charge distributions^[36].

1.2 Thesis Goals

This thesis is comprised of three main chapters that deal with three aspects of the GOCAT method. Two of them, Chapter 5 and 6, are actually a direct continuation of the work of DITTNER, with both points already being discussed as part of the closing words to his PhD thesis^[37].

Chapter 5 deals with an issue that arose quite early in the GOCAT approach. When GOCATs are optimized for an arbitrary reaction, the required complexity (e.g., the *number* of point charges) is unknown prior to the optimization. This means that the EA can easily arrive at solutions that are more complex than necessary. The reason for this is simple: in point charge GOCATs, the number of charges should in principle be bounded by two factors: the absolute value of the electrostatic potential (ESP) required to satisfiably catalyze the reaction, as well as the overall shape this potential needs to have in order to be effective. Thus, with GOCATs that are limited to an absolute charge value of 1 e per point charge, a lower limit that is unique to the reaction in question must exist. If a GOCAT is optimized with fewer charges, the globally optimal solution is not obtained. On the other hand, if it is optimized with more, the simplest optimal solution *could* be found, but it is statistically very unlikely, as virtually infinite combinations of charge positions and values could arrive at the same, globally optimal ESP. DITTNER points out that this could be solved via additional constraints and tricks in the fitness function of the EA. For example, when two point charges of opposite sign are close to each other, one could be set to zero and the other set to represent their effective point charge. This approach would certainly have massive advantages; it would completely alleviate the need to worry about charge number. The lower bounds in complexity would automatically and naturally emerge during global optimization, such that the GOCAT automatically is the simplest possible solution given the bounds set on the absolute charge value. In practice, however, it is extremely difficult to formulate constraints such as these and have them be

compatible with the global optimization algorithm. Instead, a two-step process is proposed in Section 5.1: for a GOCAT that is too complex, the ESP can safely be assumed to be optimal within the given parameters. It can then be used as a target value for the generation of GOCATs that have the absolute minimum required number of charges.

Chapter 4 deals with a new form of GOCATs that were proposed previously, but not implemented. In his work, DITTNER has already thoroughly explored the optimization of electrostatic interactions, but there is another way the electronic properties of molecules can be influenced: For example, by applying external forces in the form of dispersion terms, the equilibrium molecular structure can be altered and new minima found. In the ideal case, these can then react to the desired product with less activation energy. Distortions of molecular structure have, in fact, been proposed as an important mechanisms in some enzymes^[38–40]. Other fields of application that come to mind are mechanochemistry^[41,42], where mechanical force is applied to molecules in controlled milling processes, as well as molecular machines^[43–45] that can convert externally supplied energy into effective impulse toward, e.g. a specific reaction. The approach is also consistent with the idea of inverse design; essentially, a catalytic distortion of the reacting molecules is stabilized via external influence and can act as a guideline for finding ways and structures that can perform this distortion.

Lastly, Chapter 6 deals with the optimization of GOCATs containing actual molecular fragments. In many ways, this is quite similar to the approach in the DENOPTIM program package^[46], which was published in 2019. It is also utilizing an evolutionary algorithm for global optimization of molecular structures. The fitness function there is provided as an external script and can thus also serve many different purposes. The problem class, on the other hand, differs slightly; DENOPTIM is specifically meant to *modify* structures and end up with explicit molecules, which are then scanned for their properties. This is, by the definition used prior, not an inverse design approach. This is an interesting and important point to discuss when moving from abstract point charges or dispersion entities to molecular fragments in the GOCAT concept. Two distinct uses for this transition are introduced and examined which have also been published in a simpler form in a recent paper^[47]: Firstly, the use of solvent molecules to elucidate how much a perfectly optimal solvent shell can catalyze a reaction and secondly, the use of amino acid side chains to automatically design optimal active sites for arbitrary reactions. The latter point is very much similar to the *theozymes* described earlier, but with no preconceptions added into the mix except the selection of biologically available amino acids. In both cases, it is certainly *less* abstract than before. However, the optimized problem remains the same. The *influence* of the embedding is optimized, and it can be shown that it is still, in the under-

lying picture, the same optimization principle, but with different external constraints. It thus, arguably, stays an abstract optimization problem, with a closer relation to actual real-world molecules.

2 THEORY

2.1 Chemical Reactions

The traditional view of a chemical reaction is shown in Figure 2.1. Given a certain set of ingredients or *reactants*, there is a path along a very specific reaction coordinate ξ that leads to the *products*. In the picture of the Born-Oppenheimer approximation, where nuclear

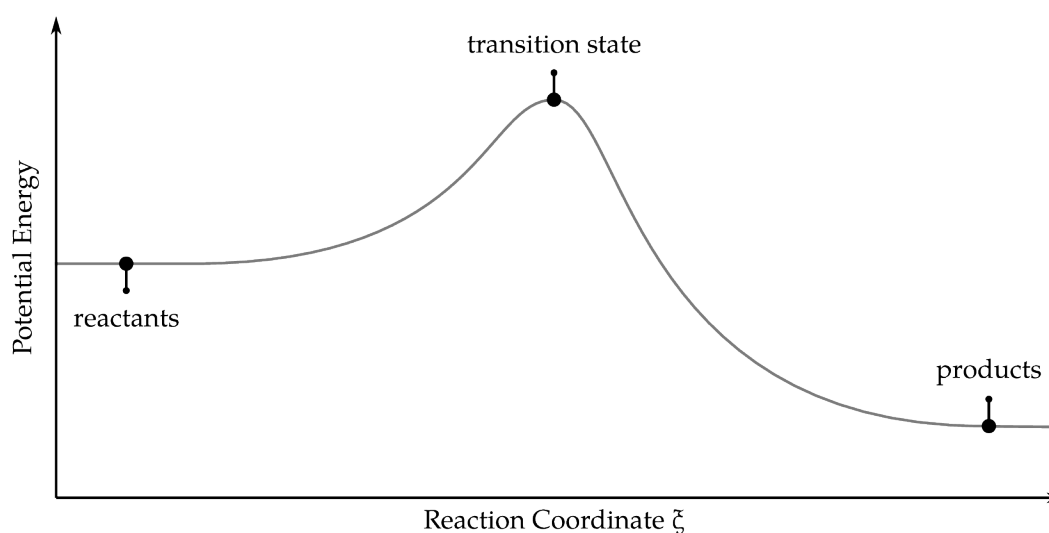


Figure 2.1: This plot shows a sketch of an arbitrary chemical reaction energy profile.

coordinates are separate from electronic coordinates, this can be seen as a specific path traced along the electronic potential energy surface (PES) of a single electronic state. The coordinate ξ is then a combined mode of nuclear motion connecting the two minimum structures in a minimum energy path (MEP). In photochemistry, where the electronic ground state is generally left behind and the Born-Oppenheimer approximation breaks down, this is obviously insufficient. However, for many reactions induced by thermal influx, this assumption can be valid and allows a direct access to the energetics of a reaction just by considering ground state structures^[48].

Unfortunately, the implication of a constant electronic state is not the only gripe one could have with Figure 2.1. For systems with more than a few degrees of freedom (DOFs), at finite temperatures, a variety of different transition structures can exist between many

different interconnected minima, and these coexist with, but are obfuscated in this singular energy profile. Additionally, the total energy of a molecular system in motion is generally higher than the potential energy for the particular reactive coordinate ξ . But important for the transition from reactant to product state is the energy stored in only this specific reactive mode, which can dynamically vary. These two points makes an intuitive connection between the height of the barrier and, e.g., the rate at which the reaction occurs, quite difficult. The dynamic variance and coupling to other DOFs is vital to accurately predict transition rates for chemical reactions and requires molecular dynamics simulation to capture properly^[48,49].

This, however, brings with it some new challenges, as computational resources are limited: Chemical reactions are *rare events*, which is a problem because simulation times are limited. This ultimately requires additional tricks to steer simulations towards reactive trajectories^[50]. In addition, there is a limit in the system size, depending on the level of theory employed in the simulation. A simulation of high quality ab initio dynamics can be very expensive and prohibitive for large systems, which is why various approaches such as reactive force fields^[51,52], semiempirical solutions (s. Section 2.2.2) and multiscale models (s. Section 2.2.3) have been developed to overcome this challenge.

A very popular alternative that avoids many of these issues is an approach from statistical thermodynamics, the transition state theory (TST)^[53]. It allows an approximation of reaction rates from the equilibrium structures of reactants, transition states and products, without the need for expensive dynamics simulations.

2.1.1 Transition State Theory

In *transition state theory* (TST), or alternatively called *activated complex theory* (ACT), one adopts the picture illustrated in Figure 2.1 in its most literal shape. By assuming that a reaction proceeds only along this single minimum energy path and never re-crosses once the transition state has been reached, one can reasonably easily estimate reaction rates for the reaction in question. The reaction in this picture could, e.g., be written like this:



Here, A and B are the reactants that react via a transition state $(AB)^{\ddagger}$ to the product C . In order to do apply transition state theory, it is necessary to precisely determine the electronic energies at the reactant and transition state structures. Then, assuming equilibrium between reactants and transition state, the equilibrium constant for the reaction can

be written as:

$$K(T) = \frac{[(AB)^\ddagger]}{[A][B]} \quad (2.2)$$

$$= \frac{Q_{(AB)^\ddagger}}{(Q_A)(Q_B)} e^{-E_0/k_B T} \quad (2.3)$$

The second line, Equation 2.3, uses the formulation of the equilibrium constant derived from statistical thermodynamics, where the concentrations are replaced by the respective Boltzmann distributions, giving K as a function of the partition functions Q and the Boltzmann factor containing the zero-energy point difference between the reactants and the transition state, E_0 . For the sake of brevity, only the original derivation of the rate constant from this by Eyring^[53] is given here:

$$k_{\text{textupTST}} = \frac{k_B T}{h} \frac{Q_{(AB)^\ddagger}}{(Q_A)(Q_B)} e^{-E_0/k_B T} \quad (2.4)$$

Where the partition function $Q_{(AB)^\ddagger}$ for the transition state has been reduced to $Q_{(AB)^\ddagger}'$, because the reactive normal mode associated with the reaction is assumed to be treatable, in essence, as a particle in a box problem.

Since statistical thermodynamics also provides expressions for macroscopic observables that relate to the partition functions, this can also be rewritten in terms of, e.g., the Gibbs free energy:

$$k_{\text{TST}} = \frac{k_B T}{h} e^{-\Delta G_{\ddagger}^\ominus/RT} \quad (2.5)$$

Note that, in this case, the change in Gibbs energy $\Delta G_{\ddagger}^\ominus$ still refers to the difference between the reactants and a transition state that does *not* include contributions from the reactive normal mode^[54].

This formulation of TST is very useful, as it gives access to an estimate of the reaction rate from run-of-the-mill calculations that can be done quickly and efficiently for small and moderately sized systems. While it also has pitfalls built-in, such as the neglect of recrossing, the neglect of quantum effects like tunneling, and a harmonic approximation in the partition functions, these can be dealt with, and many approaches exist in literature that mend them successfully^[54]. However, this is somewhat beyond the scope of this section. The most important aspect that TST provides in the context of this Thesis is that it rationalizes the picture painted in Figure 2.1: It allows a link between energy barriers along specific nuclear motions directly to reaction rates, and thus, catalysis, regardless of the points mentioned in Section 2.1 above. Ultimately, it is the reason why energy profiles such as Figure 2.1 will be important and relevant in the discussion of how well a catalyst performs. This will also be briefly touched on in Section 3.6.1.

2.2 From Structure to Energy

In quantum chemistry, one of the key properties of any molecule or molecular system is its electronic structure. From it, essentially all relevant properties, such as its energy, can be either calculated or derived. Unfortunately, this is a problem that can be solved exactly (analytically) only for systems involving one or two particles (due to the many-body problem). Beyond that, one has to apply numerical solution strategies up to the desired accuracy.

Numerically exact solutions, however, require insurmountable computational effort for most chemically interesting systems and thus, in the past decades, theoreticians have developed a number of approximations, workarounds and tricks to break the problem down into manageable pieces. Thanks to these efforts, chemists nowadays have a large selection of methods, ranging from highly approximative and fast to very precise but slow. The method thus needs to be chosen carefully according to the problem, and each method comes with its own advantages and disadvantages^[49].

Especially relevant to this work are the force field, semiempirical and QM/MM methods. Each of them will be explained, respectively, in more detail in the following sections.

2.2.1 Force Field Methods

In one of the simplest approaches possible, force field methods describe nuclear and electronic motion exclusively via empirically fitted, efficient potential functions. This has several advantages: Since the computational cost of evaluating these empirical potentials is so negligible, the approach works superbly for large-scale systems and their molecular dynamics over long time periods. It enables the estimation of processes and properties in the macro system, e.g. density, pressure, phase transitions, diffusion^[49,55], as well as for structure prediction of large molecular assemblies such as clusters^[35] and proteins^[56].

However, on the downside, force fields cannot always reproduce all quantum mechanical effects correctly (although there exist several corrections that can be applied to recover at least some of them, like in CMD^[57] and RPMD^[58]). Additionally, estimation of properties that depend on the individual electronic structure, become challenging or impossible entirely.

The most common way to write the energy in a force field is:

$$E_{\text{total}} = E_{\text{intra}} + E_{\text{inter}}. \quad (2.6)$$

The total energy is made up of the sum of the intramolecular forces, as well as the inter-

molecular forces. Typically, intramolecular forces are:

$$E_{\text{intra}} = E_{\text{bonds}} + E_{\text{angles}} + E_{\text{torsions}} + E_{\text{improper tors.}} \quad (2.7)$$

These terms correspond to the energy associated with bond stretching, angle bending, dihedral angle torsion as well as improper (out-of-plane) torsion, respectively. The intramolecular forces are usually given by:

$$E_{\text{inter}} = E_{\text{dispersion}} + E_{\text{Coulomb}}, \quad (2.8)$$

where these terms describe dispersion and charge-charge interaction between atoms of neighboring molecules.

In this specific formulation, the force field is non-reactive, meaning that bonds are never allowed to change from the predefined state. It is also non-polarizable as each atom is assigned a single partial charge value that remains constant throughout simulation. There are however (numerous) other formulations that have additional rules for polarizability^[59,60] as well as for bond breaking and forming^[51,52,61–64]. Reactive force fields, such as the latter, typically require dedicated and hand-tailored parameterization for every reaction system. They are thus very handy for kinetic studies, reactive MD calculations and similar applications, but also fail whenever actual electron density is required to accurately describe a phenomenon, such as the influence of external electric fields on the activation barrier of a reaction.

2.2.2 Semi-Empirical Methods

Post Hartree-Fock (HF) and Kohn-Sham density functional theory (KS-DFT) methods, together with available hardware, have dramatically improved both in efficiency and computational speed over the last decades. However, for very large systems and/or applications where computational speed is paramount, they unfortunately remain too time-consuming. This can, i.e., be the case in global optimization; optimizing GOCATs especially may require thousands of energy and gradient calculations *per iteration step*, which means that these calculations should ideally take only a few milliseconds for a decently sized system.

By taking (post-)HF or KS-DFT theory and replacing computationally demanding steps by empirical correction terms, this can be achieved without losing the important quantum-mechanical description of the electrons. This is called the semiempirical approach and it has already been heavily used in the conception of the GOCAT model and continues to play a very important role throughout this work.

Prominent examples for semi-empirical approaches derived from HF are the modified neglect of differential overlap (MNDO)^[65] family of methods, with other examples such

as AM1^[66] and PMx^[67], as well as OMx^[68]. In these methods, most of the computationally extensive two-electron integrals in the Fock matrix are omitted by setting the product between basis functions pertaining to the same electron but different atoms to zero. With this approximation, all 3- and 4 center integrals vanish. Additionally, these methods rely on only a minimal basis set for each atom. This introduces a number of problems that need to be accounted for by additional, empirically parameterized, terms^[49]. A recent review of semiempirical benchmark studies emphasizes that, while semiempirics of this type are fast and can be accurate for certain systems, they fail spectacularly for others and it is thus very important to know in which cases this happens^[69].

A newer generation of semiempirical methods are the extended tight-binding (xTB)^[70–72] methods. They are developed by GRIMME, BANNWARTH and co-workers and are based on the density functional tight-binding (DFTB)^[73] method. In contrast to the often difficult pair-wise parameterization required for DFTB, however, xTB employs an atom-wise parameterization, making it feasible for a great variety of systems out of the box. For the original formulation of xTB^[70], the full energy expression is:

$$E_{\text{tot}} = E_{\text{electronic}} + E_{\text{repulsion}} + E_{\text{dispersion}} + E_{\text{XB}}. \quad (2.9)$$

They are, in order, the electronic energy, repulsion term, dispersion term and a correction term for halogen-bonding. The first term, E_{el} , is obtained in a KS-DFT like scheme. In a variational solution to the eigenvalue problem

$$\mathbf{FC} = \mathbf{SC}\epsilon, \quad (2.10)$$

the KS-like MO coefficients $\mathbf{C}$ are determined. The Fock matrix elements $\mathbf{F}$ are given as:

$$\langle \phi_\mu | F | \phi_\nu \rangle = \langle \phi_\mu | H_0 | \phi_\nu \rangle + \frac{1}{2} S_{\mu\nu} \sum_C \sum_{l''} (\gamma_{AC, ll''} + \gamma_{BC, l' l''}) p_{l''}^C + \frac{1}{2} S_{\mu\nu} (q_A^2 \Gamma_A + q_B^2 \Gamma_B). \quad (2.11)$$

The expression for the first-order Hamiltonian,

$$\langle \phi_\mu | H_0 | \phi_\nu \rangle = K_{AB} \frac{1}{2} (k_l + k_{l'}) \frac{1}{2} (h_l + h_{l'}) S_{\mu\nu} (1 + k_{\text{EN}} \Delta \text{EN}_{AB}^2) \prod (R_{AB, ll'}), \quad (2.12)$$

shows where a lot of the speedup in xTB comes from. Since the K_{AB} , k_l , h_l , k_{EN} are tabulated global and atom-specific parameters and the electronegativity difference ΔEN_{AB} , as well as the $R_{AB, ll'}$ are computed trivially from parameters, $\langle \phi_\mu | H_0 | \phi_\nu \rangle$ becomes a lookup of pre-computed overlap matrix elements $S_{\mu\nu}$. The next term in Equation 2.11 is a summation over the charge distributions p_l^C for each shell l on each atom C , which upon summation over all shells, would give the (Mulliken) partial charge on the respective atom. This is similar to a Coulomb term, as $\gamma_{AC, ll'}$ is, in essence, a parameterized interatomic distance. The third term arises from a third-order correction term involving the

Mulliken charges q_A , as well as another atom-specific parameter Γ_A . These last two terms together form the basis of the self-consistent charge (SCC) iterations, where Equation 2.10 is solved for consistent partial charges^[70,72].

The other energy term contributions in Equation 2.9 are a D3 dispersion correction^[72], a pairwise Gaussian-shape repulsion potential using atom-specific parameters, as well as the halogen correction term with parameters specific to the halogen type, which are also all quickly evaluated from tabulated parameters.

The xTB varieties currently available for general simulation are parameterized to perform well on Geometries, Frequencies and Non-local interactions, which is why they were termed GFN1^[70], GFN2^[71] and GFN0^[74], respectively. Benchmarks of these methods show impressive accuracy for these target quantities across a wide variety of chemistry^[75,76]. It has also been shown that, while not specifically parameterized to perform well there, xTB can also be used to approximate energy barriers^[77], making it a very good candidate for explorative simulations^[78,79].

2.2.3 QM/MM Methods

The GOCAT approach relies heavily on the multiscale quantum mechanics/molecular mechanics, QM/MM for short. It was developed in an effort to make simulations of biological reaction systems feasible^[80]. The simulation of enzymes under physiological conditions encompasses thousands of atoms, which is, even today, virtually impossible to simulate quantum mechanically and, ultimately, requires a force field, or MM, approach. When reactions take place in an enzyme, the reactive part is, however, comparatively much smaller than the entire simulation system and can, in most cases, be reduced to just a few atoms. By embedding this small QM system into an MM system, the two approaches can be combined to yield a reactive simulation with efficient scaling and manageable computational cost. In fact, this approach has been so successful and important in biochemistry that key researchers involved in its conceptions were awarded the Nobel prize in chemistry in 2013^[81–83].

Different implementations and variants of these methods have emerged and seen use over the years, ranging from very simple to very complex^[84]. Throughout this work, the *electrostatic embedding* scheme is used (for more details, see Section 3.6.1), with no bonds being cut by the QM/MM boundary. The total energy then is given as:

$$E_{\text{QM/MM}} = \tilde{E}_{\text{QM}} + E_{\text{MM}} + E_{\text{vdW, QM/MM}} \quad (2.13)$$

Here, the QM energy is polarized by the point charges of the MM embedding to yield $\tilde{E}_{\text{QM}}$, and an additional energy term for the dispersion interactions $E_{\text{vdW, QM/MM}}$ describes

the van-der-Waals interactions between the MM embedding and the QM embedding. The explicit implementation differs slightly from software to software, but usually, the $E_{\text{vdW, QM/MM}}$ is computed by assuming the QM part to have Lennard-Jones (LJ) parameters consistent with the embedding force field. Polarization in $\tilde{E}_{\text{QM}}$ can be done by e.g. adding the Coulomb potential of the surrounding MM point charges to the one-electron Hamiltonian of the QM method^[85].

2.3 Optimization

As already mentioned in Section 2.1, chemical reactions are usually thought of as a transition from an energy minimum to a different energy minimum via a transition state. True minima on the potential energy surface (PES) are thus of great importance to chemists. They give structural information about the equilibrium of molecules and allow for e.g. the energy ranking of isomers, determination of reaction energy barriers and many other relative properties.

For a diatomic molecule, this optimization to a minimum is trivial. There must be an optimal minimum distance along the bond vector that can simply be found by scanning the distance. However, the more DOFs enter the system, the more complex the optimization problem becomes and other, computationally efficient methods need to be applied.

Generally, an extremum on any n -dimensional PES is characterized by possessing a first derivative of zero with respect to every DOF. This is true for both minima and transition states. In the case of a minimum, the second derivatives are then all positive, and in the case of a transition state, exactly one is negative.

Luckily, force fields and many modern quantum mechanical methods have analytical first (and second) derivatives of the energy with respect to each nuclear DOF readily available. Methods such as (L-)BFGS^[86], FIRE^[87] and Eigen(vector)following (EF)^[88] are common algorithms that are utilized in computational chemistry to find these extrema on even very high-dimensional PES.

The more DOFs a system possesses, however, the more locally optimal conformations become possible. A benzene, for example, is very rigid and the lowest energy conformation is easily found from any sensible starting position. An enzyme, on the other hand, can have unthinkably many rotational and conformational local optima. In this case, a local optimization will only get one so far as to find the nearest local optimum to a given starting position. If the energetically lowest conformation is to be found from *any* starting point, other methods must be employed. This can range from conformational sampling and MD simulation to global optimization algorithms, which will be discussed in the next section.

2.3.1 Global Optimization

Local optimization yields the nearest local optimum to the starting position, e.g., by following a gradient down until it (essentially) vanishes. Global optimization (GO), on the other hand, strives to find among all local optima the best one, regardless of the starting point. For small, finite systems, this can be achieved by systematically evaluating the value at every possible point. The best candidate will then be known exactly. This is the deterministic approach and there is a reason why it only works for small systems: With a growing number of DOFs, the samples that need to be evaluated grow *exponentially*. To put this in perspective, a finite two-dimensional grid with ten possible spots in each dimension would need a mere 100 evaluations. For three dimensions, it is 1000, 10000 for four and so on. The prospect of sampling 10^N possible candidates may sound somewhat manageable with these numbers, however, for an average-sized molecule with 15 atoms, there are already 40 DOFs. This would result in a staggering 10^{40} samples, and this is not even accounting for the fact that molecular DOFs are not described by 10 discrete points each but rather by continuous variables.

For larger systems, it is thus much more practical to abandon the idea of deterministically finding the global optimum and instead rely on metaheuristics to efficiently scan for promising areas in the search space. This results in a non-deterministic approach, where the dimensionality problem is somewhat mitigated at the price of no longer knowing whether an optimum found is *actually* the global optimum. This approach requires at least two essential ingredients: a way of producing solution candidates and a way of measuring their quality as a possible solution, the fitness function.^[89]

There are several different flavors of non-deterministic search algorithms, such as random sampling (Monte Carlo), simulated annealing, estimation of distribution, particle swarm and evolutionary algorithms. While quite different in their implementation and performance, they all share a set of common qualities beyond the two essential ingredients mentioned above. First of all, they can all be terminated at any given time to give the best result found so far. This is inherent to the idea of non-determinism: These algorithms generally never *finish* and can run on forever, unless they are halted at a specific point where the solution is deemed good enough. Additionally, all of these algorithms rely on randomly generating numbers. In random sampling for example, all candidates are simply created from scratch using random numbers entirely, which is why this algorithm is certainly the simplest, but, in many cases, the worst performing out of the ones mentioned above. The other ones mentioned above use random generation to a smaller, more controlled degree, and additionally involve the search space that was already sampled in order to obtain better results^[89].

This comparison plays into a concept that is central to non-deterministic optimization, the “No Free Lunch theorem” (NFLT). The fitness function in global optimization is, at each iteration step, generally evaluated only at a single point. This evaluation, possibly together with first and even higher derivatives, gives only local information about the solution candidate. Other than in local optimization, however, this local information is (usually) not at all useful to the global optimization, as the latter requires information far beyond the local space. Consequentially, the specifics of how this information is inferred by the various different algorithms has a large *problem-specific* impact on performance, e.g. depending on how the fitness function behaves. Thus, according to the NFLT, there is no GO algorithm that can beat other GO algorithms on an average over all possible problem classes. In essence, this means that there is no one-beats-all algorithm and an appropriate choice must be made for every specific problem class^[90].

Central to this work is the evolutionary algorithm (EA). It has many benefits that make it a very practical solution in molecular optimization^[91]: EAs are quite easily implemented, flexible and easily parallelized for high performance computing (HPC). On the downside, they are quite hard to configure correctly for the specific problem and require constant adaptation and modification in order to perform to their best potential. The specific implementations and modifications, as well as fitness function formulation of the EA used throughout this work are detailed in Sections 3.1, 3.2 and 3.3 below.

3

METHODS

3.1 OGOLEM

OGOLEM is a software package that was developed in Java and published by DIETERICH and HARTKE in 2010^[35]. It solves global optimization problems via an efficiently parallelized evolutionary algorithm. Originally, it was employed heavily in the optimization of cluster structures^[92–94] and is still, to date, used in this regard^[95–98]. It has also been used to solve high-dimensional fitting problems, such as ReaxFF parameterization^[99,100]. Lastly, it is also the basis for optimization of GOCATs^[34,37,47,101], which are inherently similar to the cluster optimization problem.

All of these cases share the same, underlying base functionality: At the beginning of the global optimization process, an initial pool of solution candidates is generated via random initialization. Each of these random solutions is evaluated according to a fitness function. In the case of cluster optimization, for example, this involves a local optimization with an efficient force field or semi-empirical method, in which the associated fitness value then is simply the final (locally optimized) potential energy. For parameter optimization and other problems such as the GOCATs, the fitness function can also involve more complex operations such as dynamics calculations, comparing against experimental data and more (see Section 3.3 for the GOCAT implementation).

Once a random pool is initialized, the iterative evolutionary algorithm (EA) is started. By selecting individuals from the pool (usually biased towards the better candidates) and performing cross-over and mutation operations, new candidates are created. These are, again, evaluated according to the fitness function and, if found better than those in the current pool, included to replace worse ones. This keeps the pool at a consistent size and gradually lowers the fitness in the pool, producing better and better candidates. Depending on the specific problem, the most important of these candidates is always the best one, since this will, after convergence, (hopefully) be the global optimum.

As mentioned in Section 2.3.1, the mutation and crossover operators used are generally tailored to the specific problem class. For this Thesis, the most important operators are those that are tailored to a 3D arrangement of particles. A well established crossover operator for such problems is visualized in Figure 3.1. This operator performs a crossover between two parent individuals by cutting across a plane that leaves an equal amount of

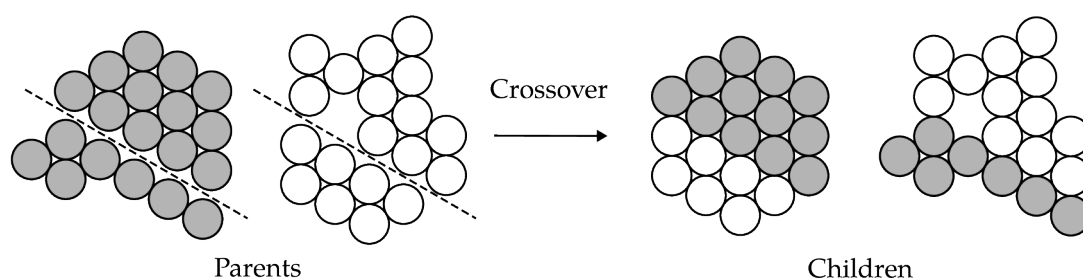


Figure 3.1: This scheme shows the main crossover operator used for particle arrangements in OGOLEM.

particles on each side for both parents, then swaps the parts above and below the plane to create two new candidates. This is a particular effective operator for such problems, as it can retain well structured particle arrangements across the crossover step. Mutations for this problem class are mostly random translations of single, some or all particles within a candidate. With charge or dispersion particles in GOCATs, this can also include randomly altering their inherent parameter values, such as their partial charge.

3.2 GOCAT Implementation

In this thesis, the vast majority of calculations done in OGOLEM are to optimize GOCATs; it is thus useful to briefly describe how this is done in detail. To optimize GOCATs successfully, each possible candidate in the optimization pool needs to have at least the following features:

1. the to-be-optimized abstract set of interaction entities,
2. a discrete set of structures on the *unperturbed* potential hypersurface leading from reactant to product,
3. a formalism to determine the energy of each of these structures on the *perturbed* potential hypersurface,
4. a fitness function that can utilize points 1-3 to ultimately score how well the GOCAT catalyzes its embedded reaction.

Point 1 is taken care of in initialization. The initial pool of candidates is generated automatically by randomly distributing a user-specified number of interaction entities around the embedded reaction. OGOLEM offers a variety of shapes for this initialization step, such as spheres, half-spheres or the combined van-der-Waals surface of the embedded reaction. Point 2 is also part of the initial human input; discrete points along a reaction

axis can be generated e.g. by NEB or IRC methods, which are briefly described in Section 3.4.

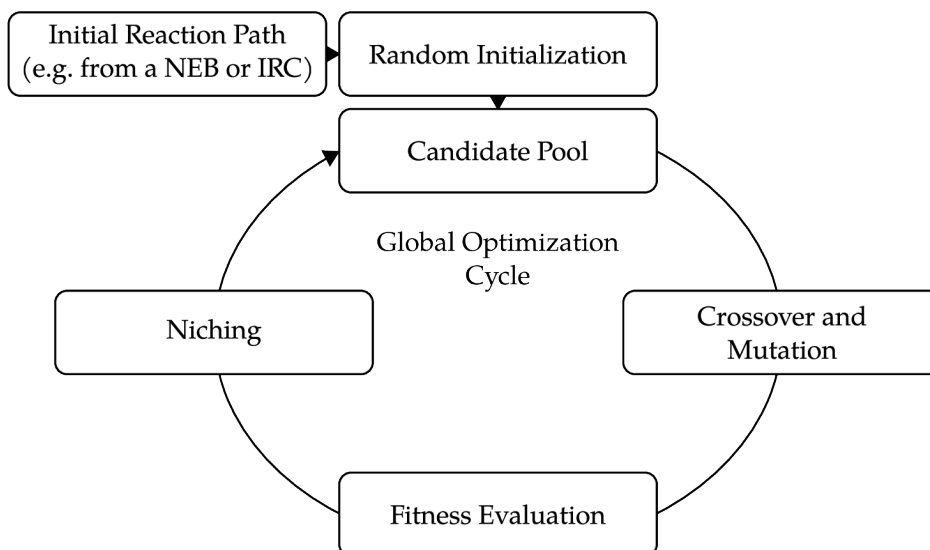


Figure 3.2: This graph shows a visualization of the global optimization cycle used to perform GOCAT optimization in OGOLEM.

Point 3 is also generally straightforward, because a QM/MM embedding functionality is already implemented in many popular quantum mechanical software packages. It usually involves passing the positions and partial charge values of the interaction entities to the software package as additional input. In practice, the global optimization in OGOLEM specific to GOCATs proceeds according to the schematic cycle shown in Fig. 3.2. Before the cycle starts, an arbitrary chemical reaction of interest is chosen and a NEB calculation is performed. The resulting discrete structures along the reaction path are then provided to OGOLEM. As a first step there, GOCATs are randomly initialized, as previously discussed. This fills the first candidate pool, usually consisting of up to 500 or more individuals. These initial solutions are, in most cases, not immediately catalytic with respect to the embedded reaction. With this first pool, the global optimization is then initiated and crossover and mutation operators are applied as discussed in Section 3.1.

Optionally, the pool can be divided into several sub-pools or niches, by applying a niching formalism. There are several ways to go about this. The two approaches that have been extensively used in this work are the following:

- **ESP Niching:** This approach works for any GOCATs containing charged interaction entities. The niche is calculated according to the electrostatic potential values generated by the charged entities at each atom in the reactant, transition state and product structures. This was also heavily employed by DITTNER in his work^[37].

- **SOAP Niching:** This approach was used primarily for the molecular GOCATs described in Chapter 6. It involves the use of the Smooth Overlap of Atomic Positions (SOAP) formalism^[102], which allows the classification of similarity of chemical environments. Further details are given in Section 3.5.

If niching is used, individuals compete only with their respective niche and thus, the pool, at any given point in time, will preserve more solution diversity. This allows further optimization in several promising catalytic regions at once, in a single pool^[94].

The cycle is run until a satisfying result is found. As previously discussed in Section 2.3.1, this does not necessarily have to be the global optimum, but is rather the best one found within a given number of iterations. To ensure a statistically good and sound solution, many different pools should be run at the same time and then bundled together at the end. Especially when niching is used during the global optimization process, the final pool can contain many potentially good solutions to the catalysis and must be treated with further, more problem-specific evaluation.

Fundamentally, the most important driver in this optimization process is the fitness function. As the later chapters will show, its formulation is very problem specific and requires a lot of fine-tuning and iterative adaptation, such that the global optimization properly converges to the desired solution. As a basis, the fitness function terms established in previous work^[34,47,101], introduced in the Section below, are modified and improved on a case-by-case basis that is described in more detail in the specific chapters.

3.3 Fitness Description

This section will briefly go over the basis fitness function Φ , which is used and modified in chapters 4-6. The term that all variants of this function include is

$$\Phi_{\Delta E} = 10 \cdot \Delta E_{\text{highest}} + \sum_i \Delta E_i, \quad (3.1)$$

which weights the largest energy barrier along the reaction path tenfold, and then adds a term that is the sum of all other energy barriers along the path, if there are more than one. Ideally, when the NEB is well optimized, there should only be a single energy barrier and thus, the term becomes $11 \cdot \Delta E$ for well-behaved candidates. This is the most important term, as it directly measures the catalysis achieved. Note that this also means that candidates are better when their fitness is *lower*.

In practice, this energy barrier term is generally not enough to measure the quality of a candidate; for any GOCAT variety that uses static, unchanging reaction paths, it is vital to keep the structures at the stationary points close to a true minimum, such that

the results remain sensible. This is achieved by introducing penalty terms involving the root-mean-squared gradient (RMSG) at the stationary points m :

$$\Phi_{\text{RMSG}}|_m = \begin{cases} \alpha \cdot (\text{RMSG} - \beta)^2 & \text{for } \text{RMSG} - \beta > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (3.2)$$

where the α is a scaling factor and β is a minimum threshold for the RMSG that needs to be crossed in order for the penalty to apply. The α is usually chosen at around 250, which makes the penalty overshadow $\Phi_{\Delta E}$ for values above β very quickly. The value for β is usually chosen to be around $10 \frac{\text{kcal}}{\text{mol} \cdot \text{\AA}}$, which is somewhat arbitrarily chosen to be considered "close" to a minimum. An additional term commonly used in the electrostatic GOCATs penalizes total energies that are higher than the reference at the reactant and/or transition state, with a similar construction as Φ_{RMSG} :

$$\Phi_{\text{stab.}} = \begin{cases} \kappa \cdot (E_{\text{R/TS}} - E_{\text{R/TS,ref}})^2 & \text{for } E_{\text{R/TS}} - E_{\text{R/TS,ref}} > 0 \\ 0 & \text{otherwise} \end{cases}. \quad (3.3)$$

In this case, the value for κ is usually chosen to be around 10^4 , such that candidates that do not fulfil this requirement are immediately classified as much worse than those that do. With these terms, DITTNER's original fitness function^[34] can be adequately summarized as:

$$\Phi_{\text{Static}} = \Phi_{\Delta E} + \Phi_{\text{RMSG}} + \Phi_{\text{stab.}}. \quad (3.4)$$

In the adaptive fitness function^[101], the fitness value itself is calculated from the same principle values. However, it is embedded into a more complex scheme that is shown in the schematic Algorithm 1. With this scheme, a new reaction path is calculated if Φ_{Static} is already good enough to be considered catalytic according to the old fitness function. This is done exclusively with the nudged elastic band (NEB) method introduced below.

Algorithm 1: Adaptive Fitness Function

```

Input: GOCAT,  $\Phi_{\text{Static}}$  (reference)
staticFitness =  $\Phi_{\text{Static}}$  (GOCAT);
if staticFitness >  $\Phi_{\text{Static}}$  (reference) then
    | return staticFitness;
end
newReactant = locallyOptimize(GOCAT.reactant);
newProduct = locallyOptimize(GOCAT.product);
if isConverged(newReactant) and isConverged(newProduct) then
    | newPath = calculateNewPath(newReactant, newProduct);
    | GOCAT.updatePath(newPath);
end
return  $\Phi_{\text{Static}}$  (GOCAT)

```

3.4 Reaction Path Modeling

One of the most important ingredients in the optimization of GOCATs is the reaction path. Reactant and product of the to-be-catalyzed reaction are generally easily defined and optimized, e.g. by semiempirical methods. On the other hand, the transition state, as well as the reaction path between the two, are not as straightforward to find. OGOLEM itself is indifferent to which method was used to generate a finished, human-input reaction path (e.g. by the popular IRC^[103] or any other method commonly used in this context^[104]). However, adaptive re-optimization of new reaction paths during the global optimization require a reliable way of automatically finding a reaction path and transition state between two locally optimized structures (a two-ended approach). Thus, something like an IRC would not work there.

The nudged-elastic-band (NEB) method is a solid solution to this problem and was therefore implemented in OGOLEM for exactly this purpose^[37]: It works by interpolating between the reactant and product structure and performing a path optimization to find the minimum energy path (MEP) along the reaction coordinate. This requires a finite set of discrete structures connecting the end points that are then connected in a band and optimized together on the potential energy surface.

As a first step, the reactant and product structures need to be interpolated to give an initial guess for the reaction path. For very basic reactions, this can be as simple as a linear interpolation scheme where the atom coordinates r_i of image i along the path are translated by a fraction of the vectors connecting the respective atoms in reactant and

product structures.

$$r_i = r_{\text{reactant}} + \frac{i}{N_{\text{images}}} (r_{\text{product}} - r_{\text{reactant}}) \quad (3.5)$$

This approach can, however, fail spectacularly for e.g. rotational and twisting motions, leading to overlapping atoms and other artefacts during path optimization. For this purpose, alternative interpolation schemes have been developed, such as the image dependent pair potential (IDPP)^[105]. In this scheme, instead of interpolating between the coordinates, the interpolation is done between all interatomic distances of the reactant and product:

$$d_i^{ab} = d_{\text{reactant}}^{ab} + \frac{i}{N_{\text{images}}} (d_{\text{product}}^{ab} - d_{\text{reactant}}^{ab}) \quad (3.6)$$

Where d^{ab} is the Euclidean distance between atoms a and b . Because these distances are essentially constraints that need to translate back to actual molecular geometry, a potential is generated from a weighted RMSD of the actual atomic distances to the idealized, interpolated distances:

$$V^{\text{IDPP}}(i) = \sum_a^N \sum_{b>a}^N \omega \left(d_i^{ab} - d^{ab} \right)^2 \quad (3.7)$$

The weight function ω can e.g. be chosen to enhance the potential additionally where distances are very small, such that very small distances are avoided in the finished path. This potential is then used to improve the linearly interpolated path^[105].

In the second step, each of the interpolated images i along the path are connected via a spring constant potential which forces the images to stay connected in Cartesian space. This connected band can then be relaxed on the true, underlying potential energy surface derived from e.g. quantum mechanical calculations, V_{PES} . During the relaxation, the force F_i acting upon each image would then be^[106]:

$$F_i = F_{\text{spring}}^{\parallel} + F_{\text{PES}}^{\perp} \quad (3.8)$$

With the force $F_{\text{spring}}^{\parallel}$ acting parallel to the band:

$$F_{\text{spring}}^{\parallel} = k [(r_{i+1} - r_i) - (r_i - r_{i-1})] \cdot \hat{\tau}_i \quad (3.9)$$

And the force $F_{\text{PES}}^{\perp}$ acting orthogonally to the band:

$$F_{\text{PES}}^{\perp} = -\nabla V_{\text{PES}} + (\nabla V_{\text{PES}} \cdot \hat{\tau}_i) \hat{\tau}_i \quad (3.10)$$

The perpendicular component, as well as the orthogonal component, are generated via projections onto the band tangent unit vector $\hat{\tau}_i$. It is given as:

$$\hat{\tau}_i = \begin{cases} r_{i+1} - r_i & \text{if } V_{i+1} > V_i > V_{i-1} \\ r_i - r_{i-1} & \text{if } V_{i+1} < V_i < V_{i-1} \end{cases} \quad (3.11)$$

This emulates always following the force down from the saddle point (the point of highest energy), which results in more stable alignment of the band^[107].

When all of the images are treated in this manner, the NEB images close to saddle points tend to optimize in such a way that they sit on the slope left and right of the true transition state, which is suboptimal if a well-optimized TS structure is desired. This can be remedied via the introduction of a climbing image (CI) at the highest energy image, which has a modified force description: by removing the parallel interaction $F_{\text{spring}}^{\parallel}$ and inverting the orthogonal component, this climbing image is maximized along the band tangent direction, which, close to a transition state, should be a good approximation to the reaction mode. This converges them to the true transition state much more reliably.

Lastly, the band can be optimized using any local optimizer of choice. The local optimization algorithms common in theoretical chemistry such as L-BFGS or FIRE all work in this context, however, throughout this work, FIRE was used exclusively. This is due to the fact that it was shown to perform very robustly and reliably in NEB optimization benchmark studies^[108].

For most of the re-optimization of NEBs in OGOLEM, a newer implementation of the NEB termed AutoNEB was used^[109], which automates how many images are interpolated between the reactant and product frame. By estimating where they are needed most and optimizing in small windows instead of interpolating and optimizing all images at once, the NEB becomes more efficient and more capable to run without user supervision and input.

3.5 Molecular Similarity

It is reasonable to assume that someone trained in chemistry can fairly easily tell whether two molecular structures are different or the same, given a molecule that is sufficiently small. For a computer however, this problem is significantly more complicated. Even for small molecules, the atoms could be ordered differently in the data files, or molecules could be translated or rotated in relation to each other, which would make a comparison between the two structures non-trivial. One of the most common and popular methods of structure comparison is the Root-Mean-Squared-Deviation (RMSD). To determine the RMSD between two structures A and B, the number and type of atoms and their ordering in both structures need to be identical; then, the center of masses of the two structures can be aligned and a rotation determined that minimizes the difference between each atom pair from structure A to structure B. This minimal Euclidian distance between the two structures is then a measure of similarity that is invariant to rotation and translation. While this is a perfectly acceptable method to compare structures that are known to be of

equal composition, it becomes a problem when the atom types vary from A to B. Additionally, the atom ordering becomes a detrimental issue once the structures are different enough that a 1:1 identity is unclear.

With the amount of data that theoretical chemists produce nowadays, and with the onset of machine learning, it has become obvious that new methods are needed. In fact, a whole host of methods has been developed over the past years, ranging from conceptually straightforward (like the Coulomb Matrix^[110]) to mathematically complex. One of the latter ones is the Smooth Overlap of Atomic Positions (SOAP) identifier^[102]. Compared to the RMSD, it is not only invariant to rotation, translation or atom ordering, but it can also compare structures of differing composition and atom number, which will be a highly important property in molecular GOCATs (Chapter 6).

To determine the SOAP identifier for a given structure, an atomic neighbor density function is constructed with Gaussian functions around the central atom with coordinates $\vec{r}$ and the other atoms with coordinates $\vec{r}_i$.

$$\rho(\vec{r}) = \sum_i \exp(-\alpha |\vec{r} - \vec{r}_i|^2) \quad (3.12)$$

This density is then expanded in a basis of spherical harmonics Y_{lm} and radial functions g_n with the respective expansion coefficients c_{nlm} :

$$\rho(\vec{r}) = \sum_{n,l,m} c_{nlm} g_n(\vec{r}) Y_{lm}(\vec{r}) \quad (3.13)$$

The power spectrum for a given molecular environment ξ is then given as:

$$P(\xi)_{n_1,n_2,l} = \pi \sqrt{\frac{8}{2l+1}} \sum_m (c_{n_1,l,m})^\dagger c_{n_2,l,m} \quad (3.14)$$

These elements (one for each of the combinations of maximum angular degree number l and permutation of radial functions n) can be put into feature vectors that describe the local molecular environment. The difference between structures can then be determined by taking e.g. the dot product of their respective normalized feature vectors, resulting in something close to 1 for similar structures and approaching 0 for structures that are increasingly different.

3.6 Choice of Semiempirical Method

GOCAT optimization, as reiterated in Section 1.1, involves a large number of potential energy gradient evaluations. Thus, a very efficient quantum mechanical method is required to minimize the time spent in fitness evaluation. In the past^[34,37,101], this role has been filled by the MNDO variant PM7, as implemented in the MOPAC2016 program package.

3.6.1 QM/MM Implementation

The QM/MM implementation in MOPAC2016 is straightforward: By calculating the electrostatic potential of all MM charges beforehand and adding them as an additional Coulomb potential on the diagonal one-electron Hamilton matrix elements $h_{\nu\nu}$, their general effect emerges during the SCF iterations^[143]:

$$\tilde{h}_{\nu\nu} = h_{\nu\nu} - \sum_{\text{MM}} \frac{q_j}{r_{ij}} \quad (3.15)$$

Simply adding a constant potential, however, implies that all orbitals centered on a given atom i experience the exact same Coulomb interaction, no matter their shape or size. Hence, the molecule is polarized as if the electric field around each atom were perfectly homogeneous. For large systems, such as enzymes, this has been shown to be an adequate approach^[111]. However, when the Coulomb potential generated by the charges neither enters a) the explicit integration of basis functions (even though it depends on the distance r to the atomic nuclei) nor b) the off-diagonal two-atom terms (which are typically non-zero even in MNDO methods), the polarization description simply becomes wrong or, at the very least, unphysical in the direct molecular picture.

This can be shown by a simple calculation: In Figure 3.3, density difference plots are shown for the three methods PM7, xTB and MP2 within two different setups. Both setups feature an oxygen molecule and two point charges with charge -1 (red sphere) and +1 (blue sphere). In the first case (linear), the two point charges are arranged in a linear fashion with the oxygen molecule, such that the potential at both atoms is the same value, but of different sign. In the second case (orthogonal), the charges are oriented orthogonally to the oxygen bond vector, such that the potential on the vector is exactly zero, but non-zero in the space around it.

Looking at the density difference with and without the point charges, all three methods show intuitive polarization in the linear case. The electron density is smaller in the vicinity of the negative charge (as denoted by the light blue isosurface) and larger in the vicinity of the positive one (as denoted by the light red isosurface). On the other hand, the polarization in the orthogonal case is simply unphysical for PM7 and xTB, as the density is completely unaffected by the strong charges in direct vicinity. For MOPAC, the reason for this is easily identified: Even though the electrostatic *field* is not vanishing at the atomic positions, the electrostatic *potential* is. Thus, the contribution to the Hamiltonian in equation (3.15) is zero.

In the xTB methods, the polarization occurs due to a Coulomb potential computed from the point charge embedding and the respective shell-wise Mulliken charges in a

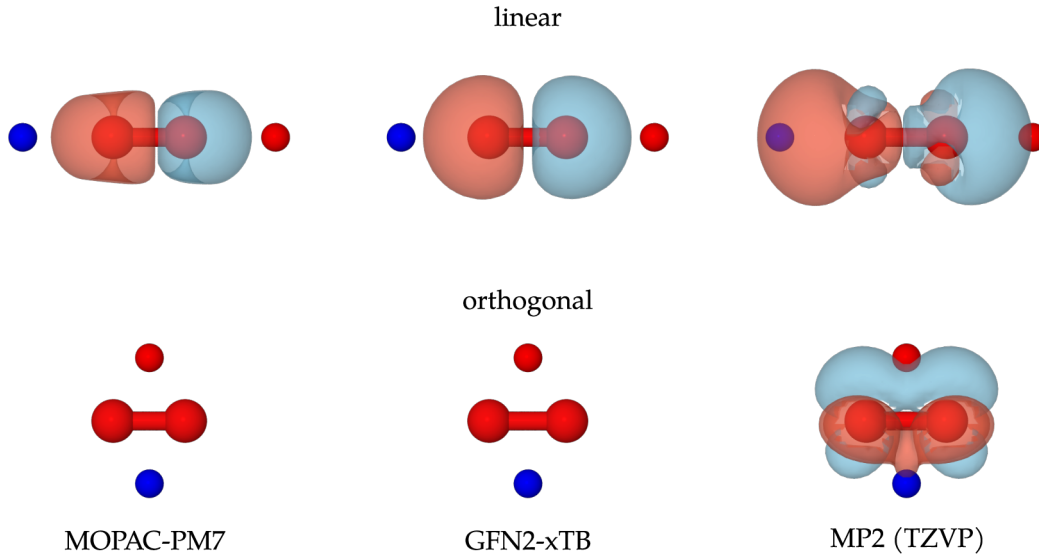


Figure 3.3: Two systems involving an oxygen molecule are shown here: In the top row, the linear setup consists of one charge of +1 positioned 2 Å away from the center of the oxygen bond in the negative x direction and another charge of -1 on the opposite side. On the bottom, the charges are positioned orthogonally to the bond vector at a distance of 1 Å. Shown for each of the three methods PM7, xTB and MP2 is the difference arising in the electron density when the charges are present vs. when they are not. For the MP2 density, the smooth MP2 density given in ORCA was used. All plots were generated by calculating the difference in each density grid, then dividing all grid values by the individual maximum and then using a cutoff value of 0.02 for the isosurfaces.

manner similar to the electrostatic energy term in the respective xTB Hamiltonian^[71]:

$$E_{\text{embedding}} = \frac{1}{2} \sum_A \sum_{p,l} q_{A,l} q_p \gamma_{A,l,p}, \quad (3.16)$$

where the sum is over all atoms A with shells l and point charges p . The point charge distance one would typically expect in a Coulomb potential is modified here according to the Mataga-Nishimoto-Ohno-Klopmann formula describing the average chemical hardness η of the given shells on atom A and the point charge:

$$\gamma_{A,l,p} = \sqrt{\frac{1}{r_{Ap}^2 + [\frac{1}{2}(\eta_{A,l} + \eta_p)]^{-2}}} \quad (3.17)$$

The $\eta_{A,l}$ in turn, are parameterized for each element and shell l . The chemical hardness of a point charge η_p is, by default in xTB, that of the carbon atom s shell. As noted, the individual (s, p and d) shells have different chemical hardnesses via

$$\eta_{A,l} = (1 + \kappa_A^l) \eta_A, \quad (3.18)$$

where κ_A^l is a shell-specific parameter. At the end of the day, this approach also works out to have the same underlying issue. Because the potential is simply assumed to be a

constant at each atom position, even for all interactions that depend on the distance from the nucleus, the description fails for this specific case.

In ORCA, the effect of point charges is implemented differently, most likely with explicit integration of the potential together with the basis functions, and thus yields correct polarization, as shown in the MP2 example¹ in Figure 3.3.

The takeaway here is that the underlying feature GOCATs rely upon, the QM/MM embedding, has its implementational issues in both xTB and PM7, and both can produce physically wrong results in some cases. This is something that should always be kept in mind when using these methods for models in which point charges are very close to the QM atoms.

A possible quick fix that could be made to PM7 would be to give the electric field, instead of the potential, at the atom positions. Then, a) the polarization would occur somewhat more uniquely given a certain point charge distribution and b) the problem outlined above, where the potential at the atoms is zero, would be fixed. Still, the best improvement would probably be to include the point charge embedding in the calculation of the basis function overlap integrals, which would require huge systematic changes to the MOPAC code base but would yield the most physical result.

In the end, the methods are used as-is throughout the work, accepting the trade-off of accuracy for speed and assuming with confidence that the polarization will nonetheless be transferable to higher-level QM calculations. Sadly, as will be shown later in Section 6.3.2, this is not always the case and can lead to issues.

3.6.2 Energy Barrier Determination

Semiempirical methods are parameterized for certain purposes and systems, so it is always good practice to verify their applicability to a specific reaction system beforehand. Both the PMx and xTB family semi-empirical methods are not especially great at correctly determining reaction barriers. Depending on the specific reaction, both can have either good or mixed results^[71]. However, especially when transition metals are involved in the reaction, the xTB methods pull ahead remarkably^[77].

The two proof-of-principle S_N2 and Diels-Alder reactions that have played a part in previous work^[37], as well as the proton exchange reactions in more recent applications^[47], warrant some attention here in retrospect. The former two will be discussed here, the proton exchange will be explored further in Section 6.3.2.

¹With larger basis functions, electron density spilling^[112] can become a problem, where the density is overpolarized due a strong interaction with the pure Coulomb charge with no Pauli repulsion. In context of minimal basis semi-empirical methods, however, this is not an issue that needs to be addressed at this point.

For the S_N2 example reaction,

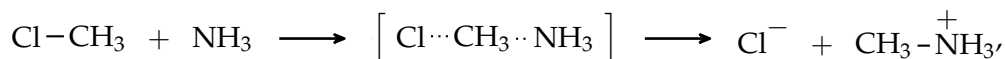


Figure 3.4 shows how the individual semiempirical and DFT methods compare to the high-quality DFT calculation. To obtain this graph, the reactant and product structures

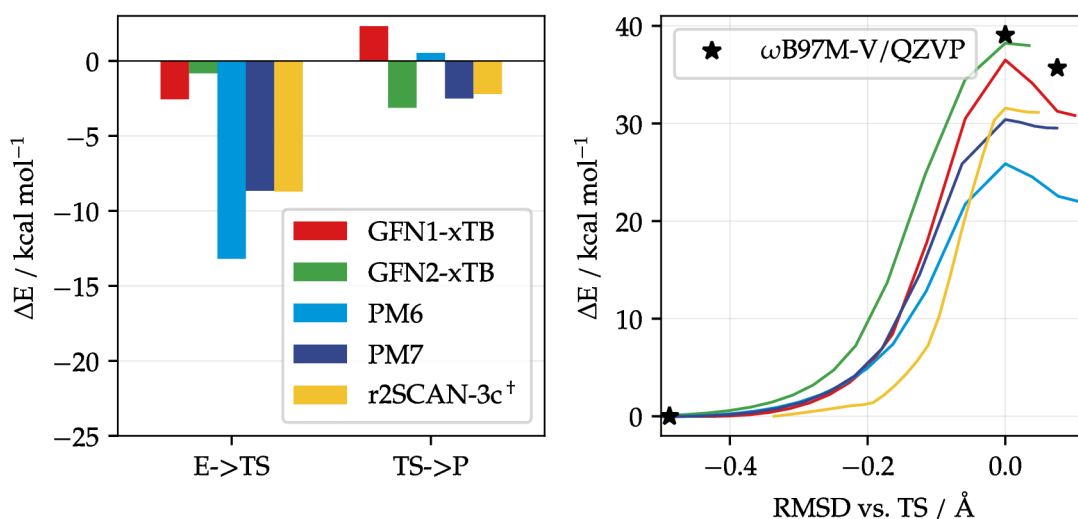


Figure 3.4: The left plot shows the difference in forward and backward energy barriers for the S_N2 reaction in relation to ω B97M-D4 (QZVP) DFT single points for different semiempirical and the r2SCAN-3c methods. The right plot shows the energy profiles for the individual methods with ω B97M-V (QZVP) energies of the stationary points inlaid. ([†] Note that the energy profile for r2SCAN-3c was generated via an IRC instead of a NEB calculation.)

were locally optimized using the respective method and then a NEB calculation was done to find the energy profile and transition state. For r2SCAN-3c, the energy profile was generated from an IRC instead, as a NEB could not be converged satisfyingly, and for ω B97M-V, no complete energy profile was calculated. Instead, the r2SCAN-3c structures at only the stationary points were taken and locally optimized to obtain accurate single point energies.

For the concerted Diels-Alder reaction between cyclopentadiene and maleic anhydride used in DITTNER's later studies^[101] (see Section 5.1 for additional details), equivalent data is shown in Figure 3.5. Just like the benchmark study done by the Grimme group^[71] suggests, the GFNx methods are outperformed by the PMx methods, and PM7 specifically, for Diels-Alder type reactions. This is also the case for this specific reaction. Interesting is the fact that r2SCAN-3c also seems to perform poorly in this case, giving worse results than PM7 compared to the ω B97M-V/QZVP reference, giving a clear edge

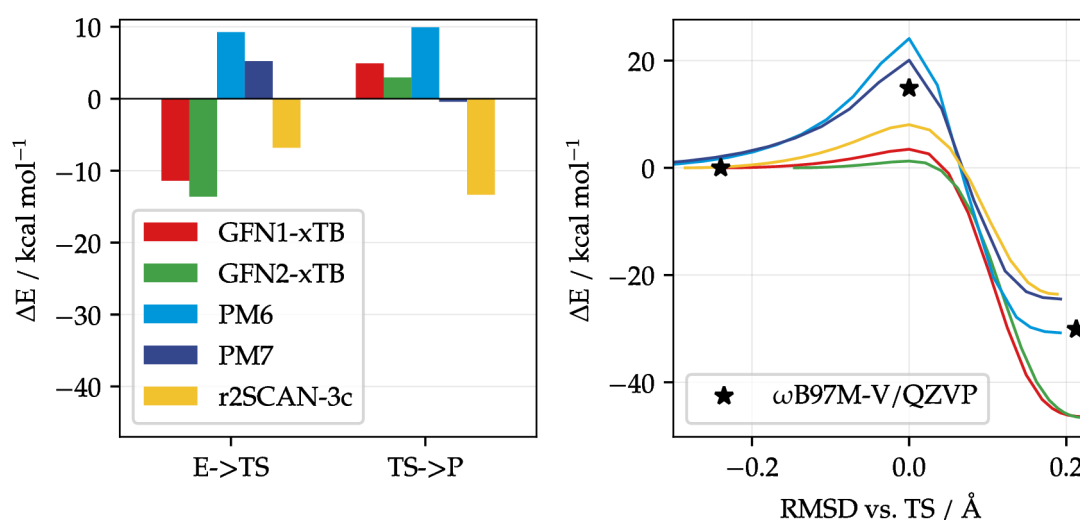


Figure 3.5: The left plot shows the difference in forward and backward energy barriers for the Diels-Alder reaction between cyclopentadiene and maleic anhydride in relation to ω B97M-D4 (QZVP) DFT single points for different semiempirical and the r2SCAN-3c methods. The right plot shows the energy profiles for the individual methods with ω B97M (QZVP) energies of the stationary points inlaid.

to the PMx methods for these types of reactions.

As outlined in Section 2.1.1, catalysis can be measured by the way it elevates reaction rate, which in turn is connected to lowering of the associated energy barrier. If one were interested in accurate predictions of the reaction rate, one would need very highly accurate (free) energy calculations. However, when optimizing GOCATs, this accuracy is not necessary. In fact, the most important features the underlying semiempirical methods need to deliver are accuracy in reaction mechanism, structure and how an embedding polarizes the reactants. This should lead to accurate enough GOCATs such that they are valid also for higher quality *ab initio* methods, from which more accurate reaction rates can then be calculated. Hence, throughout this work, the focus will lie mostly on finding clean, optimized structures and GOCATs that lower electronic energy barriers as much as possible, and not so much on the quantitatively correct reproduction of energy barrier heights. Since the comparison between GOCATs is always apples to apples, this should not be an issue, other than the QM/MM issue listed above. In many instances presented here, the GFNx methods were preferred over the PMx methods because they are more practical to work with and they perform good enough to be able to properly optimize GOCATs, even when the PMx methods perform better in reproducing barrier heights or even structures of higher quality methods.

STERIC EMBEDDING

The focus in the GOCAT model has mostly been put on the *electrostatic* picture in past works. However, the intermolecular forces in a QM/MM setting are not limited to only Coulomb interactions. Another large role is played by dispersion interactions, which could, in theory, just as well be utilized to produce a catalytic effect. The goal of this project is to elucidate whether this approach is fruitful in an abstract fashion, analogous to electrostatics, and whether or not this can complement or complete the former electrostatic-only treatment.

In the QM/MM scheme employed throughout this work, the dispersion forces do not enter the QM Hamiltonian in any way and instead are calculated in the MM part. This requires an appropriate parameterization of the QM part, as is common in an electrostatic embedding scheme. In a simple first approach to this dispersion GOCAT scheme, the method of choice was to use default OPLSAA^[113] Lennard-Jones parameters for both the QM and MM atoms.

4.1 Implementation

The global optimization procedure in OGOLEM in this case is mostly unchanged compared to the process described in Section 1.1. The only ingredients that need to change are the abstract particles that make up the GOCAT, because in contrast to the electrostatic case, they must now feature variable Lennard-Jones parameters instead of partial charges. This also necessitates some changes in the fitness function, because the static fitness Φ_{Static} as it is described in Section 3.3 does not retain the same information here. This is due to the fact that, as mentioned previously, the presence of the LJ entities does not change the QM energy or structure. Thus, having catalysis result from non-locally-optimized structures is virtually impossible. The modified fitness function is described schematically in Algorithm 2. This is almost identical to the adaptive fitness function in Algorithm 1, with the exception of the first two if statements. Instead of comparing to a reference fitness value, the candidate is discarded immediately if the initial gradient norm at the stationary points is too large (with a fairly high τ_1 of around $1.5 \cdot 10^4$). If this threshold is met, the candidate is evaluated as normal, however not compared to the reference fitness, but to an arbitrarily set τ_2 (for the aforementioned reasons) which is usually chosen to be in

Algorithm 2: Adaptive Fitness Function

```

Input: GOCAT
if RMSG(stationary points) >  $\tau_1$  then
    |   return non-converged;
end
staticFitness =  $\Phi_{\text{Static}}$  (GOCAT);
if staticFitness >  $\tau_2$  then
    |   return staticFitness;
end
newReactant = locallyOptimize(GOCAT.reactant);
newProduct = locallyOptimize(GOCAT.product);
if isConverged(newReactant) and isConverged(newProduct) then
    |   newPath = calculateNewPath(newReactant, newProduct);
    |   GOCAT.updatePath(newPath);
end
return  $\Phi_{\text{Static}}$  (GOCAT)

```

the region of 2 or 3 times the reference fitness.

Additionally, a check was added after the local optimization of the new reactants and products which ensures that no bond changes have occurred during the optimization. This is done by assigning bonds depending on the interatomic distance compared to the atomic radii. If any changes are found versus the reference, the candidate is discarded. This is important since, especially at the start of the global optimization procedure, the random candidates might have very large (high σ) LJ entities in close vicinity to bonds, effectively pushing atoms apart and invalidating the candidate. This is undesired, and sorting these out as soon as possible improves the overall efficiency of the optimization.

The variable Lennard-Jones parameters of the abstract entities are, just like the partial charge values in the electrostatic case, modified during the global optimization via a mutation operator that randomizes their value, but respects boundaries set in the input. These boundaries were usually taken as between 0 and 0.8 kcal mol⁻¹ for ϵ and 0 and 5 Å for σ . Compared to the default OPLSAA parameterization, this means the upper boundary in size is about the size of a K⁺ ion and the deepest energy well is about as deep as that of a Mg²⁺ ion.

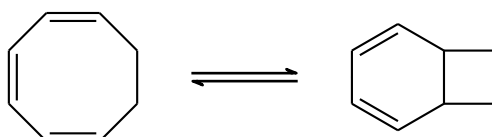
One final point to consider is the placement of these Lennard-Jones entities. As they are held in place during the entire fitness evaluation, some additional care must be taken. Significant rearrangements in the embedded molecules can only be achieved if the LJ en-

tities are positioned freely initially, even in some cases too close to the QM atoms, such that relaxation will modify the QM structure. However, in the fully relaxed structures, it would be highly undesirable to have LJ entities be too close to atoms, because then the repulsive part of the LJ potential could be able to apply unlimited force to the QM atoms, giving unphysical results. For the initial examples presented however, this turns out to not be a problem, since the penalty placed on gradient norms already prevents candidates with this particular defect. In Section 4.2.4, this will be discussed in more detail as the effect becomes more problematic with an increasing number of LJ entities.

4.2 Cyclooctatriene

In this section, the electrocyclic (or valence) isomerization of cyclooctatriene is presented as a proof-of-concept reaction. Generally, the idea behind this example was to find a chemical reaction system that has multiple similar minima across its potential energy surface which are accessible by ground state chemistry. Optimally, some of the reactions should involve heavy structural rearrangement, such that steric effects can play a major role. The expectation here is that catalysis is most easily attained when this is the case, and not as easily when it is not. Additionally, steric catalysis could offer the ability to discover other pathways between minima that make previously impossible interconversions possible and even competitive.

Cyclooctatriene is a molecule that can undergo an internal ring closure by ways of thermal cycloaddition^[114,115]:



The equilibrium of this reaction is known to be mostly on the left side at 100°C, with an estimated difference in free energy of only about 1.5 kcal/mol^[116] between the two isomers. A B3LYP/def2-QZVP DFT calculation that was done to compare the energy levels of the individual isomers of cyclooctatriene agrees with this estimate ($\Delta G = 1.61$ kcal/mol).

In Figure 4.1, the different isomers of cyclooctatriene are shown with their respective r^2 SCAN-3c, GFN2-xTB, wB97M-V/def2-QZVP, PM7 and B3LYP/def2-QZVP energies. Note that these are the electronic energies, not free energies. The structures for each of the isomers are shown in Figure 4.2. As can be seen in Figure 4.1, all methods except B3LYP/def2-QZVP predict the *cis* closed-ring isomer (cisP) to be the most stable structure, which is somewhat acceptable considering the small difference between the two. Throughout all structures, GFN2-xTB slightly overestimates the energetic difference. It

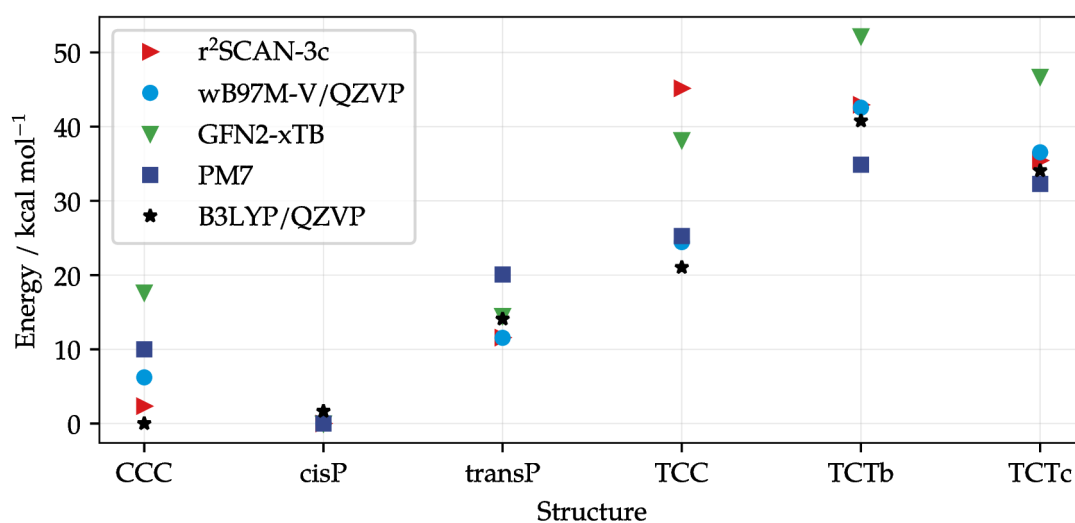


Figure 4.1: This electronic energies for all 4 open-ring isomers (CCC, TCC, TCTb and TCTc) and both closed-ring isomers (cisP, transP) of cyclooctatriene were calculated using the 5 methods r^2 SCAN-3c, GFN2-xTB, PM7, ω B97M-V/def2-QZVP and B3LYP/def2-QZVP. The energy of each isomer is given in relation to the lowest lying one.

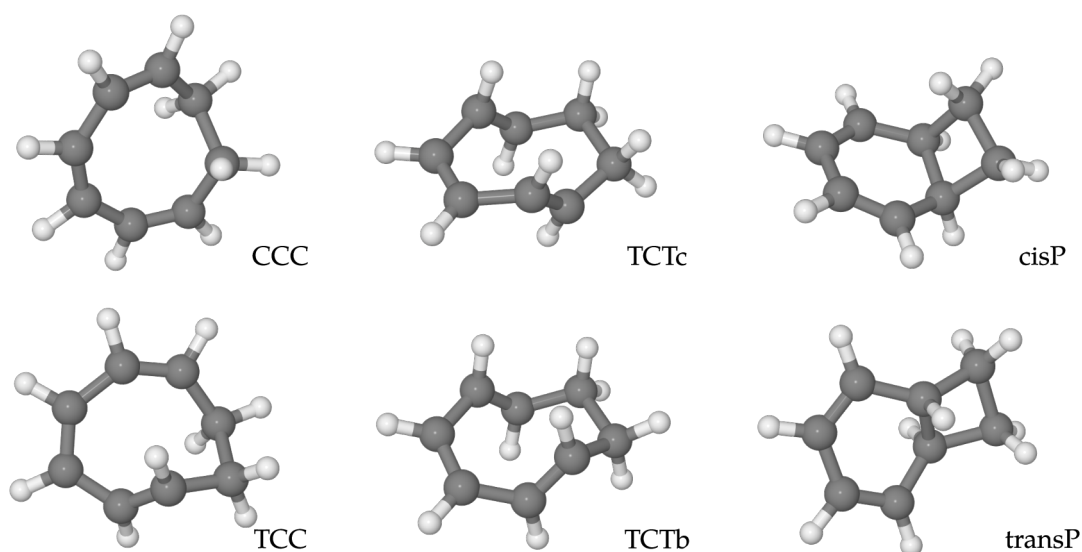


Figure 4.2: All isomer structures of cyclooctatriene shown here were optimized using r^2 SCAN-3c.

is, however, correct in their qualitative ordering, with one exception: The all-*cis* isomer (CCC), which is the most stable of all open-ring structures, is estimated to be less stable than the *trans* isomer of the closed-ring structure (transP).

After careful consideration, GFN2-xTB was chosen as the semi-empirical method here regardless. This was done specifically because all isomerization reactions mentioned in

this section have a clean transition state in GFN2-xTB and very well optimized NEBs could be obtained. This was not the case for the other semiempirical codes mentioned before (PM6, PM7 and GFN1-xTB).

The specific energy relations aside, cyclooctatriene is a good candidate considering the conditions for a good proof-of-principle reaction mentioned above. First of all, the isomerization from an open-ring to a closed-ring isomer can proceed via a host of different pathways and secondly, all open-ring isomers possess at least some flexibility in their structure that might allow steric influence to lead to successful catalysis. Before any GOCATs can be optimized however, the starting reaction path(s) must be selected. Because the optimization of GOCATs will be done with GFN2-xTB as the QM method, this was done entirely in GFN2-xTB. Table 4.1 shows all possible combinations of isomer interconversion and the respective energy barrier predicted by xTB, if applicable and a true transition state was found.

	CCC	TCC	TCTc	TCTb	cisP	transP
CCC		40.13	X	X	33.02	X
TCC	60.73		X	X	X	39.46
TCTc	X	X		40.06	X	X
TCTb	X	X	45.41		54.73	X
cisP	16.52	X	X	2.57		X
transP	X	15.77	X	X	X	

Table 4.1: In this table, all possible combinations of cyclooctatriene isomers are listed. Reaction barriers are given for a one-step thermal isomerization from the reactant listed in each column to each product listed in the rows. 'X' denotes that there is no transition state for this pair. All values were obtained from GFN2-xTB NEB calculations and are given in kcal mol⁻¹.

Looking at this table, every open-ring isomer except TCC has a lower energy barrier *towards* the *cis* closed-ring form than it has from it, cementing the fact that in GFN2-xTB, this is the lowest energy isomer. Another interesting fact is that all isomers are, in fact, interconnected by thermal isomerization reactions. For a more complete picture, Figure 4.3 shows the potential energies of all isomers, together with the respective transition states connecting them. The GFN2-xTB structures of these transition states are also shown in Figure 4.4.

For the GFN2-xTB reactive paths, the three highest energy barriers are CCC→TCC, cisP→TCTb and TCTc→TCTb. If one compares the RMSD values between each of the isomers, the most change occurs from the CCC→cisP isomerization, with an RMSD of 1.736 Å between reactant to product structure. The second highest is the TCC→transP

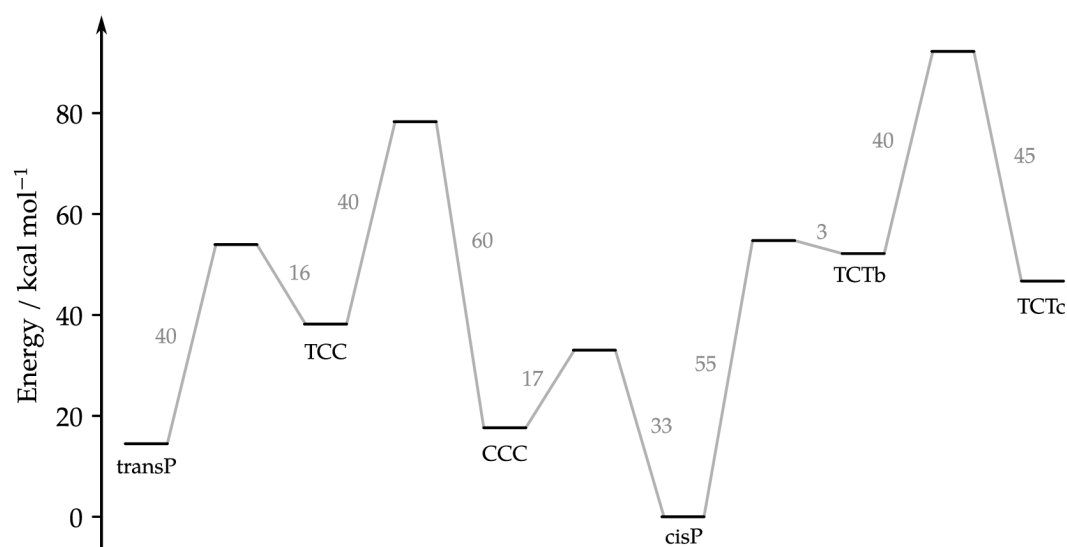


Figure 4.3: This graph shows the total predicted potential energies for all isomers using GFN2-xTB. For all isomerization reactions between them, the energy of the transition state structure, as well as the respective barrier associated with the transition, is given in kcal mol^{-1} .

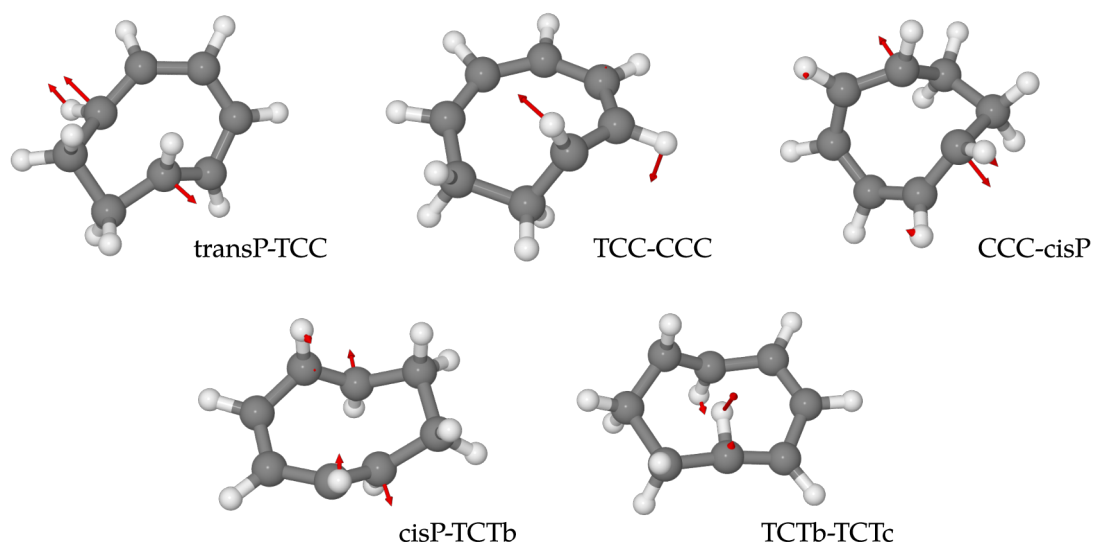


Figure 4.4: These structures are the GFN2-xTB optimized transition state structures for all isomerization reactions between the isomers of cyclooctatriene. The respective normal modes associated with the reaction coordinate are depicted as red arrows.

transition with $1.216 \text{ \AA}$ and the third highest is $\text{CCC} \rightarrow \text{TCC}$ at $1.074 \text{ \AA}$.

Three of the isomerizations were chosen for further investigation in GOCAT optimizations: First up is $\text{CCC} \rightarrow \text{cisP}$, as it involves the most structural change and the idea is to catalyze via dispersion interactions only, so the structural difference between the struc-

ture could play a large role in the efficiency of the catalysis. Next is the isomerization from CCC→TCC. It is the only pathway that can form the *trans* closed-ring isomer, and it also involves the largest energy barrier found in the connected reactive system, with far fewer structural changes. Lastly, the cisP→TCTb isomerization was chosen because the reaction proceeds very readily in the other direction, and the transition state is structurally almost identical to the product, which is not the case for the other two isomerizations.

4.2.1 Isomerization from CCC → cisP

The xTB NEB between the CCC and the cisP isomer is shown in Figure 4.5. Inset in this

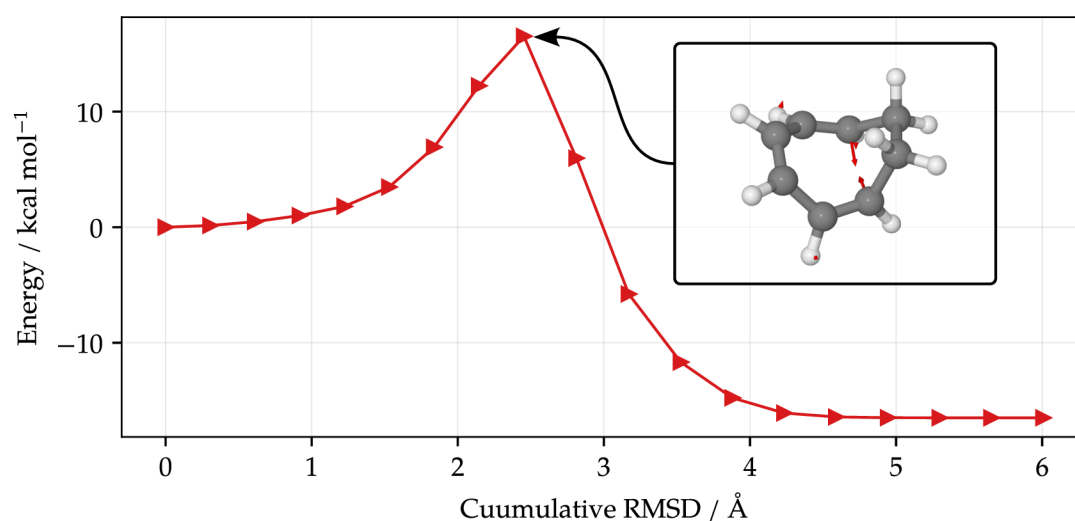


Figure 4.5: This energy path between the CCC and cisP isomer of cyclooctatriene was calculated using GFN2-xTB. The inset shows the structure at the transition state, with arrows indicating the direction of the normal mode pointing along the reaction coordinate.

Figure is the respective transition state structure. The latter is already bent in a boat-like structure, with both of the sp^3 carbon atoms bent in the same direction, unlike the relaxed CCC structure, where they are out of plane with the quasi-planar sp^2 triene carbon atoms. The RMSD between the reactant and transition state structure is 0.47 Å, between transition state and product structure it is 0.6 Å. Thus, structurally, the transition state is slightly closer to the reactant structure.

This reaction path was used as input for a first GOCAT optimization run using 10 LJ entities. For this purpose, 5 different pools containing 600 candidates each were optimized for around 10^5 iterations and then pooled back together for evaluation. Figure 4.6 shows how the three stationary structures in the final candidates have changed in comparison to the uncatalyzed reference by comparing their individual RMSD values. The graph shows

that the solutions are all very similar with respect to how they influence the structure of the reacting molecule. The reactant structure is obviously the target here, as it is changed the most out of the three structures throughout all solutions in the pool. The product structure is affected the second most, while the transition state itself remains essentially untouched. Beyond this, a quirk of the way the evaluation was done here is also visible: because 5 different pools were brought together here, at least three distinct distributions are found that have slightly varying fitness values, with most candidates having worse fitness than the respective best ones which stand out at lower fitness values.

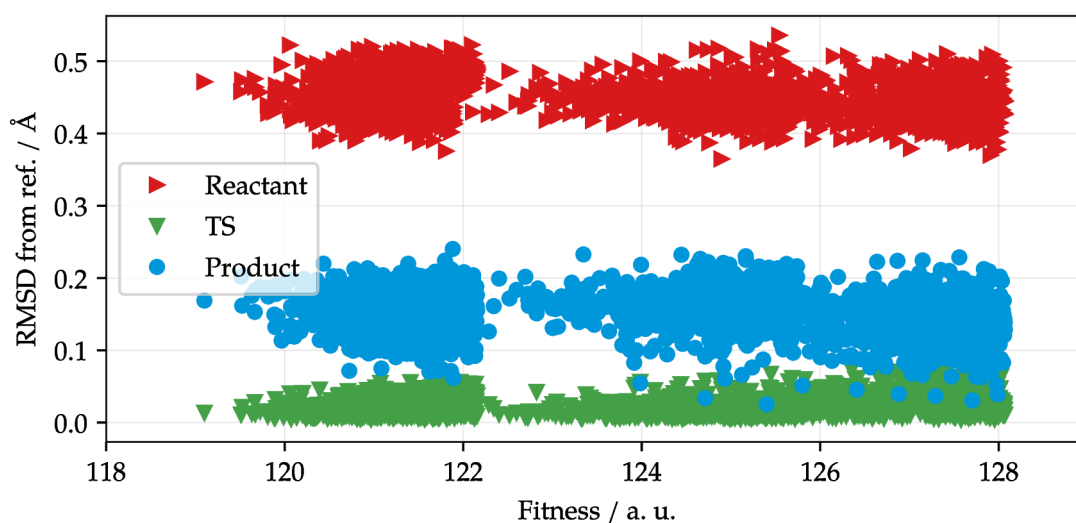


Figure 4.6: This plot shows the RMSD between reference and GOCAT embedded structure for each of the stationary points of the $\text{CCC} \rightarrow \text{cisP}$ isomerization of cyclooctatriene throughout the final GOCAT pool. The x axis is the fitness of each candidate, with a lower fitness corresponding to a better solution.

If one brings this together with Figure 4.7, which shows the energy profile of the uncatalyzed reference together with those of the best and worst of the optimized GOCAT pool, as well as with the structure shown in Figure 4.8, it paints a clear picture of how the catalysis is achieved. The embedding acts as a stabilizing agent for a new reactant structure that is structurally much closer to the transition state than in the reference case. With this new minimum, both the structural rearrangement as well as the energy barrier associated with the isomerization are lowered. In the best candidate, the energy barrier is lowered from 16.52 to 10.24 kcal mol⁻¹, while the reactant is slightly destabilized by 0.68 kcal mol⁻¹. The contributions from the LJ entities are strongest at the transition state, which is visible if one plots the best candidate's QM and LJ contributions to the total energy separately, as done in Figure 4.9. The resulting NEB is also *shorter*, as evidenced

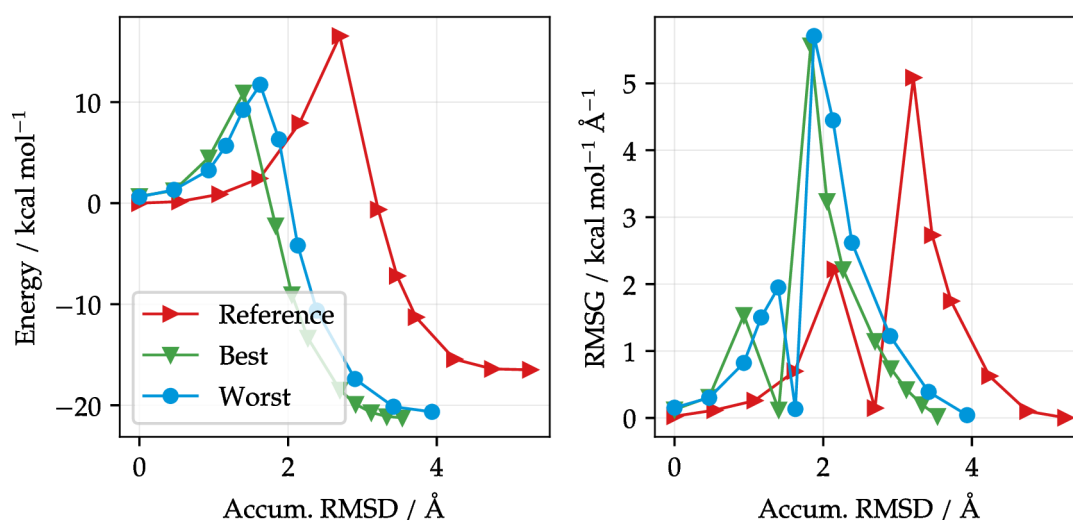


Figure 4.7: On the left, the energy paths for the CCC→cisP isomerization of cyclooctatriene is shown for the uncatalyzed reference, as well as for the best and worst candidate in the final pool. All energy levels along the paths are given in relation to the energy of the reference reactant, which is set to 0. The right plot shows the residual gradient at the structures, indicating that the equilibrium structures are indeed well relaxed in all three cases.

by the reduced accumulated RMSD required to get from reactant to product. Figure 4.8 shows, in addition to the distorted structures in the optimized embedding, the potential generated by the LJ entities. The embedding forms a very tight pocket around the bent reactant, effectively keeping it trapped for the entirety of the isomerization.

4.2.2 Isomerization from CCC → TCC

The xTB NEB between the CCC and the cisP isomer is shown in Figure 4.10. If one calculates the direct RMSD between the reactant and transition state structures, and not the accumulated one that is given in the Figure, it gives 0.78 Å, which makes the transition state of this reaction structurally slightly more similar to the product with only 0.67 Å between the latter two, which reflects with a later transition state along the reaction axis. Overall, the structural change required to get towards the transition state and away from it is notably higher than the difference between the reactant and product structure directly (for which the RMSD is 1.074 Å). This indicates that the isomerization requires a significant *detour* on the potential energy surface. Analogous to the previous example, 5 different optimization runs were done, each with a pool of 600 candidates for a total of about $7 \cdot 10^4$ iterations each. Figure 4.11 shows, again, how the three stationary points in the optimized candidate differ structurally from the reference path. In the direct com-

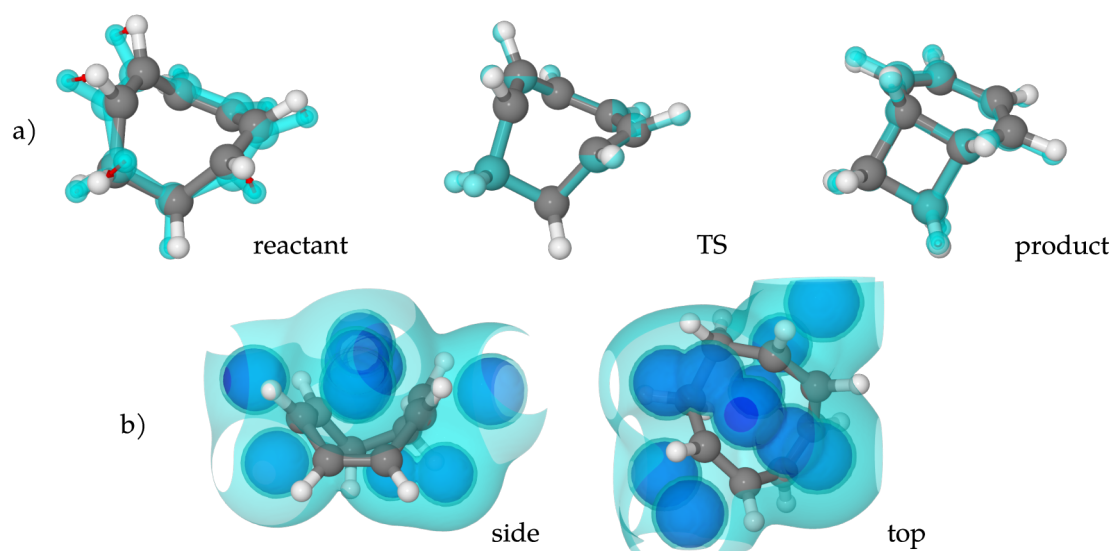


Figure 4.8: This Figure shows the structures from the best candidate found in the GOCAT optimization with Lennard-Jones entities for the CCC $\rightarrow$ cisP isomerization of cyclooctatriene. Part a) shows specifically the reactant, TS and product structures along the embedded reaction path with the original, uncatalyzed reference shown in light blue. Red arrows additionally point from each reference atom to the respective atom in the catalyzed structure. Part b) shows isosurfaces of the combined GOCAT Lennard-Jones potential, with the repulsive part shown in dark blue (cutoff 0.05) and the attractive part in light blue (cutoff 0.05).

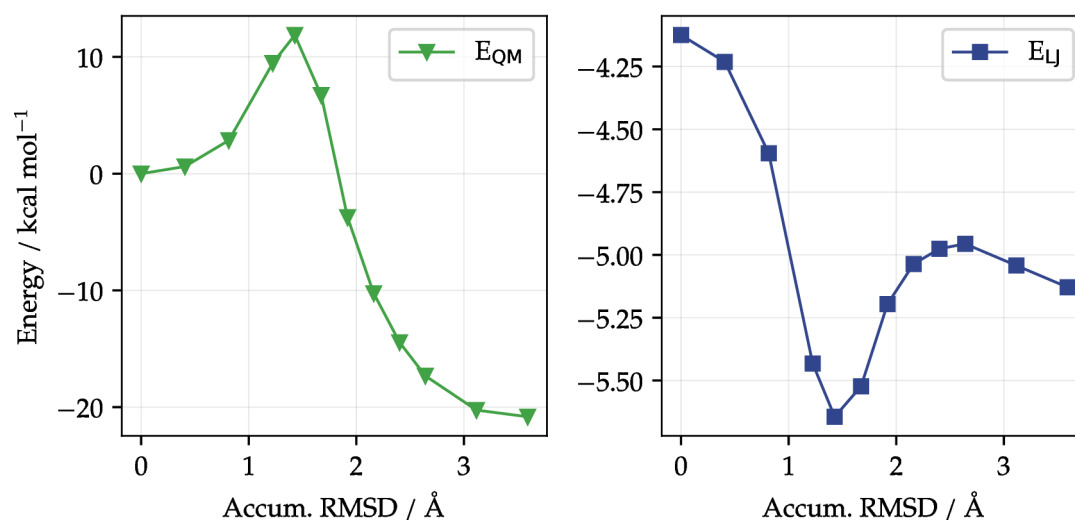


Figure 4.9: In this plot, the energy contribution terms from the PM7 hamiltonian (left) and the LJ entities (right) is plotted separately for the entire reaction energy profile. Note that here, the QM energy (left plot) is given in relation to the reactant energy, which is set to zero.

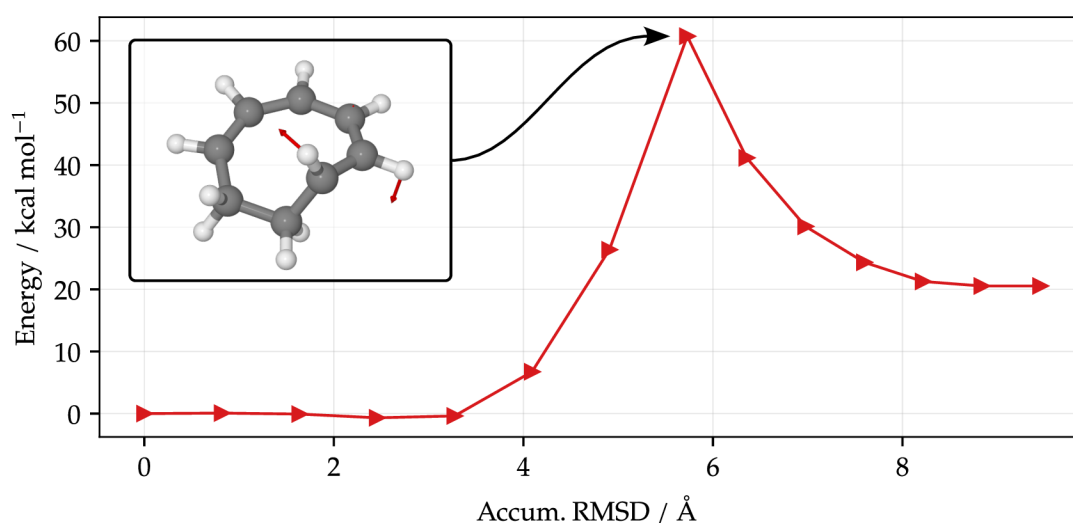


Figure 4.10: This energy path between the CCC and TCC isomer of cyclooctatriene was calculated using GFN2-xTB. The inset shows the structure at the transition state, with arrows indicating the direction of the normal mode pointing along the reaction coordinate.

parison to Figure 4.6, one can see that the solution pool is much more inhomogeneous. Additionally, there is no such clear preference over stabilization of a distorted *reactant* structure. Instead, both the reactant and product structure are affected significantly. One feature shared with the CCC→cisP solutions is that the transition state structure once again remains essentially untouched here.

In Figure 4.12, the energy profiles of the reference, together with those of the best and worst candidates in the final pool are shown. The effect of shortening the reaction path is much less obvious in this case, and the catalytic effect itself is a lot weaker. In the previous case, the isomerization of CCC→cisP, the barrier was reduced by $6.3 \text{ kcal mol}^{-1}$, which corresponds to a relative reduction of 38%! By contrast, the massive barrier of 61 kcal mol^{-1} for the CCC→TCC isomerization could only be lowered by $4.7 \text{ kcal mol}^{-1}$, which is a mere 8% decrease from the reference value.

Structurally, the catalysis appears to be facilitated by stabilizing the reactant in a structure which is closer related to the transition state. In comparison to the CCC→cisP isomerization, however, this effect is much less pronounced, and less catalytic effect results from the stabilization. In Figure 4.13, the difference in energy barrier between each optimized candidates and the reference is plotted against 4 distinct differences in RMSD: The top two graphs shows the difference in RMSD between reactant and transition state in catalyzed vs. uncatalyzed reaction, as well as between transition state and product in the second graph. Both of these are for the CCC→cisP isomerization. For CCC→cisP, there is

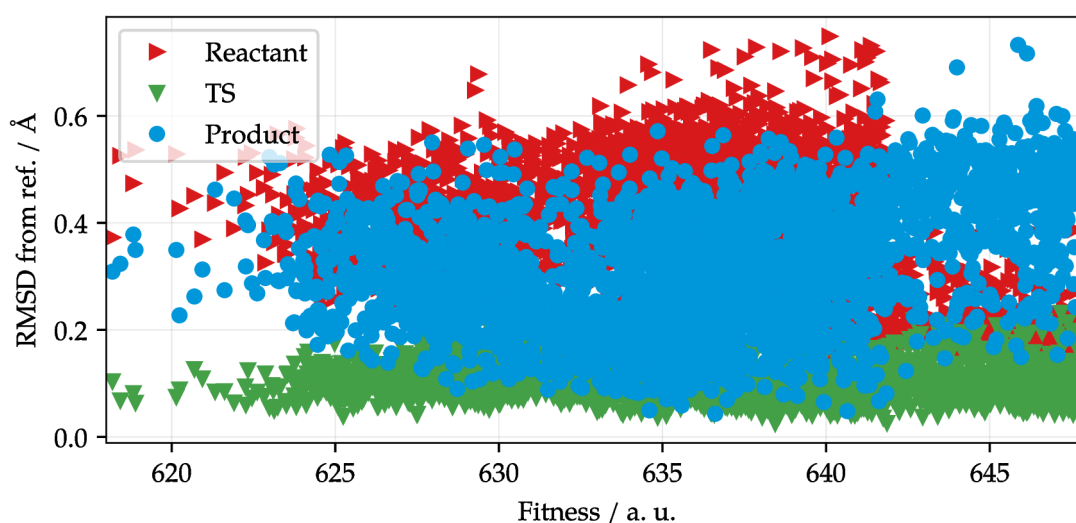


Figure 4.11: This plot shows the RMSD between reference and GOCAT embedded structure for each of the stationary points of the CCC→TCC isomerization of cyclooctatriene throughout the final GOCAT pool. The x axis is the fitness of each candidate, with a lower fitness corresponding to a better solution.

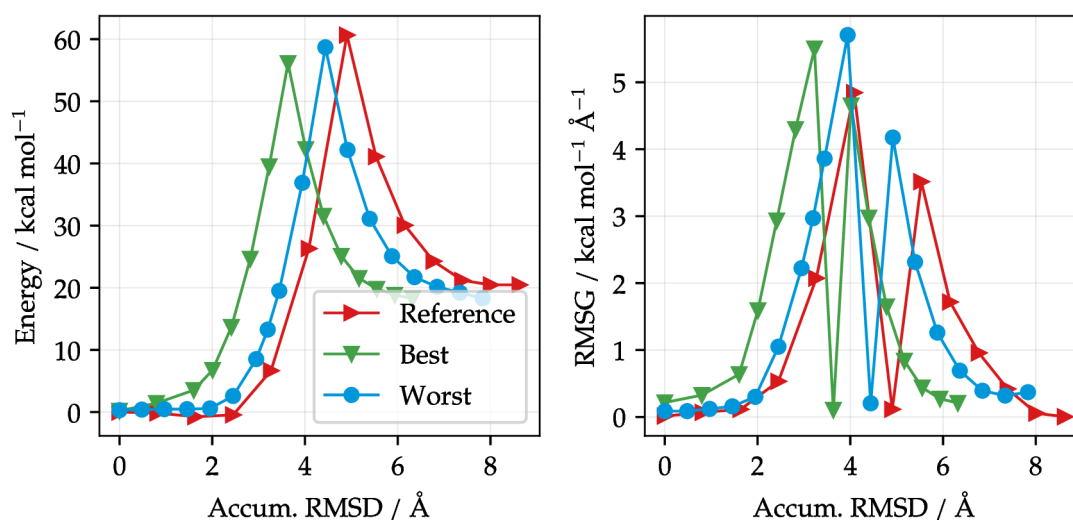


Figure 4.12: On the left, the energy paths for the CCC→TCC isomerization of cyclooctatriene is shown for the uncatalyzed reference, as well as for the best and worst candidate in the final pool. All energy levels along the paths are given in relation to the energy of the reference reactant, which is set to 0. The right plot shows the residual gradient at the structures, indicating that the equilibrium structures are indeed well relaxed in all three cases.

an obvious correlation between barrier reduction and stabilizing a reactant structure that is closer to the transition state, but not for increased similarity between product and tran-

sition state. For the CCC→TCC isomerization, this is much less clear. Both plots show essentially no clear correlation between the distortion of structures and barrier lowering, suggesting that the mechanism of catalysis here is, in fact, different.

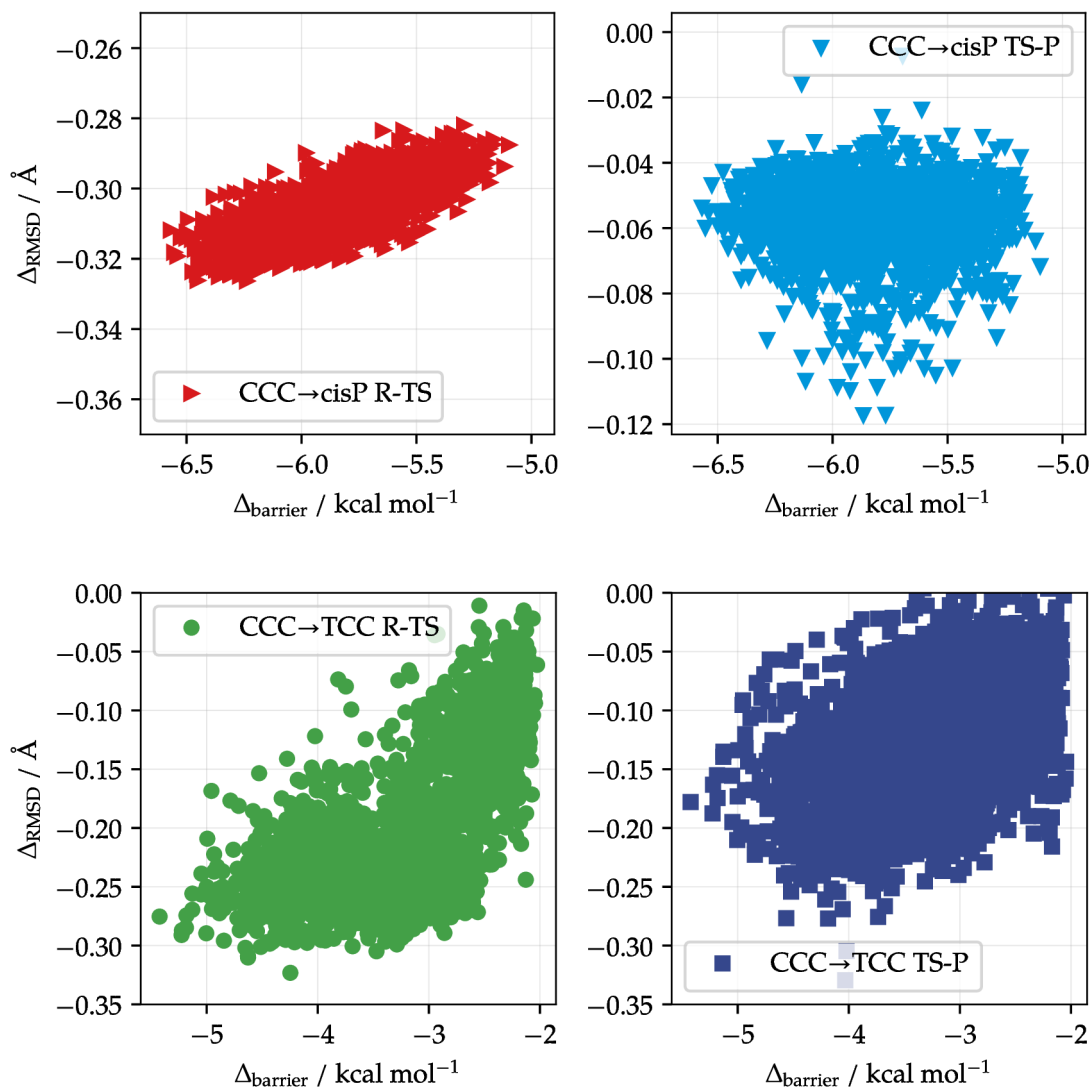


Figure 4.13: In these four plots, the difference in RMSD is plotted against the difference in energy barrier for the reactant to transition state and transition state to product structure, respectively for the CCC→cisP and CCC→TCC isomerizations.

4.2.3 Isomerization from cisP → TCTb

Compared to the other two isomerizations, this one has a notably late transition state that is very similar to the product. The corresponding structure and reaction path are once again shown in Figure 4.14.

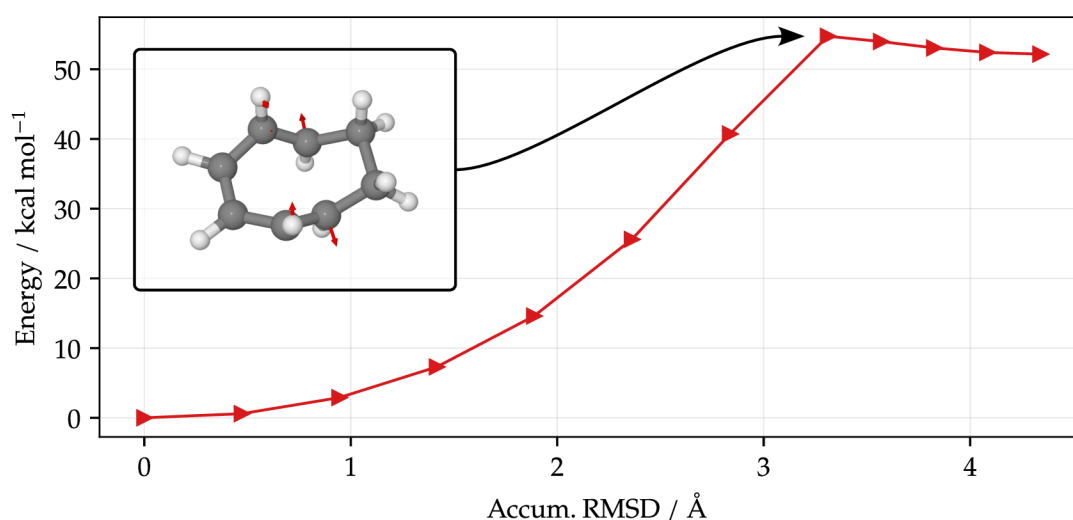


Figure 4.14: This energy path between the cisP and TCTb isomer of cyclooctatriene was calculated using GFN2-xTB. The inset shows the structure at the transition state, with arrows indicating the direction of the normal mode pointing along the reaction coordinate.

After an analogous global optimization procedure with 5 pools of each 600 candidates optimized for about $8 \cdot 10^4$ iterations each, the results are shown below. In Figure 4.15, the RMSD values from the optimized structures to the reference structures are plotted against the fitness.

Compared to the previous two cases, the deformation of the structures is even less pronounced, with overall smaller RMSD values, as well as fewer differences between the three stationary points. This is expected, since out of all isomerizations, this one has the second highest barrier, but the least difference between reactant and product structure. Additionally, Figure 4.16 shows that the minuscule deformation of the reactant and product structure towards the transition state shows only very weak correlation with the barrier reduction.

Overall, the best solution, of which the energy path is shown in Figure 4.17, lowers the energy barrier by $6.2 \text{ kcal mol}^{-1}$, which brings the transition state energy down to almost that of the product structure. For the reverse reaction, this would mean the barrier is essentially removed, making any further catalysis dependent on also further stabilizing the product structure.

The small impact on the reaction geometry combined with the comparable lowering of the barrier suggests that catalysis in this case is mostly achieved through stabilization of the transition state geometry via favorable van-der-Waals interactions. In fact, the GOCAT itself is shaped like a pipe (Figure 4.18), favorably interacting with both the product and

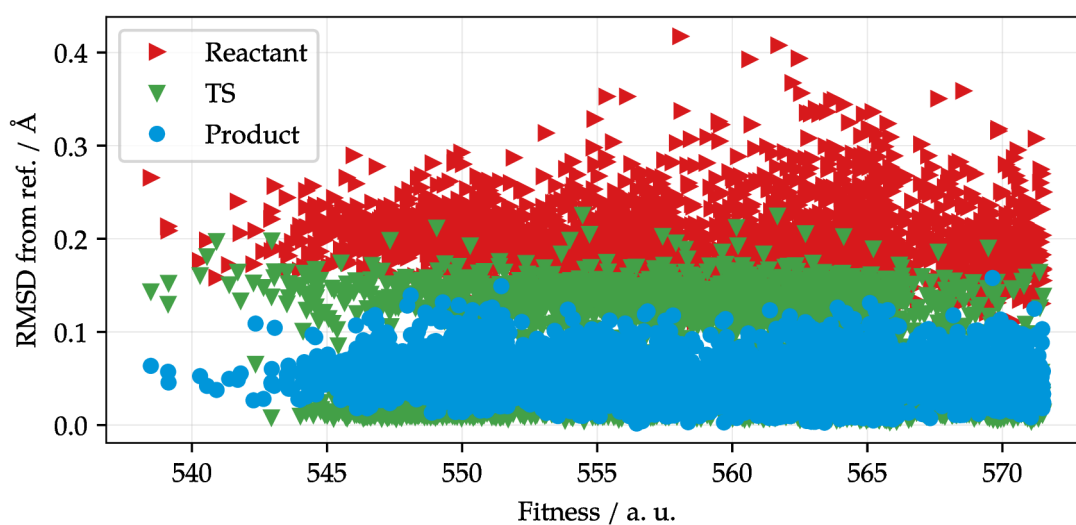


Figure 4.15: This plot shows the RMSD between reference and GOCAT embedded structure for each of the stationary points of the $\text{cisP} \rightarrow \text{TCTb}$ isomerization of cyclooctatriene throughout the final GOCAT pool. The x axis is the fitness of each candidate, with a lower Fitness corresponding to a better solution.

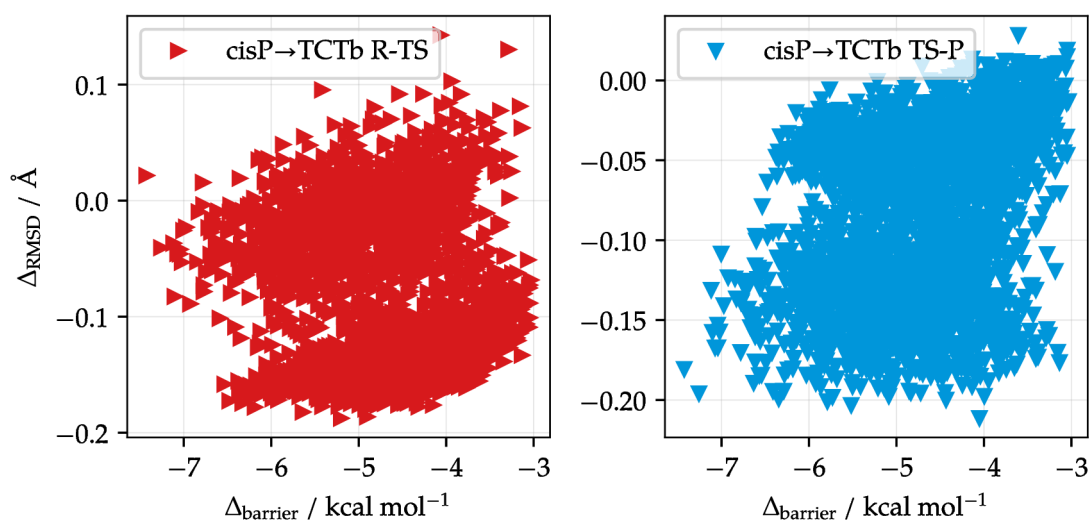


Figure 4.16: These plots show the correlation between the change of the energy barrier of the reaction (x-axis) with the RMSD between reactant and transition state (left plot) and RMSD between product and transition state (right plot).

transition state geometries.

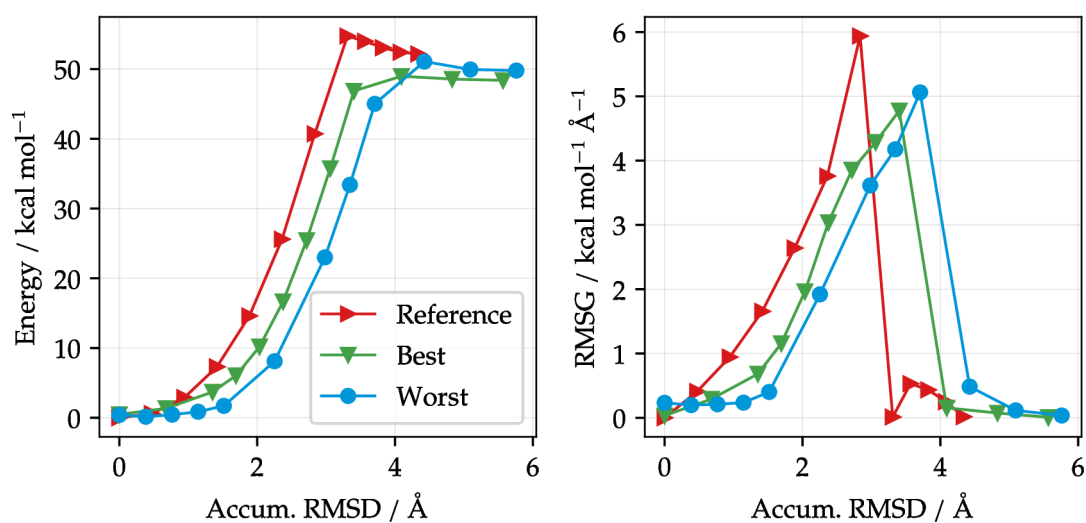


Figure 4.17: On the left, the energy paths for the $\text{cisP} \rightarrow \text{TCTb}$ isomerization of cyclooctatriene is shown for the uncatalyzed reference, as well as for the best and worst candidate in the final Pool. All energy levels along the paths are given in relation to the energy of the reference reactant, which is set to 0. The right plot shows the residual gradient at the structures, indicating that the equilibrium structures are indeed well relaxed in all three cases.

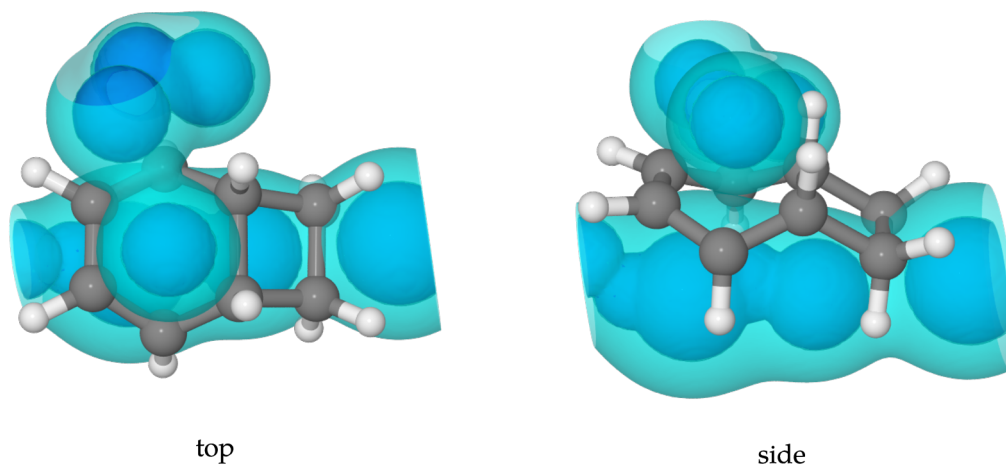


Figure 4.18: Shown are the top (left plot) and side (right plot) view of the best candidate found for the cisP-TCTb isomerization. The positive part of the LJ potential (blue, cutoff 0.05) and the negative (attractive part, cyan, cutoff 0.05) are shown in the same image. The transition state structure fits perfectly into the attractive part of the optimized cavity.

4.2.4 Surface limit

In all examples above, the number of LJ particles was always chosen to be exactly 10. Obviously, with them positioned so closely to the reactants, their exact *number* will also have a palpable impact on the catalytic proficiency of the resulting GOCAT. More particles means that more energy in the system can be utilized towards catalysis, up to the limit where any additional particles placed would be too far away to have any effect on the reactants.

To visualize this effect, for the CCC→cisP isomerization, 3 pools of 600 candidates for each of the entity amounts from $N = 10$ to $N = 20$ were optimized for $8 \cdot 10^4$ global optimization steps and then pooled together for evaluation. Figure 4.19 shows the best fitness found for each pool. While the trend is somewhat softened by the statistical vari-

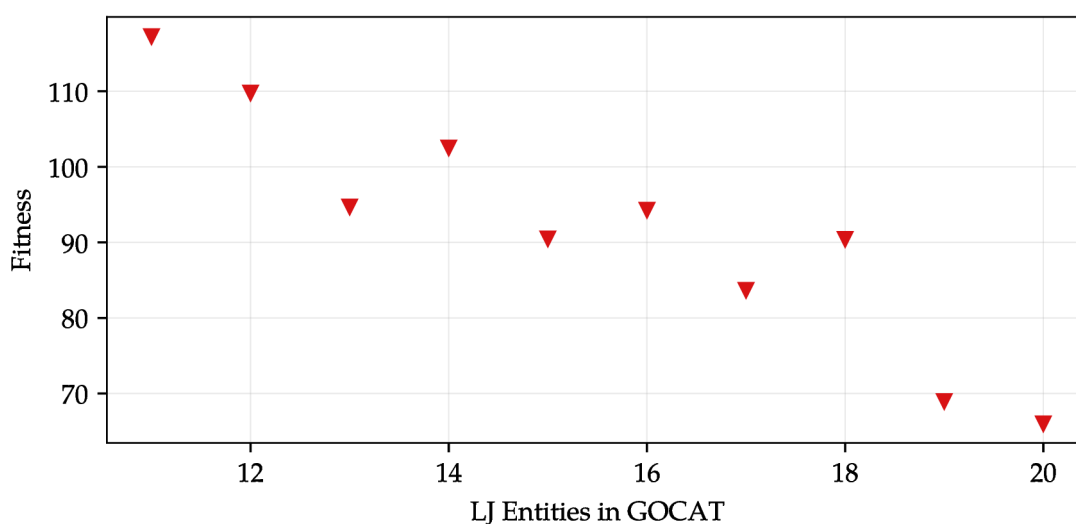


Figure 4.19: This graph shows the respective best fitness GOCAT for 11 to up to 20 LJ entities.

ation between runs, it is clear that GOCATs with more LJ particles can achieve superior catalytic effects compared to GOCATs with fewer particles. This is not due to a general difference in mechanism or structure (they are exactly the same for 10 and 20 entities), but rather due to the fact that more entities generate a stronger combined potential acting upon the reactant atoms. This can also be visualized by taking into account the total combined LJ energy contribution in the system as shown in Figure 4.20. The correlation is not 1:1 however, as can be seen in the case of the best GOCAT for 14 particles. The total energy contribution there is higher than the ones for 13 and 15 respectively, however both have a better fitness. Some other factor might thus also play a role here, like the distinct shape of the LJ cavity the combined particles are able to form around the reactant, or in

more abstract terms, the *distribution* of the LJ energy terms across the reactant atoms.

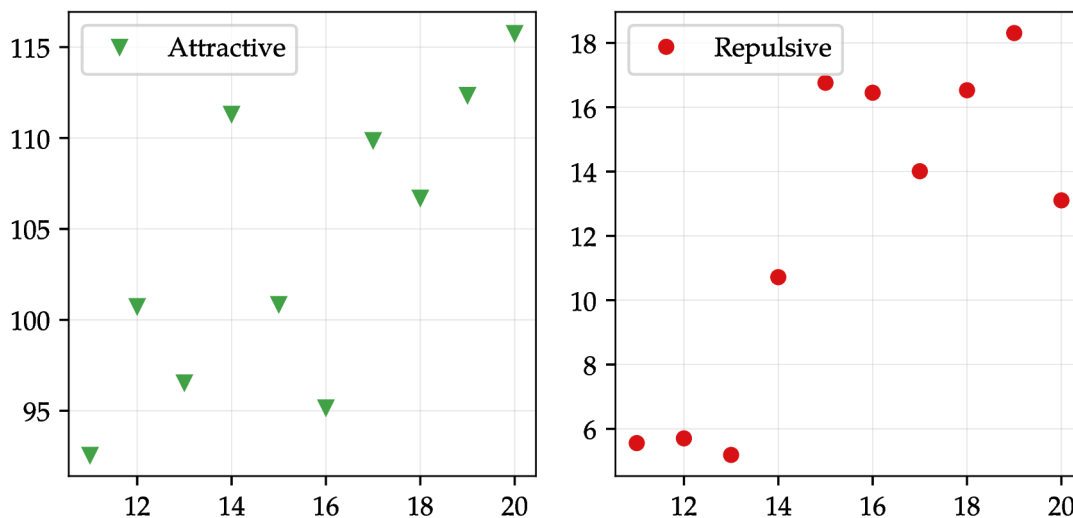
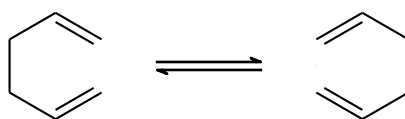


Figure 4.20: These two plots show the total attractive (left) and repulsive (right) LJ potential contributions in the respective best GOCAT for each number of LJ entities. The contributions shown here are the reaction path image-average over the total LJ potential summed over all atoms.

4.3 Bachelor's Project: KIRCHNER

During her Bachelor's thesis^[117], KIM KIRCHNER employed this dispersion catalysis method as it was being developed in a first exemplary study, parallel to further testing and refinement on the general setup. Her thesis deals with a very similar, closely related reaction system, the Cope rearrangement:



This isomerization can, depending on how the reactant is arranged, react via a boat-like or chair-like transition state, with the energy favoring the chair-like pathway. The hope was that, starting from the boat-like pathway, the dispersion catalysts would be able to stabilize the reactant in such a way that the transition state would become chair-like and the catalysis would be facilitated in this way. However, just like in the cyclooctatriene case, this could not be achieved. Instead, the results are pretty much in line with the observations already detailed in the preceding section.

Perhaps more importantly, the thesis contains a first direct comparison between an electrostatic and a dispersion-only GOCAT. For the Cope rearrangement, the electrostatic

GOCATs are clearly more catalytic. The best electrostatic GOCAT that was found in the Bachelor’s thesis reduces the GFN2-xTB reference barrier (for the boat-like reaction path) of 28.62 kcal/mol to 17.36 kcal/mol (39% reduction), which is more than a factor of two better than the best dispersion GOCAT at 24.7 kcal/mol (14% reduction).

What KIRCHNER additionally found is that when both charge and LJ parameters are allowed to change during global optimization at the same time, the resulting behavior is almost identical to the behavior in a charge-only case. This is an important observation, since it means that point charges can be assigned LJ parameters to yield more atom-like behavior without drastically altering how the global optimization operates. This is a discovery that also plays a role for the quasi-molecular embedding mentioned briefly in Section 5.2 and the entirety of the molecular embedding in Chapter 6.

4.4 Conclusion

In conclusion to the dispersion-only catalysts, a few key observations can be made: First of all, this mode of catalysis is obviously possible within the GOCAT scheme, and it can have distinct influence on the energy barriers of reactions.

The approach rediscovers well-known and established concepts of catalysis. For one, there is *preorganization*; by providing a rigid cavity that forces the reactant into a more transition-state like conformation, the necessary reorganization energy needed towards the transition state is lowered. Intuitively, this effect gains importance whenever the reactant structure must reorganize significantly towards the transition state. This has been shown with the $\text{cisP} \rightarrow \text{CCC}$ isomerization of cyclooctatriene, where GOCATs are predominantly changing the structure of the reactant minimum towards the shape of the transition state.

The $\text{CCC} \rightarrow \text{TCC}$ isomerization has the largest energy barrier in the reaction system, involving a very strained rotation around a double bond. In contrast to the $\text{cisP} \rightarrow \text{CCC}$ reaction, there is no clear indication of how dispersion interactions can catalyze this reaction; all three stationary points look fairly similar and there is no clear directionality to exploit. Nonetheless, a tangible catalysis is achieved, by providing a rigid pocket that reduces the overall intramolecular rearrangement necessary to achieve the isomerization.

In the last example, $\text{cisP} \rightarrow \text{TCTb}$, *transition state stabilization* also enters the picture and plays a major role. Rather than by stabilizing favorable reactant geometries, it predominantly achieves catalysis by introducing favorable interactions with the transition state structure.

Sadly, none of the GOCATs found an alternate, energetically more viable pathway via a previously unstable transition state. This might, however, be something that can

eventually be made possible with additional modifications to the fitness function, since the stabilization of heavily distorted structures is obviously already possible.

When compared to the already established electrostatic GOCATs, the LJ-only variety presented in this chapter is at a systematic disadvantage. Obviously, LJ interactions are inherently much weaker than Coulomb interactions, and mixing the two in a GOCAT seems to mostly result in electrostatic catalysis. This is certainly something that will have to be investigated further in the future.

Lastly, if the cavity is very tight and strained, how does the reactant even find its way into one such cavity? In fact, there are ways the molecule could indeed be trapped into a strained structure, e.g. by induced fit or structural rearrangements in the active site due to secondary reactions, as was mentioned in the introduction. This will also make a good entry point for further studies.

ELECTROSTATIC EMBEDDING

The (purely) electrostatic approach to the GOCAT is the original and best researched approach to date. The work of DITTNER has thoroughly explored its ins and outs^[37]. However, there are still a few loose ends that need to be explored and a few additions that have to be made before proceeding with the method.

One of the open issues that still remain is the fact that the number and shape of point charges used in GOCAT optimizations is (somewhat) arbitrarily chosen. There is no way to determine the “good” or “right” amount for a given reaction other than by trial and error. This will be explored in the first section of this chapter.

A second section will then briefly discuss the augmentation of electrostatic GOCATs with van-der-Waals parameters to obtain an abstract quasi-atomic model and possible applications.

5.1 Simplification Techniques

As stated in the introduction to this chapter, whenever a chemical reaction is chosen as a GOCAT optimization problem, a number of charges or entities have to be introduced as the embedding. The question here would, in essence, be: How many entities are enough? In the past, this has been somewhat arbitrarily chosen. A more elegant way to solve this problem would be to start from a solution that is safely known to be *complex enough* and then reduce the complexity in a step-wise fashion, preserving the electrostatic features, until the solution quality deteriorates significantly.

In DITTNER’s adaptive GOCAT model^[101], the catalytic field is generated by 81 point charges on a sphere around the reactants. As a small recap, the structure, as well as the energy profile, are shown in Figure 5.1, which shows that, starting from a concerted Diels-Alder, a GOCAT was obtained that stabilized a two-step mechanism instead, resulting in impressive catalysis. This example could be argued to represent such a maximally complex solution, appropriate for further refinement. It will thus serve as an important example throughout this chapter.

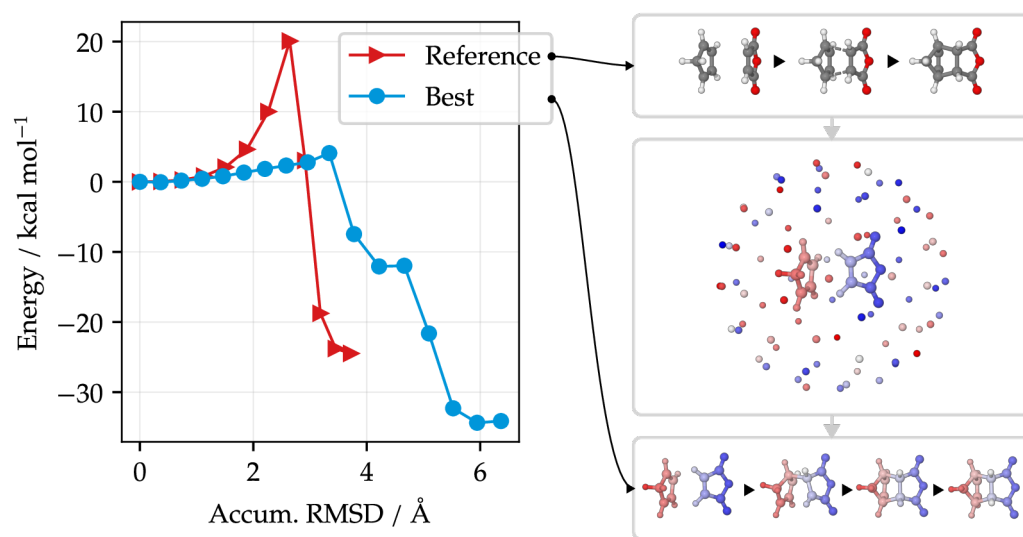


Figure 5.1: The left plot shows the energy profiles for both the uncatalyzed Diels-Alder reference reaction (structures on the top right) and the best 81-charge-GOCAT found by Dittner. For this GOCAT, the right side shows the structure of the charge sphere and the structures of the stationary points. Note that the point charges are color coded according to their partial charge value (red is negative, blue is positive). The atoms are color coded according to the electrostatic potential at their respective coordinate (again, red being negative and blue positive potential).

5.1.1 General Approach

The goal of simplification seems clear: For a reference GOCAT a with N entities, find a corresponding GOCAT a' with $N - m$ entities, such that m is maximized while the difference in fitness $\Delta_{\text{fit}} = \text{fitness}(a') - \text{fitness}(a)$ between the two remains minimal. The easiest way of obtaining this would be to use the same global optimization cycle that produced a , but instead of the original fitness function, define a new one. This fitness function must adequately describe how any new GOCAT found differs from a in its catalytic properties. This can be approached in two ways.

The first would be to globally minimize the difference in *fitness*, however this might lead to completely unrelated GOCATs that do not share any of the spatial or electrostatic information. Additionally, this requires a re-evaluation of the fitness at each step, slowing it down with additional energy and gradient calculations.

The other way would be to globally minimize the difference in the Coulomb potential itself. If the latter is sufficiently reproduced in the vicinity of the reactants, one would assume that the fitness will also be sufficiently reproduced. This approach does not require any energy or gradient calculations and will thus be faster by several orders of magnitude, making it perfectly suitable for global optimization and, ultimately, the preferred choice.

For a given set of point charges q_i , the Coulomb potential ϕ_j can easily be calculated as a scalar field for any given point in Cartesian space $\vec{r}_j$ from the GOCAT point charges q_i :

$$\phi_j = \sum_i \frac{q_i}{|\vec{r}_j - \vec{r}_i|} \quad (5.1)$$

Calculating ϕ_j in a reasonably dense grid around the reactants for a reference GOCAT yields a reference potential that is suitable as global optimization target. One such reference potential is visualized in Figure 5.2. However, the square grid that is obtained

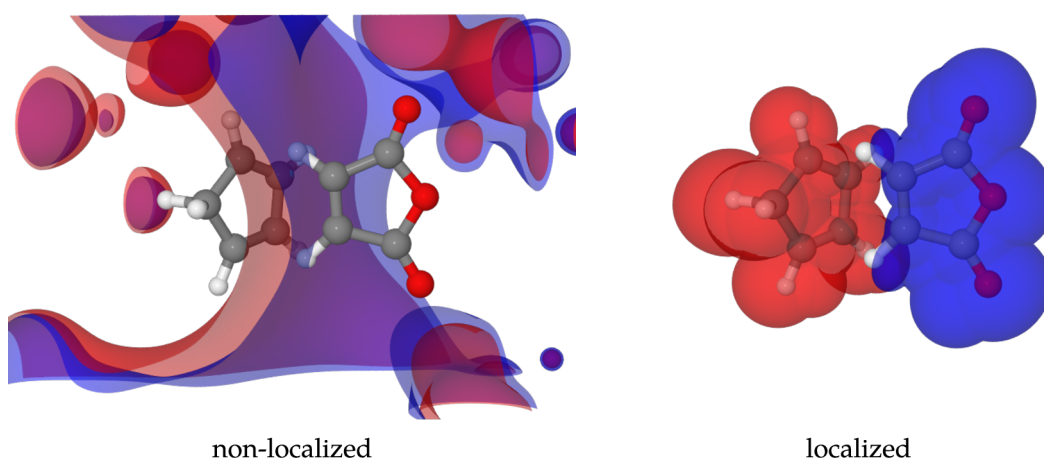


Figure 5.2: On the left, the raw ESP calculated for the Diels-Alder example used by Dittner^[101] is shown with red denoting a negative, blue denoting a positive sign. On the right, the same ESP was localized via atom-centered exponential decay. Both isosurfaces are shown at a cutoff value of 0.025.

in this fashion often contains a lot of information outside of the relevant space (around the reactant atoms). A simple way of dealing with this is letting the scalar field decay to zero the further it gets away from any relevant atoms. This is done by checking for each discrete cell the distance r to the closest reactant atom, and then applying an exponential decay function $\delta = e^{-\beta r}$. Figure 5.2 shows the comparison between the localized and non-localized version of the same reference potential.

In the global optimization, the fitness function Φ is then simply the difference between an ad-hoc calculated potential of a new individual and the reference potential, given by the RMSD:

$$\Phi(a) = \sqrt{\sum_{x,y,z} (\phi_{\text{ref}}(x,y,z) - \phi_a(x,y,z))^2} \quad (5.2)$$

In contrast to the original GOCAT fitness function, this fitness function is simple enough

such that an analytical gradient can be implemented to the effect of Equation 5.3. Note that each charge q_i has 4 degrees of freedom (x_i, y_i, z_i, q_i) , which each need to be treated respectively.

$$\frac{\partial \Phi(a)}{\partial x_i} = \frac{1}{\Phi(a)} \sum_{x,y,z} (\phi(x, y, z) - \phi_{\text{ref}}(x, y, z)) \frac{\partial \phi(r)}{\partial x_i} \quad (5.3)$$

$$\text{with } \frac{\partial \phi(x, y, z)}{\partial x_i} = \frac{q(x - x_i)}{((x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2)^{\frac{3}{2}}} \quad (5.4)$$

$$\text{and } \frac{\partial \phi(x, y, z)}{\partial q_i} = \frac{1}{\sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2}} \quad (5.5)$$

(The derivative of ϕ can of course be equivalently formulated with respect to y_i and z_i but this is omitted here.) In the current implementation, this gradient performs adequately for local optimization using, i. e. L-BFGS^[86] or FIRE^[87]. Figure 5.3 shows some benchmark timings for the different tasks involved in generating the grids. While all tasks scale linearly with the number of atoms, they scale *exponentially* with increasing grid resolution. Especially all tasks related to gradients require an additional degree of freedom and thus blow up rapidly. This, in turn, raises the question of how much resolution is actually necessary for a given problem. The answer requires an actual example for benchmark studies, which will be introduced in the following section.

5.1.2 Revisiting Diels-Alder

As a first example, the adaptive Diels-Alder GOCAT, described by DITTNER^[37] in his dissertation, will be used. The reference individual is the best individual from the respective pool, the same that is shown in Figure 5.2. Before any new, simplified individuals can be optimized, two separate but fairly independent parameters need to be adjusted: the resolution used for the electrostatic potential grid, as well as the number of charges used in the new individuals. By choosing a resolution that is certain to be sufficient, one can determine the minimum number of charges necessary for reproduction of the field. The grid cell size that was taken here is 0.75 Å (which corresponds to a resolution of 1.33 Å⁻¹, which is the unit that will be used from here on out). Then, 5 global optimization runs for each simplification step $81 - m$ ($m = 1, 2, 3, \dots, 80$) with 1000 individuals per pool and active niching were run for 10⁵ iterations. The averaged results are shown in Figure 5.4.

It is clear that fewer than 12 charges are not enough to reproduce the reference fitness on re-evaluation. This hard boundary most likely has two reasons: First of all, the restriction of $|q| = 1$ e still applies, such that a certain amount of charges must be available so that their combined presence can produce a strong enough field. Secondly, the grid is localized around the entire molecule, where it is almost homogenous throughout

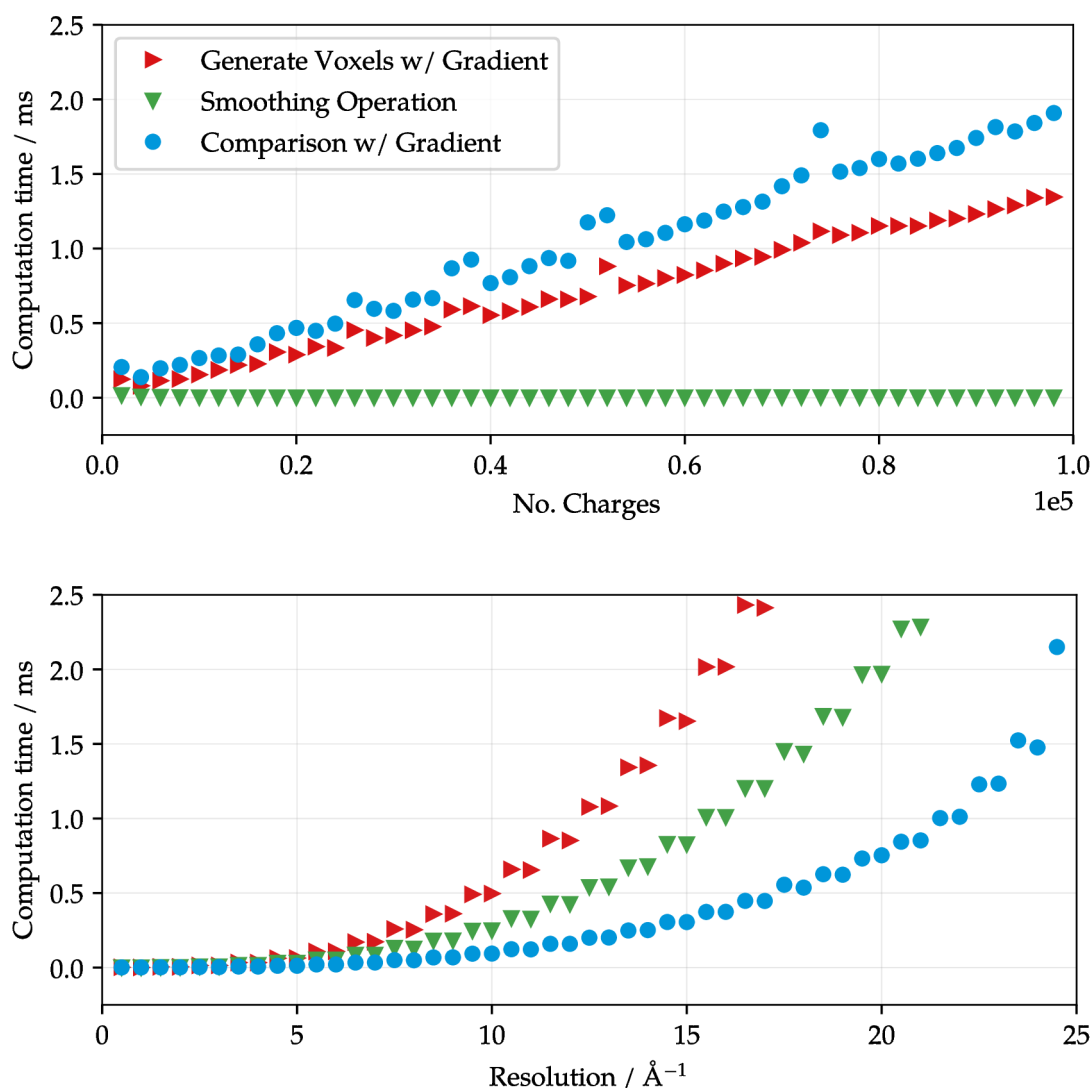


Figure 5.3: In the top graph, the time spent on three different parts of the computation is shown in relation to the number of charges. In the bottom, the relation is to the resolution (or cell size) used.

(which is clear when considering Figure 5.2). This essentially requires the point charges to be at least a little spread out around the molecule to reproduce the actual size of the homogenous field correctly.

With this information, the sufficient resolution can then be determined by redoing optimizations in an analogous fashion with a fixed number of 12 charges at varying resolutions, of which the results are shown in Figure 5.5. The results show three things: In general, 12 charges, which was taken as the lower limit above, is apparently still insufficient to *reliably* reproduce the reference fitness, so slightly more should be used for this

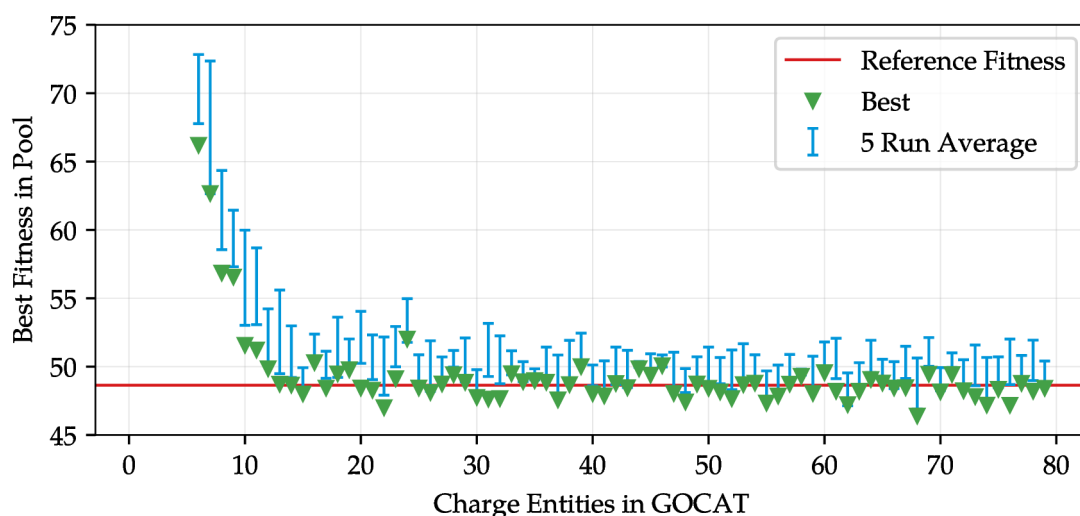


Figure 5.4: In this plot, the 5 run average of the best fitness for the Diels-Alder reaction are shown for each simplification step.

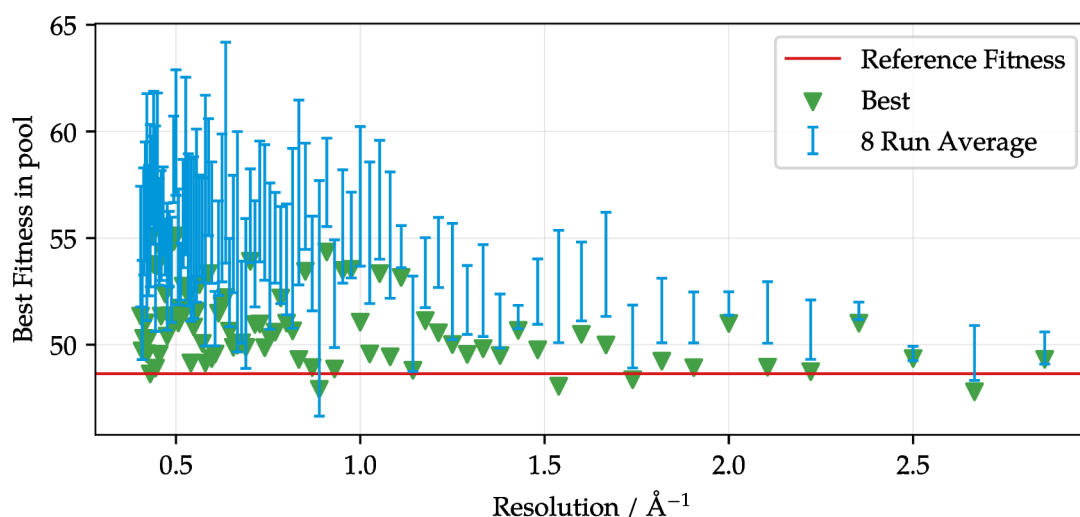


Figure 5.5: This plot shows the fitness averages for 12 charge GOCAT optimization runs for the Diels-Alder reaction versus the resolution of ESP grid used.

specific reaction. Secondly, there seems to be a general trend towards more reliable fitness with higher resolutions, however it seems small compared to the statistical variance of the global optimization at these settings (pools, iterations). Due to the exponentially scaling nature, it is advisable to choose a resolution as coarse as possible. Resolutions above $1.0 \text{ \AA}^{-1}$ already slow the computation down significantly. Additionally, the results at resolutions of less than $1.0 \text{ \AA}^{-1}$ show that it is possible to find good solutions for even

fairly coarse grids, meaning that global optimization iterations should definitely be given priority over resolution here. The 10^5 global optimization iterations done here are clearly not enough to produce good solutions reliably, and more steps (in the region of 10^6 and more) are absolutely necessary.

For final production runs, three different charge entity amounts were chosen: 9, 16 and 30. For each of the three, 10 pools of each 500 individuals were globally optimized with a resolution of $1.0 \text{ \AA}^{-1}$. The very best individuals found respectively have the fitness values 56.511, 49.003 and 46.068, which brings them all very close to the original reference value of 48.641. The 30 charge version even surpasses the original individual due to slightly lower gradients at the stationary points. Figure 5.6 shows that each of the three variants reproduces the original energy path reference almost exactly.

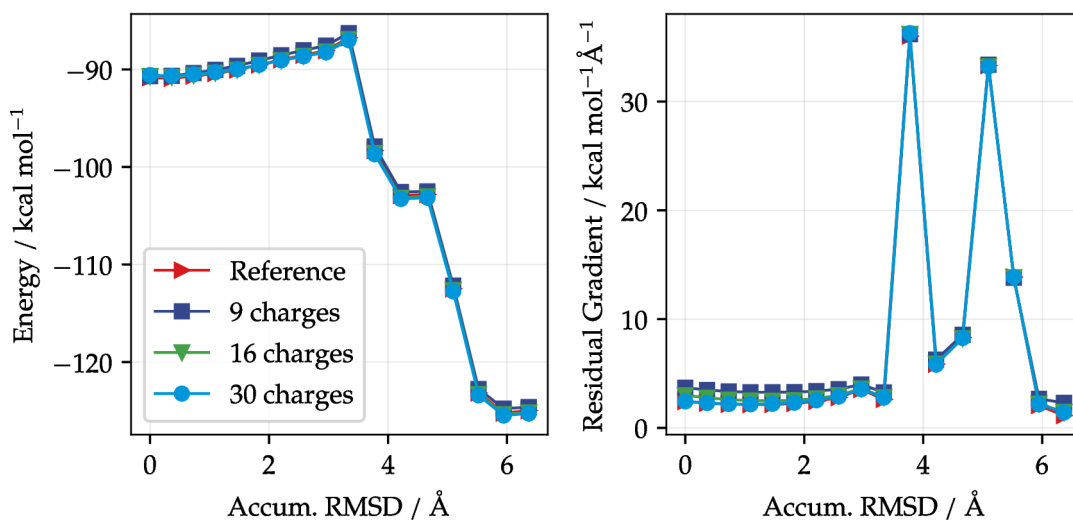


Figure 5.6: This plot shows the energy paths (left) and gradients (right) of the 9, 16 and 30 charge GOCATs built to reproduce the original reference.

Structurally, the very best individuals are very similar as well: There are negative charges distributed along a band almost orthogonal to the diene plane and positive charges distributed along a band in-plane with the anhydride. However, this picture gets less clear in the 30 charge variant, as here the charges once again begin to compensate each other like in 81 charge case, leading to a very complex, opaque scheme.

5.1.3 Revisiting S_N2

The method applied to the Diels-Alder reaction in the previous section can also be readily applied to any other GOCAT model using charged entities, such as the result from the original 2018 paper^[34]. The solution found therein is very symmetric apart from a charge

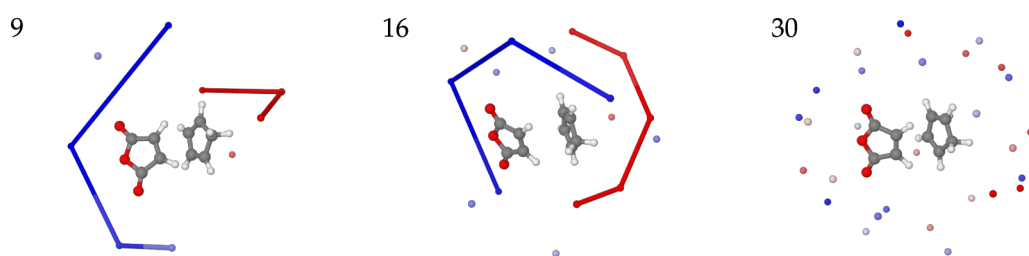


Figure 5.7: The structure of the 9, 16 and 30 charge GOCATs. In the 9 and 16 charge structures, bonds have been added to the peculiarly placed lines of strong charges around the reactants.

positioned slightly off-center. This can be seen in the reference potential shown in Figure 5.8.

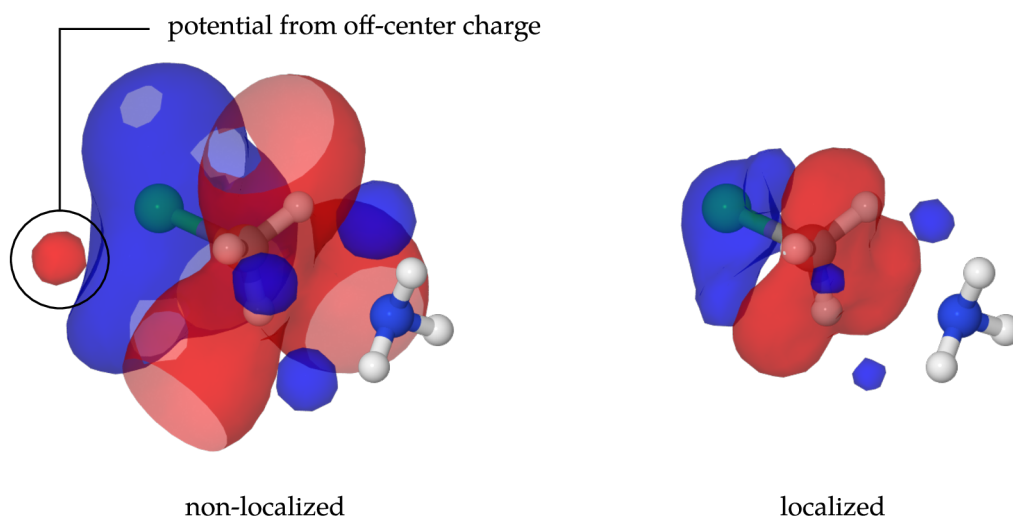


Figure 5.8: On the left, the raw ESP calculated for the S_N2 example used by Dittner^[34] is shown with red denoting a negative, blue denoting a positive sign. On the right, the same ESP was localized via atom-centered exponential decay. Both potentials were generated at a grid resolution of $2.0 \text{ \AA}^{-1}$ and both isosurfaces are shown at a cutoff value of $0.02 \frac{\text{kcal}}{\text{kcal}\cdot\text{e}}$.

Again, for final production, 10 pools of each 500 individuals were optimized with a resolution of $2.0 \text{ \AA}^{-1}$, for 9 charges and 6 charges, respectively. Compared to the reference fitness value 153.16, the newly found individual with 9 charges at 166.47 is very close in its final fitness value. For 6 charges, the individual is reproduced significantly worse, with a fitness of 255.76. This is mirrored when looking at the energy profiles in Figure 5.9. For 9 charges, the profile is almost identical, with the product being ever so slightly more stabilized, however both the 6 and 9 charge individuals have higher residual gradients at the stationary points compared to the reference.

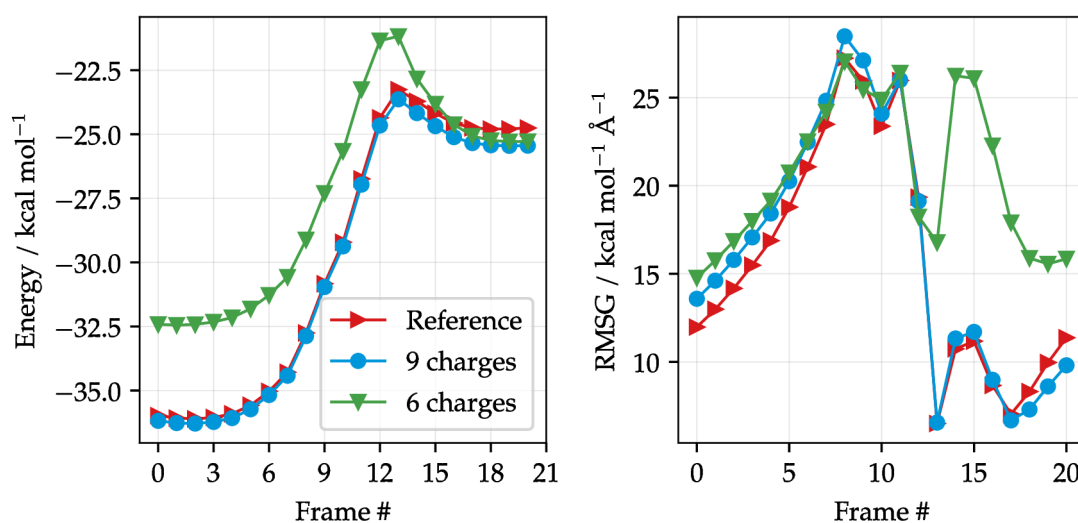


Figure 5.9: The left graph show the individual energies of the frames along the S_N2 reaction path, each for the original GOCAT and the 9 charge and 6 charge simplified variants. The right graph show the corresponding RMSGs.

Nonetheless, the 9 charge example shows that the simplification has worked as expected and the reaction can, without any issue, be catalyzed without the 10th charge. This suggests that the latter, indeed, could just have been an artefact resulting from the fact that a 10th charge is not necessary to construct the optimal point charge arrangement for catalysis of this reaction.

5.2 Quasi-Molecular Embeddings

When DITTNER developed his adaptive GOCAT scheme^[101], his successful attempts at optimizing a GOCAT with new, on-the-fly NEB calculations were confined to embeddings that were situated far from the reaction center, e.g. on a spherical shell. This is due to the fact that a NEB requires optimized start and end points. With significantly charged point charges placed very close to the atoms, the probability of oppositely charged point charges and atoms collapsing into one another is fairly high, leading to convergence issues and a very inefficient global optimization.

This problem can be remedied by adding Lennard-Jones potentials to the respective point charges. The resulting quasi atoms, or ghost atoms, then remain abstract in nature, but allow for the search of embeddings that can sit close to even very complex reactants and (hopefully) be much more refined and easier to interpret. They allow full relaxation of reactant and product structures and can thus also capture the required flexibility required in the embedding to go from reactants to products.

These quasi-atoms were implemented in OGOLEM such that they interact with the QM atoms via their charge value (as before in the point charge GOCAT model) and via an additional LJ-12-6-potential, with parameters mimicking an sp^3 -hybridized carbon atom. The advantage of this approach is that the charges remain on the surface of the QM molecules throughout the reaction, as their equilibrium distance is well defined. For the interaction term between the quasi-atoms, a very simple repulsive quadratic potential was used depending on their distance r_{ij} and the respective σ value product¹.

$$E_{ij} = \begin{cases} -r_{ij}^2 + \sigma_i\sigma_j & \text{for } r_{ij}^2 < \sigma_i\sigma_j \\ 0 & \text{otherwise} \end{cases} \quad (5.6)$$

If the interactions between the quasi-atoms had Coulomb or dispersion character, they would, at best, cluster together and make the global optimization operators inefficient. At worst, attractive terms could draw them too close to atoms and bonds and generate QM/MM artefacts.

It is also very important to note that there is a distinct difference compared to the dispersion-only particles introduced in Chapter 4. There, the effect of distorting the geometry was specifically desired and thus, the particles were introduced very close to the reacting molecules and held in place. In the case of the quasi-atoms here, they are always locally optimized to their equilibrium distance and either kept fixed in the one-embedding scheme, or optimized together with the reacting molecules in the NEB to get a frame-wise embedding scheme. These two different approaches will also become important for the molecular GOCATs in Section 6.2.3, where the implementational details and consequences for the two routes are discussed in more detail.

5.2.1 Bachelor's Project: REINHARDT

Some early testing of this approach was done by REINHARDT in his Bachelor's thesis^[118]. The goal in his work was to elucidate whether or not an approach using quasi-atoms placed close to the reactants would be more beneficial for adaptive GOCATs. In order to have a good reference to compare against, the Diels-Alder model used by Dittner (discussed earlier in context of simplification techniques) was chosen as a basis. It was then modified from a spherical to both a one-embedding and framewise-embedding scheme. In the one-embedding scheme, 16 charged quasi-atoms were placed on the van-der-Waals surface of the reactants and then held fixed throughout the fitness evaluation process, including reactant, product and NEB optimization. In the frame-wise scheme, the embedding was then allowed to move for all three optimization procedures, generating an

¹In this case, since all were equally taken to be sp^3 carbon atoms, $\sigma_i = \sigma_j = 3.55 \text{ \AA}$.

embedding that can account for the movement of the reactants during the reaction.

In even very early results, this led to GOCATs with essentially zero energy barrier, as shown in Figure 5.10. Compared to the spherical 81 charge variant used by DITTNER,

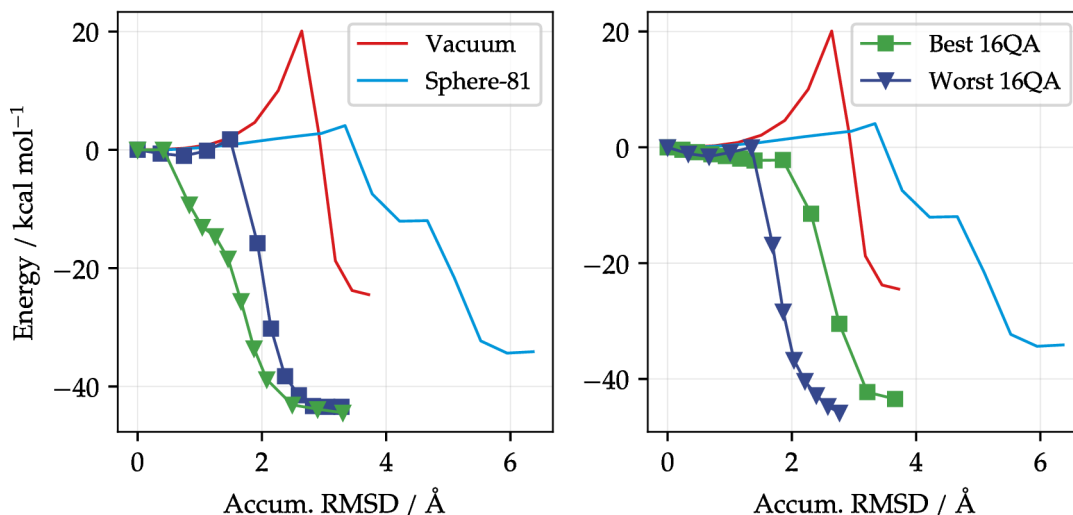


Figure 5.10: The left plot shows the energy profiles for 16 quasiautomatic (16QA) GOCATs optimized with the one-embedding scheme. The right plot shows the corresponding energy profiles for the frame-wise scheme.

the potential difference at the carbon atoms involved in the asymmetric first reaction step of the best individuals (which are also found within the first 20000 global optimization steps) is by a factor of around 1.5 larger, which helps to explain this extreme effect. Another obvious difference is the length (or accumulated change in the system) along the reaction coordinate. The reaction is now much more compact, as less translation and rotation happens in the field, as it is in the case of the spherical GOCAT. Both one-embedding and frame-wise approach nonetheless retain the asymmetric reaction mechanism, however the second transition state in both cases is much less prominently visible.

5.3 Conclusion

With the simplification technique outlined in this chapter, GOCAT development has gained a powerful new tool. For reactions that require complex heterogeneous electric fields, the user can now develop GOCATs with enough or even unnecessary complexity, and can then later extract the most important features to yield a simple, manageable model.

However, there still remains some work to do. As shown in the Diels-Alder example, simplification is not as straightforward as it seems at first. Naturally, there are lower limits to the number of charges that can still reproduce the needed complexity and field strength.

But as of now, there is no way of telling what those are, other than by trial and error. Thus, this still requires a lot of manual intervention that could, in the future, be automated or solved by on-the-fly reduction of charge complexity within the global optimization pool.

The resolution scaling issue on the other hand seems not as problematic as one would assume. The algorithm can work sufficiently quickly at resolutions of around $1 \text{ \AA}^{-1}$, and produce adequate solutions even when the resolution is significantly lower.

In general, the methodology applied here can be useful in many aspects; It can, for example, be used to fit any kind of ESP with point charges, even that generated from outside sources. Additionally, one could best reproduce electric fields with specifically constrained charges or custom shapes, for example if a given rigid molecular embedding is to be modified, i.e. metal organic frameworks.

The quasi-molecular embeddings introduced at the end of this chapter allow the optimization of GOCATs with charges that are placed directly on the van-der-Waals surface of the reactants and still allow for optimization of the reaction path, possibly leading to much more nuanced polarization and more manageable GOCAT structures. While early results show great promise with this approach, more data will have to be collected and more thorough analysis conducted to verify and prove that this is indeed a better approach than placing naked point charges far away from the reactants.

MOLECULAR EMBEDDING

In the previous two chapters, electrostatic and dispersion interactions were considered mostly separate from each other, and in an abstract fashion. But since both approaches share the basis of the QM/MM approach, which is specifically designed for the representation of actual molecules, the steps that need to be taken to go from an abstract embedding to a molecular representation are, in fact, minimal.

If molecules are to be used, there are multiple ways of approaching this, similar to the quasi-atom embedding in Section 5.2: As a static embedding for the entire reaction path (fitting for very fast reactions such as proton transfers), or as a per-frame embedding for slow and sterically involved reactions. Additionally, one could use only a predefined set of molecules (as in the case of pure or mixed solvents) or a library of molecules that can produce embeddings in the pool that differ in their composition from individual to individual.

All of these approaches will be discussed in further detail in the following sections, after a brief description of the OPLSAA implementation that was used throughout the chapter.

6.1 OPLSAA Implementation

In order to ensure that all the software backends have full control over which atoms and terms enter the total energy calculation at which point in the program, it was deemed necessary to implement a force field directly in the Java code base of OGOLEM. Early tests were still using an interface to the TINKER^[119] software package. However, for a single gradient calculation, TINKER had to be called 2-3 times because of atom group configurations, multiplying the overhead and making the process very inefficient. However, the newly implemented force field is inspired by the TINKER implementation and the latter was used as a benchmark to ensure consistent energy and structure calculations. The Java implementation uses the same terms as the OPLSAA in TINKER and can be run using the same force field parameters. The total energy term is given as:

$$E = E_{\text{bond}} + E_{\text{angle}} + E_{\text{dihedral}} + E_{\text{improper}} + E_{\text{dispersion}} + E_{\text{Coulomb}} \quad (6.1)$$

For the bonds and angles, a simple harmonic term is used:

$$E_{\text{bond}}(r) = K(r - r_0)^2 \quad (6.2)$$

$$E_{\text{angle}}(\theta) = K(\theta - \theta_0)^2. \quad (6.3)$$

In TINKER OPLSAA, the force constants K are given in kcal/mol, or $\frac{\text{kcal}}{\text{mol} \cdot \text{rad}}$ in the angle case, while the r_0 and θ_0 are given in Å and degrees, respectively. The improper and dihedral angles use a Fourier term:

$$E_{\text{dihedral}}(\varphi) = \frac{1}{2} (\alpha_1 [1 + \cos(\varphi)] + \alpha_2 [1 - \cos(2\varphi)] + \alpha_3 [1 + \cos(3\varphi)]), \quad (6.4)$$

where the α_i are again given in kcal/mol. For the intermolecular forces, there is the dispersion term:

$$E_{\text{dispersion}}(r_{ij}) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right], \quad (6.5)$$

which is a standard Lennard-Jones potential with the parameter σ_{ij} and ϵ_{ij} in units of Å and kcal/mol, respectively, determined from the geometric mean of the individual atom parameters $\sigma_{ij} = \sqrt{\sigma_i \sigma_j}$ and $\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}$. Lastly, the Coulomb term:

$$E_{\text{Coulomb}}(r_{ij}) = 332.06 \sum_{ij} \frac{q_i q_j}{r_{ij}}, \quad (6.6)$$

which has a peculiar conversion term that arises from unit conversion to $\frac{\text{kcal}}{\text{mol} \cdot \text{e}}$. It gives a proper energy upon insertion of the charges q_{ij} which are given as fraction of elementary charge e in the force field parameters. These last two terms are also used for 1,4 intramolecular interactions, albeit scaled by a factor of $\frac{1}{2}$.

All of these terms also have their analytical first derivatives implemented. Furthermore, the implementation can be very specifically instructed which terms to calculate by assigning individually to three groups: *deactivated*, *passive* and *QM/MM*. Atoms in the first group are simply ignored for all energy terms. Interactions with the members of the second group are one-sided, meaning that all active atoms are calculated as normal, but the passive atoms receive no energy or gradient terms. Lastly, the QM/MM group implements an embedding scheme which calculates energy terms as normal for all MM atoms, and only QM-MM dispersion terms, treating the QM atoms effectively deactivated otherwise, such that they can be handled by any other QM software, in line with the electrostatic embedding scheme described earlier.

As benchmark, this implementation was tested for single point calculations, gradient calculations and local optimization of butane, benzene, a mixture of small alkanes and all amino acids parameterized in OPLSAA. In all cases, it gave exactly identical results to TINKER using the same parameters, showing that the implementation holds up to expectations.

6.2 Solvation Shells

As mentioned in the introduction to this chapter, going from abstract quasi-atoms to explicit molecules is straightforward. The only additional step required is to provide the MM force field with the appropriate atom types and bond lists.

A first demonstration of this approach will be shown here, where it will be used to optimize explicit solvent shells around reaction centers. In a way, this circles back to one of the earliest GOCAT-like approaches^[120], where water clusters were optimized around an S_N2 reaction frame-by-frame, in order to assess an upper bound for the influence of solvents on the reaction. There are, however, two key differences:

- Instead of optimizing the energy of the candidate frames specifically, the optimization target is directly the energy *barrier* of the embedded reaction.
- The GOCAT approach allows for the inclusion of the explicit embedding into the NEB calculation, resulting in a reaction path on the combined potential energy surface, rather than independent minima that need not necessarily be coherently connected.

In this chapter, the approach to molecular embeddings will be built up stepwise and the individual steps will then be compared each in order, starting with a completely static approach that is similar to the original GOCAT formulation^[34], then going from there to a half-adaptive setup and eventually, to a fully adaptive optimization scheme^[47].

6.2.1 S_N2 - Static Reaction in Static Embedding

For the initial setup, GOCAT optimization was done with a basic, static scheme:

- The input reaction path is the gas-phase path, which is not allowed to change during the global optimization.
- The GOCAT entities are OPLSAA acetone molecules.
- Prior to fitness evaluation, the OPLSAA embedding is optimized in an MM-only local optimization together with the (fixed) transition state of the embedded reaction.
- After this local optimization, the static fitness Φ_{Static} is evaluated as described in Section 3.3.

For the last step, in this case, the MM embedding is assumed to be a charge-only embedding, meaning that for the purpose of fitness determination, the acetone atoms behave

just like fixed point charges. This rudimentary approach has a few advantages and disadvantages: first of all, the results are very easily comparable to the original ones that were generated with entirely *abstract* point charge models^[34]. Secondly, the global optimization process is very fast and efficient, due to the fact that only a few semi-empirical single-point calculations are needed for fitness determination. The obvious disadvantage is the fact that the embedding is, for the purpose of fitness determination, fixed into the locally optimized configuration for the transition state. Thus, the dynamic movement of the solvate shell from reactant to product structure is not captured in the process. Additionally and more importantly, the reaction path is locked into its gas phase form, which is most certainly incorrect. This will also be further discussed in the more advanced versions of this model.

For the calculation itself, 10 pools of each 320 candidates were optimized with this procedure for each 1 million iterations and then pooled back together for evaluation. First of all, as expected, the electrostatic picture is very similar to the known results from charge-only GOCATs. The left side of Figure 6.1 shows the best candidate's electrostatic potential on the atoms of the S_N2 reaction transition state as an inset to the reference and catalyzed reaction paths. As expected, the chlorine atom experiences a net positive, while the nitro-

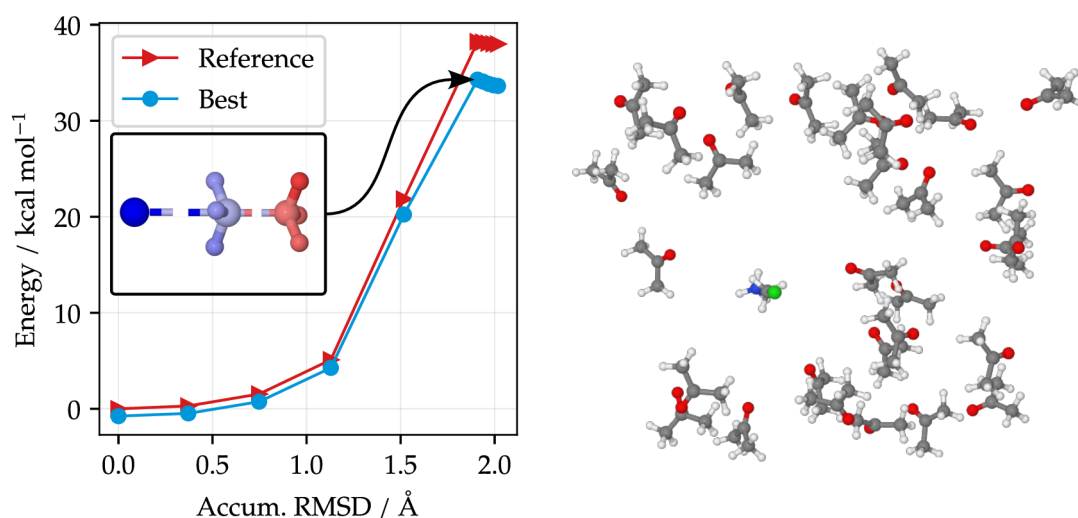


Figure 6.1: The left plot shows the energy paths of the reference and the best candidate found in the static embedding / static reaction path system, using the unmodified 2018 fitness function. Inset is the transition state structure, with the atoms colored according to the ESP values generated by the acetone embedding at their respective positions, from negative (red) to positive (blue). The right plot shows the structure of the best embedding found, including the transition state of the embedded reaction.

gen atom experiences a net negative electrostatic potential.

Looking at the structures themselves, however, unveils another major issue with this setup. The right side of Figure 6.1 shows the best candidate's structure from the respective pool, using the fixed point charge fitness function. Immediately obvious is the huge empty gap between the reactants and the embedding, showing that the reactants have not been solvated, but are instead avoided by the solvent molecules. This is very much a defect of the assumptions made: Because the fitness function employed here only includes the QM-MM Coulomb interaction, but not the MM-MM Coulomb or dispersion interaction, it automatically prefers solutions where the acetone molecules are far from the QM center. This is due to the fact that, just like in the original GOCAT approach, the QM/MM system is never locally optimized as a whole and all structures therefore have a large gradient norm. The fitness function merely ensures that it is not too large. The penalty employed to do this is obviously minimized when the charged particles are further away from the center, weakening the interaction and additional gradient introduced.

One possible fix to this problem is considering the entire QM/MM system for fitness determination, including the MM-MM and QM-MM interactions – however, the setup then requires additional tweaks. In the Φ_{stab} term(s) of the static fitness function (Equation 3.3), a comparison to a reference containing no MM atoms system will always create an offset that stems from the introduction of these atoms into the system. Consequentially, these terms would either have to be left out or, in addition to the energy of the uncatalyzed path, a separate MM-only system containing just the embedding needs to be evaluated prior in order to quantify the offset. Adding this offset should then leave only the difference due to interaction of the QM/MM interface, which is acceptable in terms of the fitness function. In the ideal case, the acetone molecules will orient themselves optimally around the reaction center. Thus, the separate reference system for offset determination should ideally also be in an optimal configuration.

To this end, an MM-only system of 30 acetone molecules was globally optimized using the cluster optimization mode of OGOLEM and using the OPLSAA force field. The system was run on one pool containing 120 candidates for a million iterations, yielding the structure shown in Figure 6.2. The offset energy of this optimum is $-540.6306 \text{ kcal mol}^{-1}$, which was added to the reference energy values of the isolated QM system.

With these tweaks, once again 5 pools with each 600 candidates were run for 1 million iterations and pooled together for evaluation. With the improved setup, the electrostatic picture remains qualitatively unchanged, however the structures are now much more intuitive and chemically reasonable. This can be seen in Figure 6.3, where, again, both the reaction path, as well as the structure of the best candidate are shown together.

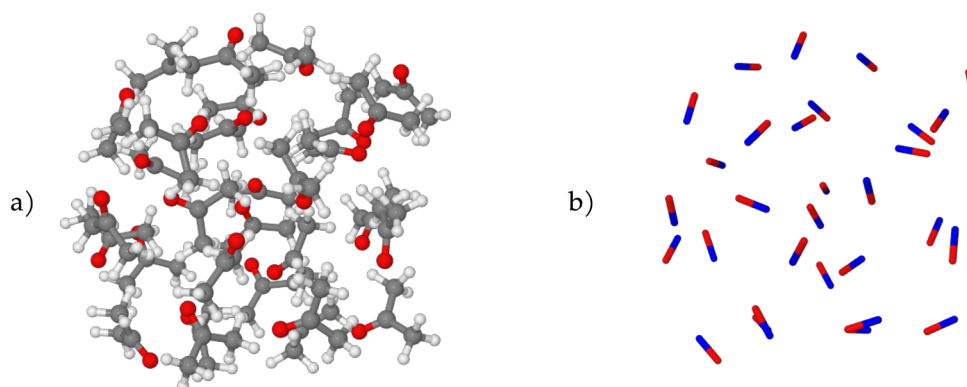


Figure 6.2: This plot shows a) the globally optimized acetone cluster with 30 acetone molecules that was generated to determine an appropriate offset in reference energy, as well as b) only the isolated carbon (blue) and oxygen (red) atoms of the acetone as an approximation to the dipole arrangement in the cluster.

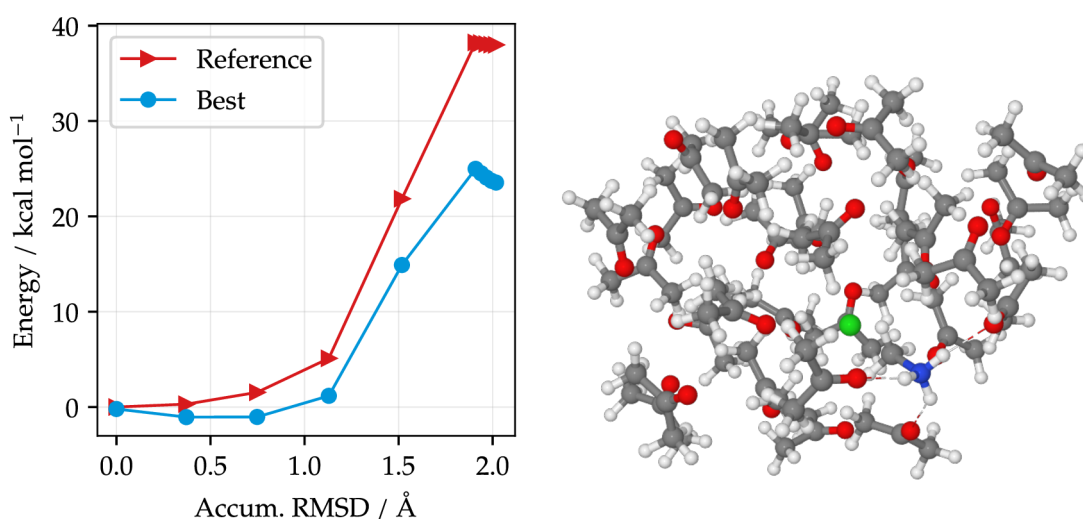


Figure 6.3: The left plot shows the energy profile of the reference and the best candidate found in the static embedding/static reaction path system, using the modified setup that includes all QM-MM and MM-MM interactions, as well as the reference energy offset determined from the globally optimized acetone cluster. On the right, the structure of the best embedding found is shown, including the transition state of the embedded reaction.

Not only does this improve the structure dramatically, but the higher proximity of the acetone dipoles also improves the catalytic effect by a factor of 3. An issue that persists in the static case, however, are the residual gradients at the minimum, transition state and product structures that need to be simply accepted and penalized such that they remain, at the very least, small enough. This means that, in essence, the embedding structure

remains only a rough estimate of the truly optimal arrangement.

6.2.2 S_N2 - Adaptive Reaction in Static Embedding

The highest residual gradients in the completely static model emerge, unsurprisingly, at the embedded QM structures. This is due to the fact that they are completely fixed into the gas phase path. Just like in the purely electrostatic case, this can be remedied by allowing acceptable candidates to re-calculate a new reaction path on the fly during the global optimization. In this setup, a slightly modified variant¹ of the adaptive fitness function is used (s. Section 3.3). The embedding itself is, just like in the static case above, only locally optimized once prior to fitness evaluation on the MM level and is held fixed in this structure.

To test this setup, once again 5 pools of each 600 candidates were optimized for 10^5 iterations each and pooled back together for evaluation. The best energy profile for the best candidate and structure are shown in Figure 6.4.

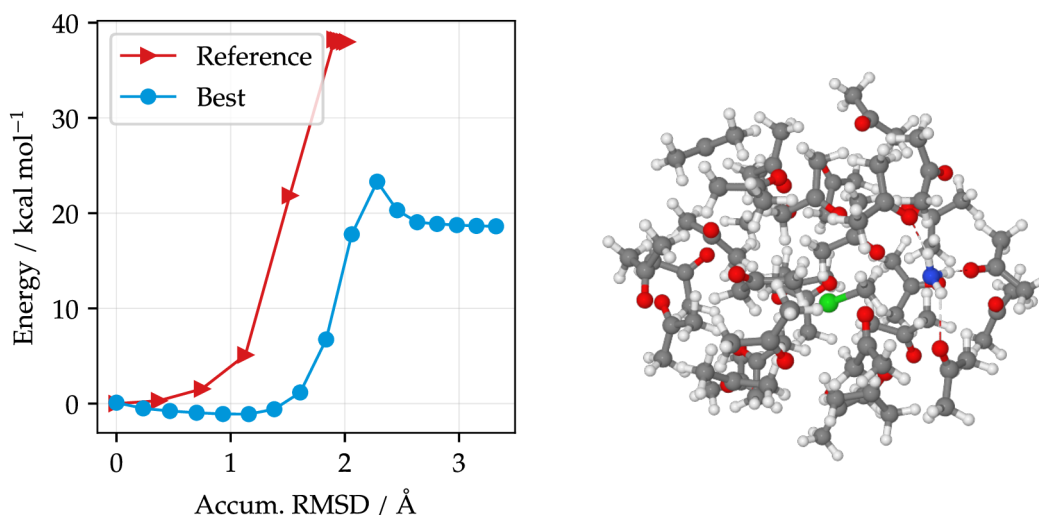


Figure 6.4: The left plot shows the energy paths of the reference and the best candidate found in the static embedding/adaptive reaction path system. On the right, the structure of the best embedding found is shown, including the transition state of the embedded reaction.

In direct comparison to the rigid, fixed setup, this approach delivers a much improved, smooth energy path that much more closely resembles the structure of the solvated reaction. The acetone arrangements themselves are actually comparatively similar. Just like in the rigid case, the nitrogen atom sits at the cluster boundary with three acetone oxygen

¹The modifications that needed to be done are only of implementational kind; Because the setup now deals with molecules instead of point charges, the local optimization process becomes slightly different.

atoms each forming a hydrogen bond to the ammonia hydrogen atoms, and the chlorine atom is surrounded by acetone molecules with their oxygen atoms generally facing away from it.

While the residual gradients at the QM atoms are greatly reduced in this approach, the rigidity of the embedding still imposes gradient norms of more than $5 \text{ kcal mol}^{-1} \text{ \AA}^{-1}$, which should ideally be much lower for true minima and transition states.

6.2.3 S_N2 - Adaptive Reaction in Adaptive Embedding

Getting rid of the residual gradients is a matter of locally optimizing the embedding together with reactant and product before a new reaction path is calculated and thus, include the embedding in the NEB. This brings with it a few implementational considerations: The immediate consequence is that the *one* embedding scheme previously used for GOCATs is replaced by a *frame-wise* embedding. Obviously, this requires a re-evaluation of the global optimization process: in the global optimization operators, the candidates in the pool are assumed to have a unique embedding that is used to calculate mutations as well as crossovers between the respective candidates. If they now possess an embedding that differs from frame to frame, either a re-formulation of the operators is in order, or some other heuristic needs to be found such that each candidate again only retains one unique embedding.

In practice, the simplest solution to this is to retain only the information about the *initial* embedding that each candidate has, *before* it has entered the fitness evaluation and local optimization procedures. Thus, the candidates in the pool have only one unique embedding, and the frame-wise version needs to be regenerated anew every time a candidate is checked for fitness. This ensures that the global optimization can proceed as normal and gives results that are reproducible upon calculation of the reaction path².

Again, this setup was tested with 5 runs of each 600 candidates for 10^5 global optimization iterations. The results are shown in Figure 6.5. As can be seen, with such a small change in the setup, the results are massively different in many aspects. For once, the catalytic effect itself has been improved by another 5 kcal mol^{-1} . Additionally, the reaction path seems a lot longer, however one should keep in mind that the RMSD used as the x axis now includes deviations from the entire system, including the rearranging acetone molecules. Lastly, the residual gradients at the stationary points are now much

²Technically, there is a small caveat: When a new embedding (e.g. from a crossover) is evaluated, the respective parent's reaction path is used in a local optimization with the new, unique embedding. The resulting, new reaction path is then saved as the child's reaction path, but the original new embedding is retained; thus the finished child has a different reaction path / embedding combination than the one used in the very first fitness evaluation. In practice, this difference, however, turns out to be small and non-problematic.

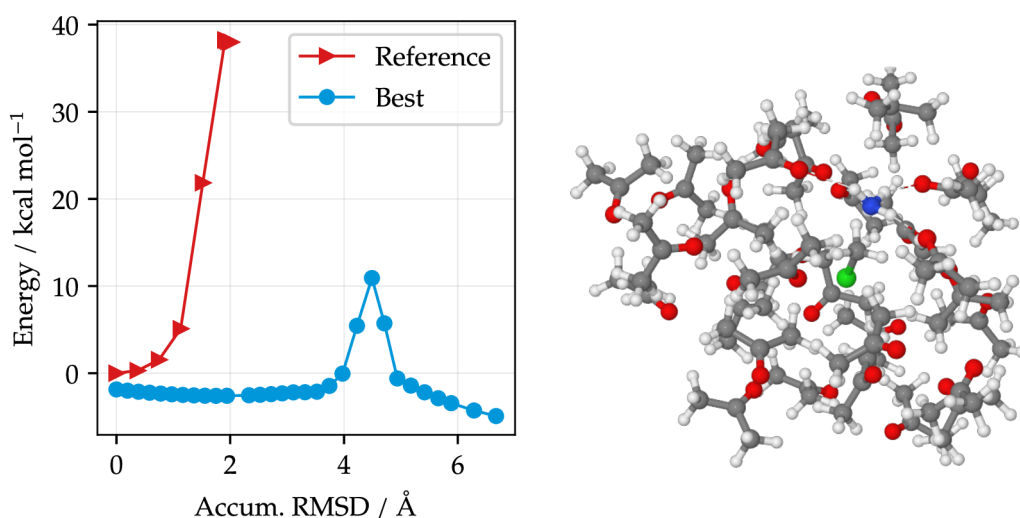


Figure 6.5: The left plot shows the energy paths of the reference and the best candidate found in the adaptive embedding/adaptive reaction path system. On the right, the structure of the best embedding found is shown, including the transition state of the embedded reaction.

lower (at around 1 to 3 kcal mol⁻¹ Å⁻¹), which can now be properly considered to be actual stationary points on the potential energy surface. Another noteworthy feature is the massively stabilized product structure compared to the previous approaches. This is a consequence of the fact that now both the reactant and product structure are properly locally optimized in the full QM/MM system before interpolation and calculation of a new reaction path. This leads to potentially much lower energy optima being found as start and end point for the new reaction path.

6.2.4 Comparing the Setups

Of the four different approaches discussed in the sections 6.2.1 to 6.2.3, the three viable setups will be compared and discussed here. The first viable setup is the static model where all QM-MM and MM-MM interactions are included. Its advantages and disadvantages are as follows:

- + Very quick and efficient global optimization, as only a few energy/gradient single points are calculated per fitness evaluation.
- + Simple setup, as fitness function has fewer options and parameters that need to be considered and adjusted.
- The input reaction path is static, thus changes to the reaction path due to an embedding are assumed to be negligible.

- The embedding is only locally optimized once in presence of the reaction transition state and is then used for all structures along the reaction path, making it effectively rigid. This is equivalent to assuming the embedding to always be oriented perfectly for the transition state structure during the reaction.
- With the two last points combined, large residual gradients are accepted at reactant, transition state and product structures, making the structure prediction ultimately unreliable.

In summary, this setup might be a good choice if the other two approaches are too computationally demanding, i.e., if the QM part is too large, and whenever a reaction occurs so fast that significant rearrangement of the embedding is not to be expected, e.g., with a proton transfer. However, the gas phase reaction path staying as is seems to be such a detriment that even though many more global optimization iterations can be made in the same time frame due to the efficiency, the results are usually worse at the end of the day.

Allowing the reaction path to be recalculated during global optimization alleviates some of the disadvantages of the fully static approach, however it introduces new disadvantages listed below:

- + The reaction path is dynamically adjusted to fit the embedding, allowing both new reaction channels to be discovered, as well as giving a much better approximation of the actual structure during reaction.
- + Better catalysis due to tighter packing of embedding and embedded reaction
- The embedding is only locally optimized once in presence of the reaction transition state and is then used for all structures along the reaction path, making it effectively rigid. This is equivalent to assuming the embedding to always be oriented perfectly for the transition state structure during the reaction.
- Due to this, large residual gradients can still occur at reactant and product structure.
- The recalculation of reaction path requires thousands of energy/gradient single point energy evaluations, making the process computationally very demanding.
- More parameters need to be adjusted that deal with how the automatic NEB algorithm calculates and evaluates new reaction paths.

This setup improves significantly on the fully static model in every aspect except the computational demand; While the fitness function itself becomes a lot more complex, with several additional parameters and settings, testing indicates that these parameters have

reliable defaults that are mostly transferable in between different reaction and embedding systems. One issue that persists with this approach is that the reactant and product structure can still be significantly displaced from the true minima. Figure 6.6 shows the structures obtained from each of the three approaches described previously, colored according to the gradient norm at each atom. As can be seen, the fully static approach has significant gradients at both the reactant, as well as the embedding atoms in close proximity. The adaptive reaction path setup improves only minimally on this.

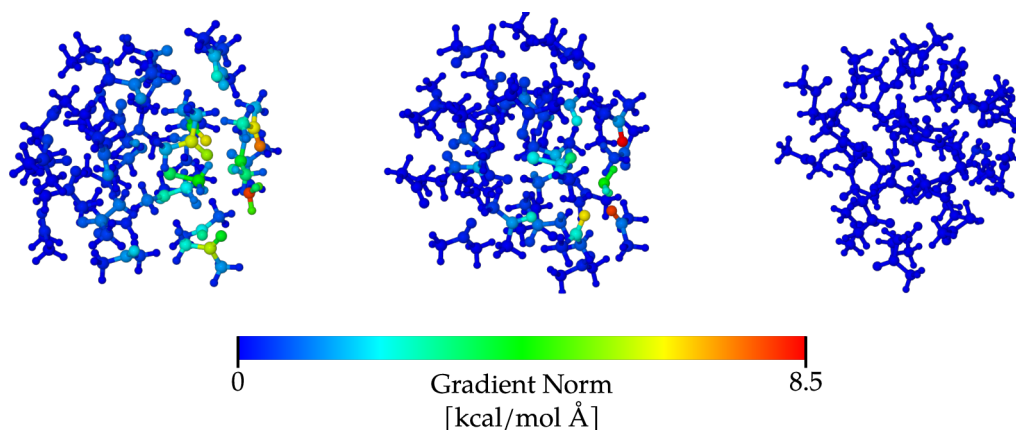


Figure 6.6: This plot shows the gradient norms at the reactant structure of the fully static (left), adaptive/static (middle) and fully adaptive (right) setup. The structures are each the best candidate from the respective pool.

Finally, the last setup, in which both embedding and reaction are adaptive, improves upon the previous once again, now additionally capturing the rearrangement from the optimal reactant embedding via the optimal transition state embedding to the optimal product embedding. As a consequence, the stationary points are now actual extrema of the PES, and the residual gradients are essentially negligible, as is also visible in Figure 6.6. When it comes to computational cost, this setup is virtually identical to the previous, with much improved gradients and, as has been shown previously, improved catalytic efficiency. It thus seems to be the most promising variant for further studies involving adaptive molecular embeddings.

6.3 Active Site Design

With just a small conceptual addition to the approach, namely the introduction of a dynamic molecular library for the embedding instead of a pre-set system of molecules, this setup can be used as a versatile and powerful active site design tool. When the molecular library is e.g. chosen to include amino acid side chains, model enzyme active sites can be generated for any arbitrary chemical reaction, with as little of an input as the reactant and

product structure. Additionally, owing to the implementation into the usual workflow of global optimization and re-optimization of reaction paths, the results are globally optimized solutions, with the complete reaction path immediately included. Naturally, this approach can not create the template for an entire enzyme out of the box, since no backbone is included in the resulting active site. The problem of protein design and folding would still have to be delegated to a follow-up step.

While the advantages of this approach are clear, the implementational cost and disadvantages are not immediately obvious; the inclusion of such a molecular library would eventually lead to candidates with differing numbers and types of atoms sharing a common pool, which is something most of the crossover and mutation operators implemented in OGOLEM were not inherently built for. Additionally, and more importantly, all previous iterations of the GOCAT model that allowed for recalculation of the reaction path made sure to only do this for *promising* candidates. In those cases, it was possible due to the ability to measure the fitness of a candidate in comparison to a reference, with a known and consistent number of atoms and respective atom types. If every candidate has differing atoms and molecules, this becomes challenging.

One solution to this would be to simply recalculate the reaction path of *every single* candidate, which is, sadly, egregiously inefficient computationally. A small pre-check was instead reintroduced, similar to the one described in Section 4.1, making sure that at least candidates that are very far off of true minima (with really large gradient values at the stationary points) and/or have very large energy barriers before recalculation are discarded.

6.3.1 Molecular Library

Breaking the uniformity of the interaction entities during global optimization requires a consistent list of all molecules that are available for new candidate generation and modification. This list is simply defined in the input file to OGOLEM for each run and consists of coordinate files containing the force field types and bond lists of each unique interaction molecule.

For protein active site design, this list totals 18 entries, including all canonic amino acids sans glycine and proline. The reason for this is that all amino acids were capped at the backbone carbon atom, which is impossible for proline without breaking the ring structure and glycine would simply turn into methanol. Note that, additionally, leucine and isoleucine become, in essence, the same molecule when capped in this way.

All of these were then optimized with the OPLSAA force field and added as a list into the OGOLEM input. Figure 6.7 shows an overview over the entire library of structures.

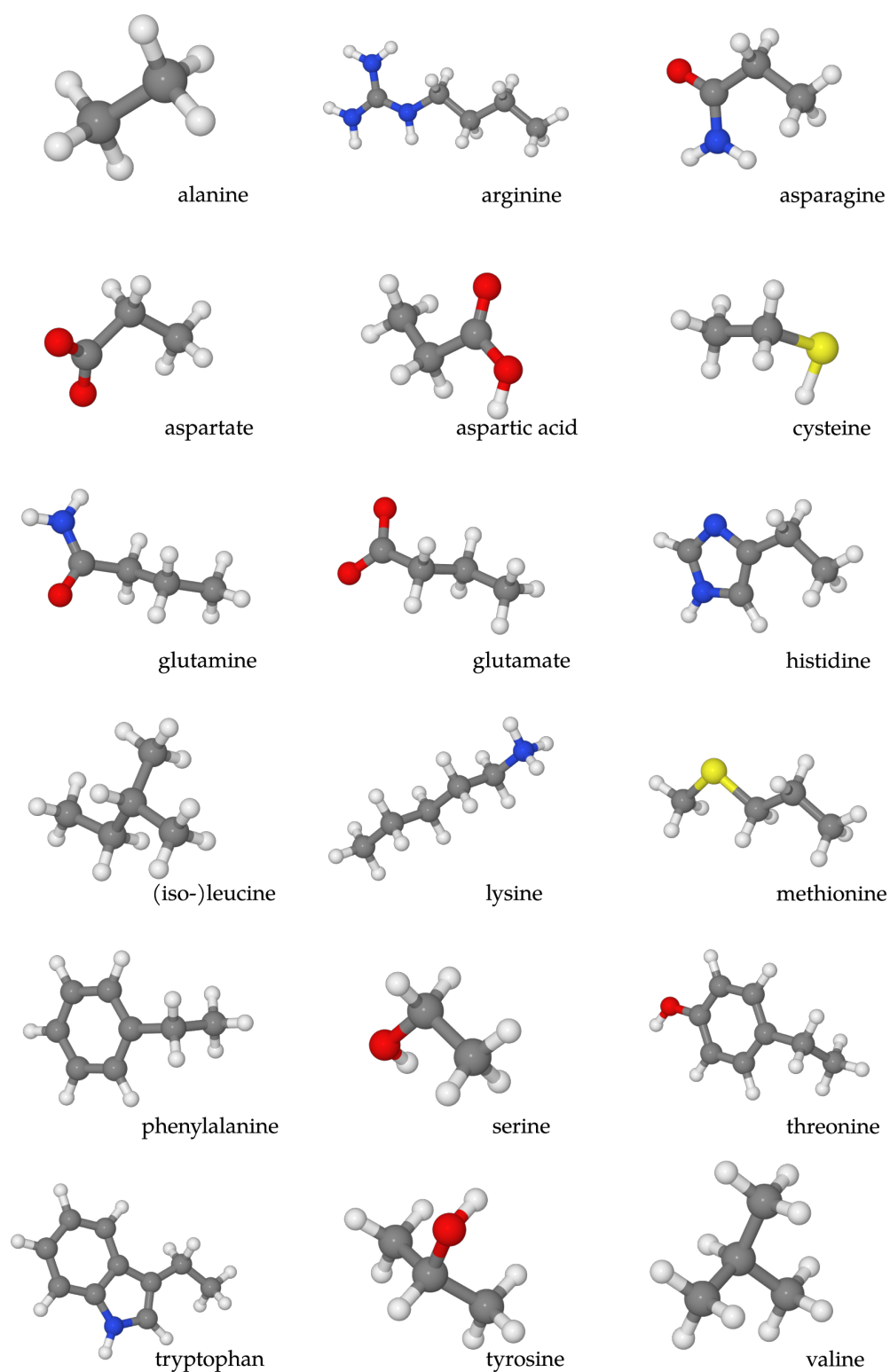


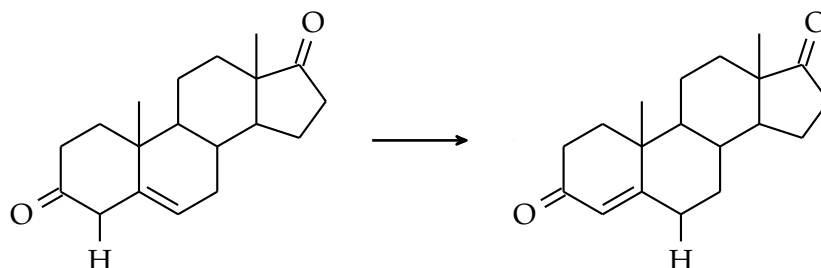
Figure 6.7: The OPLSAA optimized molecular library used for active site design. It includes capped versions of all canonical amino acids, excluding glycine and proline.

6.3.2 KSI

The setup dealing with KSI was already shown in an initial version as part of a publication^[47], however the view provided there is quite brief and early into the setup's development. This section here will thus provide additional details, insights and developments that have taken place since then.

As a short recap, KSI is an example of a very highly efficient enzyme with one of the fastest turnover rates known for enzymes (essentially being diffusion-controlled) and a catalytic proficiency of more than 10^{15} for its optimal substrates, meaning that the catalyzed reaction proceeds 15 orders of magnitude faster than the uncatalyzed variant^[121]. Due to the efforts of FRIED and BOXER^[26], who performed Stark spectroscopy to evaluate the electric field generated within KSI and related it to its catalytic proficiency, it has become a very attractive model system for electrostatic catalysis.

Consequently, this also makes it a prime candidate for a proof-of-principle study for the active site design approach introduced in this chapter. One example of an optimal substrate for this enzyme is the 5-androstene-3,17-dione, in which the enzyme abstracts a proton adjacent to a keto group and reinserts it at the 1,4 position:

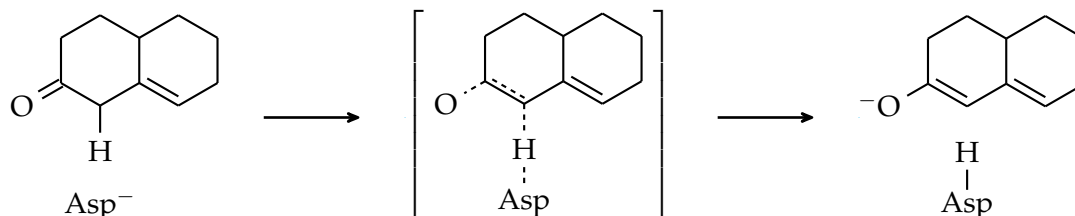


It does this in two steps: First, an aspartate abstracts the proton in the 1,2 position. Then, it must rotate only minimally to donate the same proton back to the correct position in the substrate^[121]. The first step is in this case the most problematic, since the abstraction of a proton in a 1,2 keto position is usually associated with a large activation barrier. For this reason, it is reasonable to assume that this reaction will be the rate-limiting one and catalysis efforts should be centered around it.

Reference

Because the androstene is a fairly large molecule, it was approximated with a much smaller system that includes only the two six membered rings on which the reaction takes place. This ensures that the quantum mechanical system will later be small enough for reasonable computation times, while (hopefully) sacrificing little accuracy. The aspartate involved in the reaction was also chosen to be part of the quantum mechanical system, since

a base is needed for the reaction and will give consistency with the reference. There are, however, downsides to this approach that will be discussed later. Lastly, the following scheme shows the reaction as it will be used throughout this section:



The idea here is to optimize GOCATs using the molecular library approach and benchmark them against the quasi-optimal active site of KSI as a reference. In order to do so, first the active site of KSI needs to be extracted from the enzyme structure, then a fitting semi-empirical method needs to be found and lastly, the GOCATs need to be analyzed and compared to this reference.

To setup the reference active site, the crystal structure of the wild type^[122] (PDB 1OH0) was taken as a basis. The amino acids closest to the active site were isolated, making sure that all side chains known to be involved directly in the reaction are included. Then, the amino acids were capped in the same manner as the ones used in the molecular library (Figure 6.7). This process yields a system of 328 atoms, which is visualized in Figure 6.8.

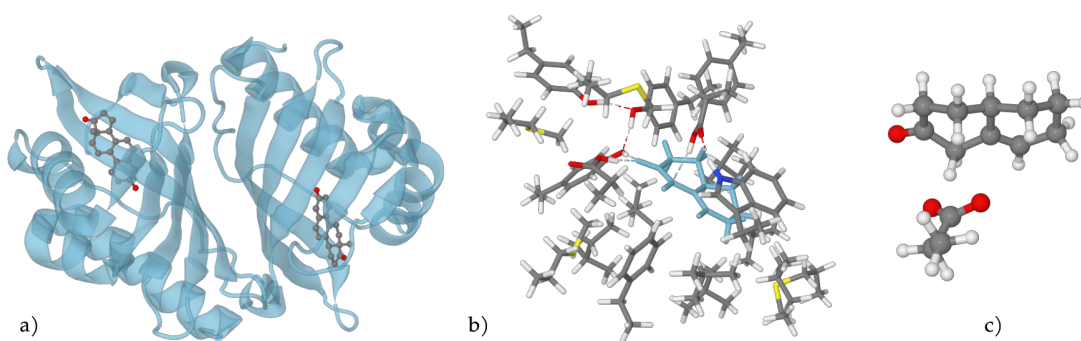


Figure 6.8: Steps taken to setup the KSI model, with a) 1OH0 with inhibitor molecules. From this the extracted and capped active site b) was generated that is shown here with the simplified substrate molecule in light blue. In c), only the substrate is shown with the aspartic acid that acts as the base during the reaction. This is also what was used as QM part for the QM/MM calculations.

While in single geometry optimizations and even isolated NEB calculations, this is still a manageable size for an explicit semiempirical treatment, this is too many QM atoms for use in GOCAT optimization itself. On the other hand, it allows for an easy check of QM/MM consistency by converting the pure QM ansatz into QM/MM and observing

closely what this does to the system accuracy. This was done for PM6, PM7, GFN1- and GFN2-xTB. However, while consistency varies dramatically for all of these methods, it was possible only for GFN1- and GFN2-xTB to optimize reasonable reaction pathways for a full-MM system, in which all side chains except the Asp acting as the base are converted to MM.

For GFN2-xTB, the QM treatment of the entire active site model works very well in vacuum. The reference structure and energy path for the reaction are shown in Figure 6.9. Problems arise however when the system is converted to QM/MM. With all side chains

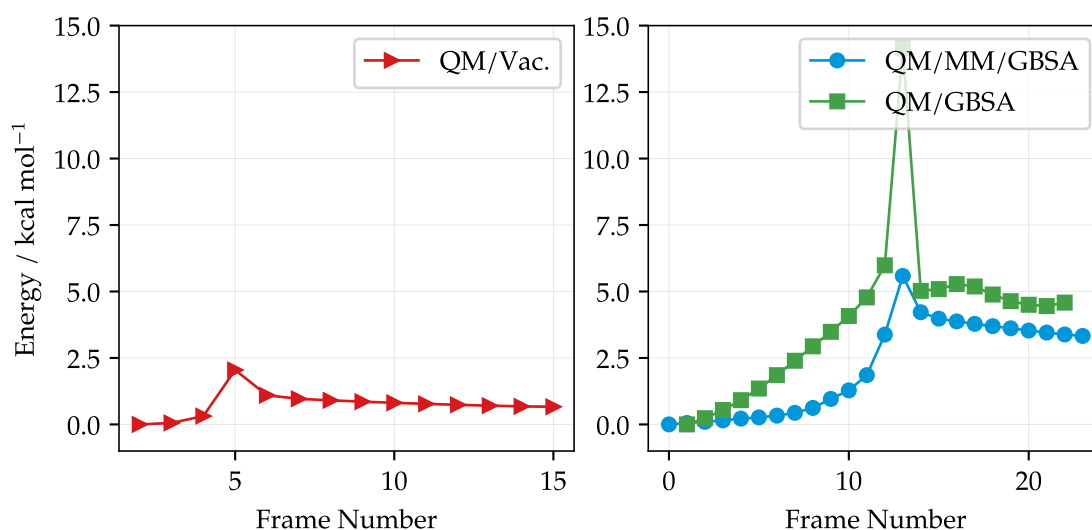


Figure 6.9: The plot on the left shows the reference energy path for the full-QM GFN2-xTB vacuum model of the KSI active site, on the right the energy path for QM/MM systems are shown (s. main text for details).

except the aspartic acid acting as base converted into MM molecules, the product becomes unstable because the aspartic acid is not basic enough any longer. This can be remedied by moving over to a system that is additionally implicitly solvated (in water). This is a reasonable approach, given physiological conditions, and allows a clean QM/MM reaction path optimization. On the flip side, this again leads to inconsistencies in the full-QM approach: With the implicit solvate, the aspartic acid forming the hydrogen bridge to the substrate carbonyl oxygen actually donates its hydrogen to the substrate, stabilizing the anion in this manner instead, which helps to explain the huge energetic discrepancy found at the transition state in the all-QM model.

These results are not entirely discouraging, since an adapted workflow will still enable a comparison to these references, when the GOCAT optimization takes place in the QM/MM setup with implicit solvent enabled. This makes it possible to generate properly

comparable results to the QM/MM KSI reference. Afterwards, when the GOCAT results are converted to all-QM, the implicit solvent is deactivated for comparison against the all-QM reference.

Fitness Function

The setup in this case requires a few unique additions to the fitness function to be able to efficiently find viable solutions. The main problem is that without any MM or QM embedding, the aspartic acid cannot abstract the substrate proton in GFN2-xTB, even in implicit solvate. Randomly generated candidates at the start of the global optimization thus have only a very slim chance of immediately hitting a good embedding where this is possible, leading to a tough start for the optimization algorithm. Thus, instead of completely disallowing any change in the bond list upon local optimization of the product, whenever the change is due to the proton not being bonded to the aspartic acid, a penalty is applied instead. This penalty is (arbitrarily) chosen to be

$$p_{\text{O-H-C}} = \begin{cases} 5000 + 5000 \cdot (r_{\text{O-H}} - r_{\text{H-C}}) & \text{for } r_{\text{O-H}} > r_{\text{H-C}} \\ 5000 & \text{otherwise} \end{cases}$$

The constant offset ensures that faulty candidates with the proton at the substrate in the product structure are always worse than non-faulty candidates. At the same time, it allows preference of embeddings that favor a longer C-H bond (and shorter distance to the aspartic acid oxygen), such that the optimization eventually converges to only solutions that have a proper product structure. Other than this additional setting, the fitness function is essentially the same as the one used for the S_N2 reaction in Section 6.2.3.

GOCAT optimization results

For the final GOCAT optimization, 5 pools of each 600 candidates were run for 50000 global iteration steps. Of the 3000 resulting candidates, all candidates were filtered out that had an energy barrier higher than 10 kcal/mol or were otherwise unsuitable due to bond list issues or unconverged reaction paths. This left only a very modest amount of 37 final candidates. These 37 can be categorized into four potentially successful solutions to the catalysis.

The first variant (1) stabilizes both oxygen atoms using hydrogen-bonded arginine and aspartic acid side chains. The most successful candidate with this variant is also the best one found throughout the entire pool and is summarized in Figure 6.10. The figure also includes a structure and reaction path upon going from the QM/MM GBSA model used during global optimization to the all-QM vacuum model. There, a second asparagine side chain that faces away from the substrate in the QM/MM model rotates upon

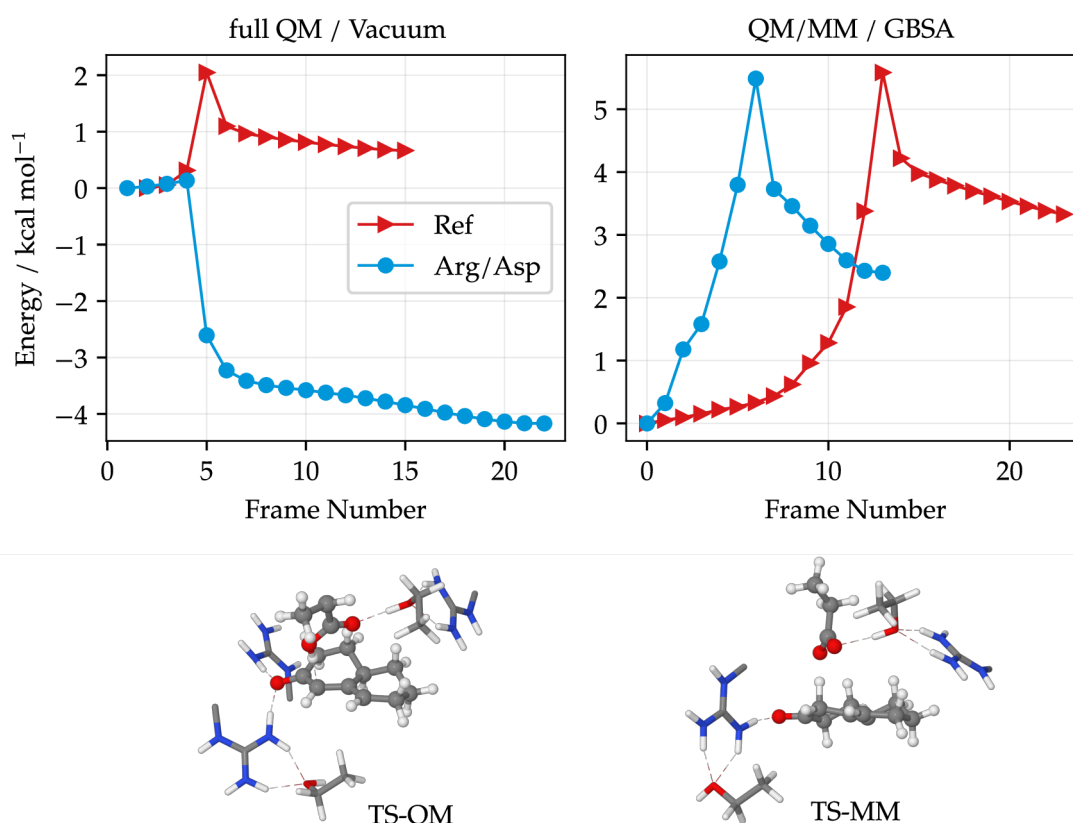


Figure 6.10: Summary for GOCAT variant 1, showing the reaction path for full-QM and QM/MM models compared to the KSI reference, as well as transition state structures. Note that for visual clarity, of the 18 side chains in the candidate, only the (truncated) arginine and aspartic acid side chains are shown that are mentioned in the main text.

local optimization in the full QM system to also form a hydrogen bond with the substrate oxygen atom. All in all, this is a very successful candidate as it not only lowers the energy barrier drastically but also allows for clean reaction path optimization and true transition state structures to be found. In fact, the QM/MM structure and the QM structure (to an even greater extent) both have very impressive hydrogen bonding structures involving asparagine, aspartic acid, glutamine and histidine side chains. This also explains why the product structure in both cases is palpably more stabilized than in the KSI reference model. What is interesting here is that the global optimization automatically discovered the necessity to also stabilize the aspartate base and thus, the aspartate remains perfectly placed to not only perform the first but also the second step in the KSI reaction system.

The second variant (2) features a lysine and histidine side chain that are hydrogen-bonded to the substrate oxygen atom. The aspartate is stabilized via a chain of hydrogen-bonded aspartic acid molecules. Compared to the first variant, this one is a little harder

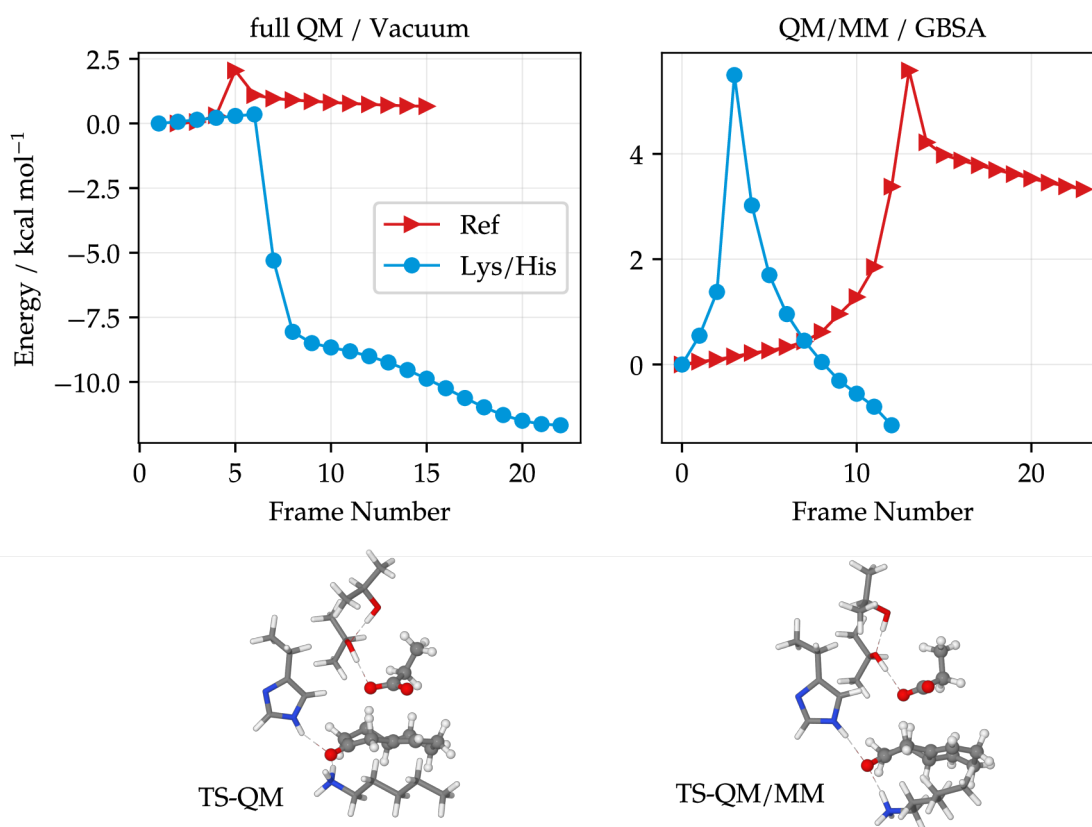


Figure 6.11: Summary for GOCAT variant 2, showing the reaction path for full-QM and QM/MM models compared to the KSI reference, as well as transition state structures. Note that for visual clarity, out of the 18 side chains in the candidate, only the lysine, histidine and aspartic acid side chains are shown that are mentioned in the main text.

to compare to reference. Although the QM/MM energy barrier is very similar to the first, there is a major issue here: Lysine is not as basic as arginine and, when allowed in the all-QM model, actually protonates the oxygen atom, slightly altering the mechanism. Figure 6.11 shows a summary of the reaction paths and structure for this variant.

The third variant (3) uses only a lysine side chain to stabilize the substrate and a glutamine side chain for the aspartate. In many ways, it is very similar to the second variant, with the same problems in replicating a clean all-QM NEB, as the lysine protonates the substrate. Additionally, this variant is worse than the first two when it comes to its catalytic properties, with both the QM and QM/MM energy barriers notably higher than the KSI reference.

The last, and actually least, (4) is very odd in the sense that in the QM/MM structure, no hydrogen bonding is present with either the aspartate or the substrate. This becomes a problem when the structure is optimized in an all-QM setting, as there are no rigid struc-

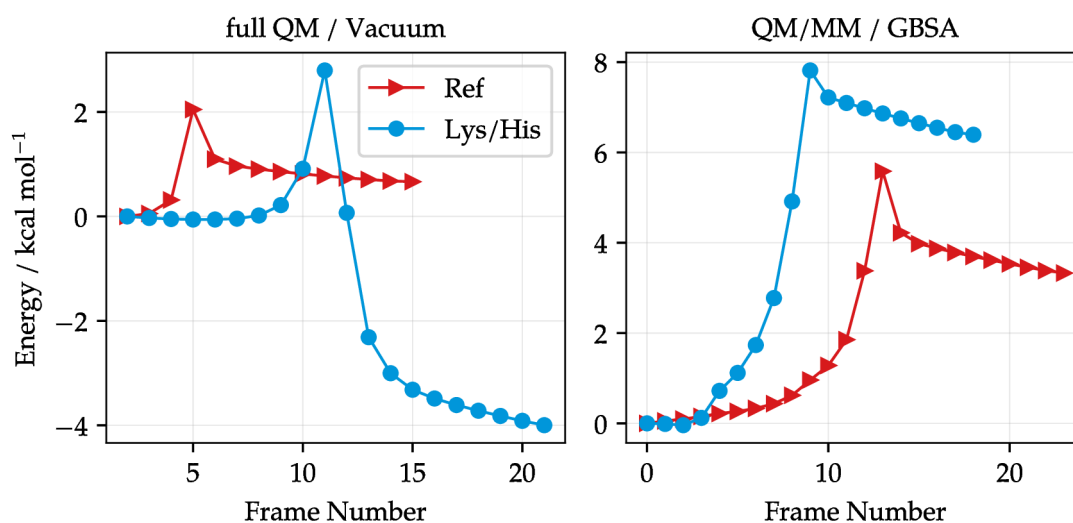


Figure 6.12: Summary for GOCAT variant 3, showing the reaction path for full-QM and QM/MM models compared to the KSI reference.

tures that keep the substrate in its optimal binding position. Upon optimization, a lysine close to the substrate carbonyl atom rotates towards it and eventually transforms this variant into one very similar to (2) and (3). However, with the xTB optimized structure, the reactant structure is no longer a stable minimum and no NEB could be optimized.

As has been discussed before in both Ref.^[47] and the sections above, the ingredients in the global optimization are actually still very similar to the abstract GOCAT models, where the catalysis in the QM/MM systems is achieved via the polarization imposed onto the QM part by the point charges of the MM part. Figure 6.13 shows how the four variants detailed above polarize the QM part compared to the KSI reference MM part. First of all, there are two very interesting features present in the reference. First of all, the net electrostatic potential is actually positive at all atom positions (around 3 kcal/mol·e at the lowest). Secondly, the oxygen atom of the aspartate that is not protonated at the transition state faces up into the substrate plane. Compared to the GOCAT variants, this is only the case in the worst variant (4). Additionally, the two best variants (1) and (2) have, in general, a much higher positive electrostatic potential acting upon the substrate. This is expected however, given the fact that compared to the charge-neutral embedding in the KSI reference, (1) and (2) have a net charge of +2, (3) a net charge of +1 and (4) is neutral. Given prior findings^[34], the absolute value of the potential acting upon the QM atoms is, however, actually irrelevant for the polarization effect. What matters is the difference between relevant atoms. In this light, it is not surprising that (1) and (2) are the superior variants: they have a very smooth polarization scheme similar to the KSI

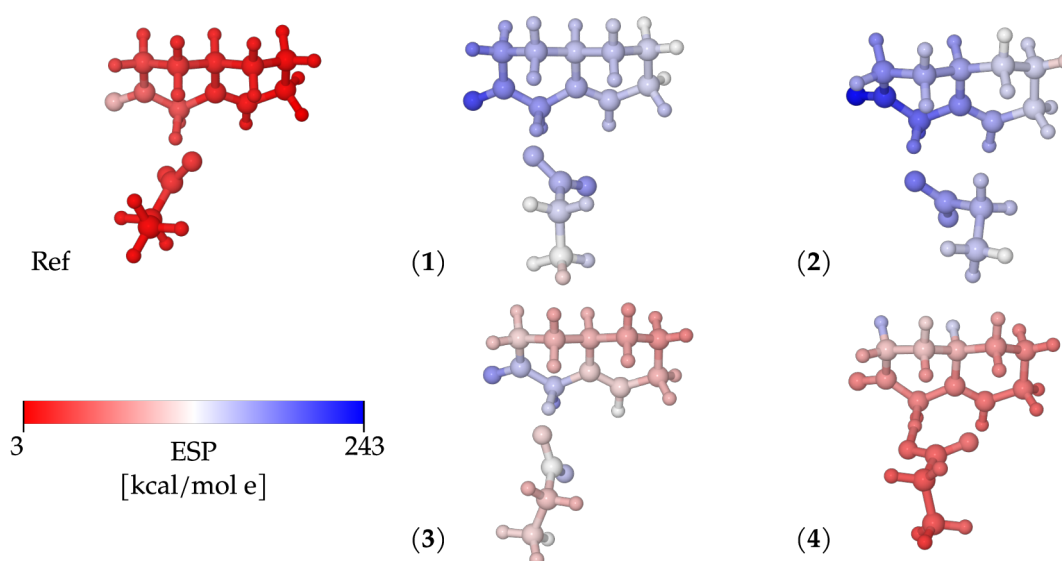


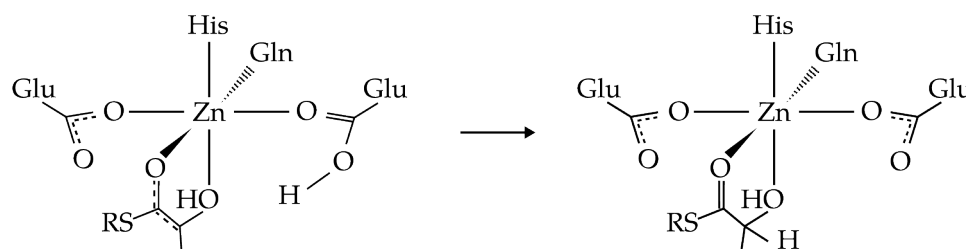
Figure 6.13: Transition state structures for GOCAT variants (1)-(4) and for the KSI reference. Note that these plots were generated from the QM/MM models.

reference, with more positive potential at the substrate carbonyl- and aspartate carboxyl oxygen atom. While (3) and (4) also share this trait, the potential is less regionally specific and more erratically distributed across the substrate molecule. As pointed out already, another interesting feature that seems to be missing in the GOCATs (1)-(3) is the oxygen atom pointing upwards into the substrate plane. The reason why this is interesting is, again, the second step of the reaction: with the carboxyl atom positioned in this way, it gives a very easy rotation axis (around the oxygen atom) that immediately puts the abstracted proton in the exact position necessary for the deposition, which might not be as easily done in the GOCAT variants. This would, however, require additional investigation of the second reaction step and how the candidates perform there and will be discussed further in the conclusion below.

6.3.3 Bachelor's Project: LEMINSKY

In his bachelor's thesis, LEMINSKY investigated the possibility of using transition metal cations in the aforementioned setup. Many enzymes contain metal ions in their active sites^[123] and it would certainly be beneficial to include this possibility in the design approach, such that a greater variety of reactions become accessible within the framework. For any system where the transition metal is going to be involved in the reaction itself, it is obviously necessary to include the metal cation in the QM part. Additionally, it might be wise to include the ligand molecules that bind to the metal cation as well, as the electronic structure of the complex is relevant to the reactivity^[124]. In an approach similar

to the one taken for KSI above, first a reference reaction was chosen. In this case, the choice fell on Glyoxalase I (GloI), which is part of the glyoxalase system that converts methylglyoxal to lactic acid. Because the intermediate, lactoylglutathione, is associated with cancer growth, the mechanism and structure of GloI have been well researched in an effort to find an efficient inhibitor^[125]. The reaction of GloI involves a zinc cation that is bound in the active site, and the rate determining step of the reaction is the following proton transfer^[125,126]:



This proton transfer was successfully modeled with GFN2-xTB, with a reaction barrier of around 19 kcal/mol with the closest 15 amino acid side chains found in the crystal structure included. For the subsequent GOCAT optimization, some adjustments had to be made to the fitness function. A method was implemented which considers the atoms closest to the zinc ion. Whenever the zinc ion has an oxygen, nitrogen or sulfur atom in its close vicinity, the side chain this heteroatom belongs to is then set to be included in the QM, else it is set to be MM. With this change, some early results could be generated by Leminsky that showed that the approach can generate functional and catalytic individuals, however just as in the KSI case, the reference could not be rediscovered. Ultimately, this approach requires a lot of additional refinement and computation time than even the approach without transition metals, due to the dynamic inclusion of additional QM regions. Consequently, while including transition metals is certainly possible within the methodology, it slows down an already quite inefficient global optimization even further, and might ultimately require a different approach.

6.4 Conclusion

The original idea behind the GOCAT approach was to find abstract representations of an ideal catalyst and then worry about how it is synthesized or constructed in a second step. In this chapter, it was shown that abstract representations are not limited to point charges or dispersion entities alone, they can also take the shape of larger molecular fragments. In the solvate shell example, it can be argued that this deviates from the abstract idea originally proposed, however, it was done as a proof-of-principle study on the feasibility

of molecular GOCATs and emerged as more of a side-product and improvement to the early studies conducted before the inception of the GOCAT model.

With molecular fragments, the GOCAT model gains the ability to work with systems that have limited building blocks which are explicitly known. It improves the accuracy of what is achievable within the given framework and can help bridge the ideal abstract catalyst and its realization. The great strength here lies in the flexibility of the fitness function. Since it is so versatile, it can easily be modified to include additional restrictions imposed on a specific reaction. In the KSI case for example, the enzymatic reference does a lot more than just lower the energy barrier of the proton transfer. It also seems to destabilize the intermediate just enough so that recrossing is less likely and positions the base exactly such that the second reaction step is just a minimal rotation away. While the GOCATs presented in the respective section did not reproduce these features, it would certainly be possible to impose these conditions through the fitness function, resulting in more precise and efficient GOCATs.

The model has its advantages, but it also comes with its disadvantages. Compared to the electrostatic or dispersion models, it is much harder to formulate specific criteria with which to determine whether to optimize new reaction paths or not. This is detrimental because these additional NEB calculations take up a lot of time comparatively and hurt the efficiency of the global optimization. But this is not something that is entirely without remedy; In fact, one approach could be to pre-filter via an ESP reference as it was described in Section 5.1. By optimizing an ideal electrostatic embedding for the reaction first (which can be done more efficiently) and then only attempting new reaction path optimizations when within a given threshold to this reference in the molecular fragment optimization, additional efficiency can be gained. This actually goes in a direction more in line with the original idea to establish a two-step process and go from an ideal embedding to a molecule best fitting this ideal. However, the result in this case would still be a somewhat abstract in-between solution that bridges the gap between the two approaches.

Eventually, to fully utilize and take advantage of this methodology, more data needs to be collected on how this model performs across multiple reaction systems, and molecular libraries. For example, it could be tested not only for other enzymatic reactions, but also for reactions in e.g. porous host molecules like MOFs or zeolithes. It would be fairly easy to incorporate constraints on the molecular fragments binding them to very specific sites around the reactants, making this an interesting endeavor worth pursuing.

SURFACE DOPING

In industrial processes, one of the most versatile and popular catalysts is the heterogeneous catalyst^[7,127]. It is, among other advantages, very easily integrated into flowing systems with high throughput. As a consequence, heterogeneous catalysis and surface science strive towards further understanding and improvement of solid and surface catalysts^[128–130].

While a vast majority of commonly employed catalysts in heterogeneous catalysis consist of transition metals, there are also other well-researched surfaces known to have catalytic properties. One such example is the graphene surface, which allows for doping with nitrogen, boron, and other elements (e.g. oxygen, phosphorus, silicon or various metal atoms in defect positions)^[131] and has been shown to enable metal-free catalysis of oxidation reactions^[132].

This, together with the fact that graphene can be easily modeled with available semi-empirical methods such as GFN2-xTB, makes graphene an attractive first candidate to establish an automated workflow for finding catalytic doping schemes for any arbitrary reaction.

7.1 Implementation

The automated doping workflow was implemented in OGOLEM, taking full advantage of its efficiently parallelized global optimization and it thus functions very similar to the previously described optimization classes. The individuals in the optimization pool are implemented as their own separate problem class termed a `DopableSurface`. It contains information about the surface grid coordinates, lattice parameters and atom symbols at the individual grid positions.

For exploration in the optimization space, a simple algorithm is implemented that substitutes a random atom within the grid with a different type of atom, resulting in a random doping of the surface. For exploitation, a crossover algorithm is implemented which works just like the one described for particles in Section 3.1. By cutting two incoming source surfaces along a common orthogonal plane and then exchanging halves between them, it exchanges structural, as well as doping information between the individuals.

7.2 Reproducing GOCAT references

For the initial setup, the goal was to dope a graphene surface in such a way that it reproduces the electrostatic potential of a reference point charge embedding GOCAT. Functionally, this requires at least the following steps:

1. Calculate a reference electrostatic GOCAT for the desired reaction.
2. Make sure the reaction proceeds as expected in the presence of a pristine graphene surface (desirably, unchanged from the gas phase path used to calculate the reference GOCAT).
3. Starting from a pool of randomly doped graphene surfaces, globally optimize to find the best fit of electrostatic potential above the graphene surface to the reference GOCAT electrostatic potential.
4. Evaluate the catalytic effect by comparing the energy barrier of the reaction on the doped surface to that on the pristine.

The global optimization in step 3 itself proceeds in a scheme similar to previous global optimization problems in OGOLEM. The starting population is initialized by reading in a reference graphene lattice and saving both lattice parameters as well as each of the atom coordinates and symbols. Then, each of these candidates is randomly doped by up to 4 hetero atoms, by simply replacing the atom symbol at a randomly selected atom site.

Subsequently, the candidates enter the fitness function. There, the doped lattice is locally optimized first, such that the electrostatic potential remains consistent when the lattice is used in the eventual evaluation in step 4. Afterwards, the Mulliken charges obtained from the locally optimized geometry are used to calculate the ESP above the surface, and compared to the ESP provided by the reference GOCAT from step 1. This comparison uses the same underlying functions that were also used in the GOCAT simplification approach in Section 5.1. Finally the difference (calculated in form of a RMSD) is augmented by a penalty term if the total energy of the doped surface is significantly higher than the undoped one. This helps prevent chemically questionable results that can happen on very highly doped lattices.

Finally, the resulting individuals are then evaluated by placing the reactant onto the doped surface, followed by local optimization. The product structure is generated from this locally optimized reactant structure and locally optimized as well. Then, a new reaction path is calculated via a NEB calculation and compared to the pristine reaction path.

For most of these steps, an efficient method is required that can locally optimize and calculate partial charges for the graphene surface quickly and can handle periodic lattices.

In this case, the GFN2-xTB implementation in DFTB+^[133] was used, as it fulfils these requirements.

7.3 Bachelor's Project: PFLAUM

As a very first attempt at exploring this new approach, PFLAUM investigated the keto-enol tautomerization of acetophenone on a graphene surface in his Bachelor's thesis^[134]. For this purpose, he optimized electrostatic GOCATs with the point charges arranged on a 2D plane below the acetophenone, as opposed to the established 3D variant. He used this as a reference for both manual, as well as automatic doping of the graphene as detailed above. The reference systems that were used are summarized in Figure 7.1. As can be seen, the

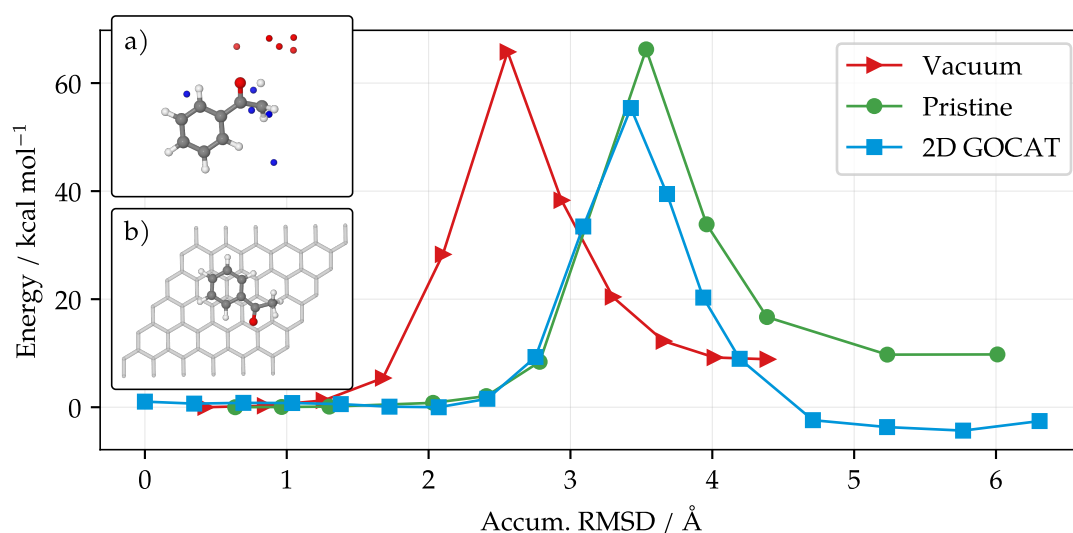


Figure 7.1: This plot shows all the reference energy profiles generated by PFLAUM for the acetophenone reference. Inlaid are a) the GOCAT structure and b) the pristine graphene surface with the acetophenone on top. Both are shown at the transition state.

pristine graphene surface does not influence the reaction, which is virtually the same as in vacuum. The GOCAT is very linear, with essentially only negative charges on one side and only positive on the other. While it does catalyze the reaction, the difference is not massive. Some of the results of trying to dope the graphene according to this reference GOCAT can be seen in Figure 7.2. The reaction mechanism remains untouched for these examples and, while the effect is quite small, all three are actually catalytic with respect to the pristine reference. In conclusion, the approach has been shown to work, and it would be interesting to try this with other reaction systems. The example here is not a reaction typically catalyzed on a graphene surface, thus moving over to reactions that are known to happen on graphene surfaces might give better results. Also, in interest of expedience

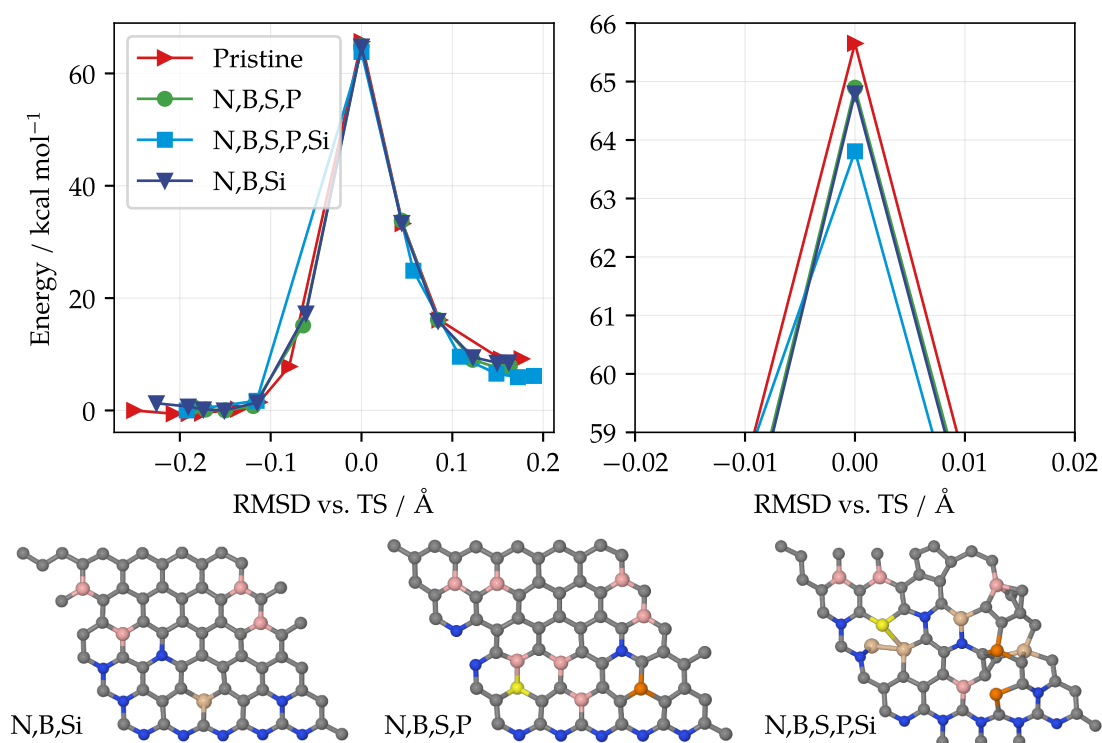


Figure 7.2: The top plot shows the energy profiles and structures for selected doped surfaces, obtained using the global doping optimization algorithm. On the bottom, the respective surface structures are shown.

in the calculations, the graphene unit cell was chosen to be very small, barely containing the acetophenone. This obviously suggests a surface that is quite densely packed with acetophenone and it might be wise to choose a bigger cell size. Lastly, this approach could also easily be transferred to 3D grids such as carbon nanotubes, which might also give interesting new results.

CONCLUSION AND OUTLOOK

As an inverse design method, the biggest unsolved issue in the GOCAT model is how the ideal catalysts can be transferred into actual chemistry. There are several avenues that have been explored and will need to be explored further in the future: First of all, the electrostatic GOCATs are already useful by themselves; similarly to the OEFFs by SHAIK and co-workers, as well as the catalytic fields put forward by SOKALSKI and co-workers^[33], they can very certainly be employed to find sites in enzyme scaffolds that, upon alteration, give improved electrostatic conditions for the respective reaction. As has been shown in the doping optimization, this can also be transferred to the modification of any suitable framework. Apart from modifying existing structures, projects such as DENOPTIM^[46] and GAMMA^[36] also hold great promise that, eventually, de novo molecular structures can be generated from GOCAT templates at the push of a button and have inherent catalytic properties, inching ever closer to the holy grail mentioned in the introduction.

The electrostatic GOCAT variant is very fleshed-out and it can be stated with confidence that the next step here should indeed be its application to real-world problems, ideally with experimental verification of its predictions, or at the very least molecular dynamics simulations. The versatility that it has gained with the introduction of simplification and alteration protocols (s. Section 5.1) allows for ultimate flexibility in this regard. For any arbitrary reaction, only a single GOCAT must now be optimized and it can be recast into any possible shape or complexity. In fact, early testing suggests that even the transmutation between electrostatic and molecular GOCAT models (Chapter 6) is possible without too much trouble.

One of the interesting points that is distinct to the implementation of GOCATs is that, with the addition of re-calculating reaction paths, new connected stationary points on the potential energy surface can be found for any arbitrary reaction. Depending on how the settings are tuned, these can be very close to known vacuum or solvated structures. However, as can be seen in the case of the dispersion GOCAT variants (s. Chapter 4), they can also differ quite a bit. GOCATs are thus able to explore new structures which could provide valuable insights or even discover previously unknown catalytic pathways. This has already been shown to be possible with electrostatic GOCATs in the Diels-Alder example by Dittner^[101]. For the dispersion GOCATs, the initial attempts at finding such a case have not borne fruit yet. This might eventually happen down the line and certainly makes

for an exciting prospect. This is also something that sets GOCATs apart from similar approaches such as WEYMUTH and REIHER's gradient-driven methods or SOKALSKI's catalytic fields, both of which presuppose a given mechanism and structure.

This is not the only interesting aspect of the dispersion GOCAT variety. While dispersion is, inherently, much weaker compared to Coulomb interaction, it might hold an interesting value that only becomes apparent once molecular dynamics are simulated. With reactants that are large and flexible, it could be interesting to encapsulate them in optimal pocket-like structures which reduce their entropy and/or favor vibrational energy distributions beneficial to the reaction. It could certainly be argued that electrostatic interactions can also do this very same thing¹, however in the interest of MD using force fields, a dispersion-like description might be more useful. This will ultimately require (manual or automatic) interfacing of the GOCAT methodology with an MD program to elucidate the effect. This could either work by introducing MD simulation into the fitness function or by using it as an evaluation tool for finished GOCAT candidates.

Lastly, the molecular GOCATs introduced in Chapter 6 could, in the future, be useful additions in catalyst design. As has already been mentioned in the introduction, this is an approach that takes a step away from the inverse design approach towards actual chemistry. It is only a step because it is neither the one nor the other completely; it is still an inverse design approach in the sense that the *influence* of an embedding is optimized, within the given parameters. The molecular identity, with its fixed bonded structures, point charge patterns and their combined potential, as well as its steric influence, serves as an additional constraint to the optimization. Under the hood, it remains the optimization of the electrostatic influence on the embedded reaction. However, this does not make it a real-world chemistry model. There is no way to arrange solvate molecules optimally "by hand" and no guarantee that active sites designed using the model can be synthesized in exactly this specific arrangement. There is also no way of knowing beforehand whether the reaction within will proceed in exactly the predicted manner. Molecular GOCATs are, in essence, an attempt to constrain electrostatic GOCATs to a strict molecular picture. This is both in terms of structure, as well as in terms of how strong an electrostatic potential can possibly be generated within a given molecular context, such that the reaction in question can proceed optimally. At minimum, they could be used as a way to generate more natural/realistic electrostatic GOCATs by transforming the molecular GOCAT via the simplification algorithms detailed in Section 5.1. Or, ideally, they could be used as-is as inspiration of new enzymatic or enzyme-like active sites, much in the vein of *theozymes*.

¹In fact, dispersion interactions are electrostatic in nature, so this is true either way.

Throughout all chapters, only a few remarks were made towards code implementation, but the details and intricacies have remained mostly in the background. Just to give an idea what stands behind this: Since the start of this Thesis in 2019, around 20000 lines of code were either added or changed in the OGOLEM source code alone (estimated from a `git diff`) to implement features presented here. In addition, around 5000 lines of python code were written for evaluation or utility purposes, not counting the miscellaneous software introduced in Chapter 9, below. Because the project has already branched from the official OGOLEM code on `GITHUB` back in 2016, its state is fairly problematic. It currently contains a large amount of code that was implemented for hypothesis testing, as well as features that were never used in production. A large amount of methods and classes have no unit-tests, which makes implementation of new features cumbersome and inefficient. In the future, it is highly recommended to take only the key features of GOCATs and re-implement them, carefully and cleanly, in the current release branch of OGOLEM, such that the project is conserved and remains usable for future research.

ADDITIONAL PROJECTS

The Chapters 4 to 7 deal with GOCATs and related subjects. This chapter, on the other hand, will deal with two somewhat unrelated software development projects, which are more of a personal addition to the repertoire of computational chemists. Both have, as of now, remained (formally) unpublished. Thus, their concept, implementation and description will be published as part of this dissertation. The two projects are `ASCIIMOL`, a `PYTHON` program that can visualize 3D molecules directly on UNIX terminals and `UIGOLEM`, which is a graphical user interface (GUI) for `OGOLEM`. The latter is also written in `PYTHON` and can generate input files, submit them to `OGOLEM`, and track their progress dynamically. Both projects are open-source and freely available on `GITHUB`^[135,136].

9.1 Visualizing Molecules On UNIX Shells: `ASCIIMOL`

The program code of `ASCIIMOL` was first introduced as open-source software on `github` in 2020. Following a recognition in 2021 by Andrew White^[137], the mini-program has gained some (rather unexpected) popularity.

When connected to a remote computer via a UNIX shell, it is often very inefficient to use X11 tunneling (or similar approaches) for visualization of molecules. In workflows that necessitate looking at small molecules frequently to check their general structure and identity, this can become very cumbersome. Since previously, to the author's knowledge, no software existed that could visualize molecules in a terminal-exclusive setting, a new one had to be developed. The program needed to satisfy three key criteria, namely:

1. **Ease of Install:** The program should be very quickly and easily obtainable on any given UNIX system without the need to compile, install third-party software or manual configuration.
2. **Simplicity:** The program should be very light-weight and have only basic functionality, so that it remains intuitive and quick in its use.
3. **Availability:** The program should be globally available, free and open-source.

With these criteria in mind, `PYTHON` was chosen as the main development language and the source code was published on `GITHUB`. Additionally, the code is provided in the `PyPI` economy, making the installation trivial on any machine that can run `PYTHON`.

The program uses the `NCURSES` library that comes with any `PYTHON` distribution. This simple library can be used to provide an interactive graphical interface on the console. It allows the capture of keyboard input, as well as dynamically displaying text on the screen.

The main application module of `ASCIIMOL` has two components: On the one hand, a configuration object that handles the file management and holds the currently active molecule information. On the other hand, a renderer object that uses the information stored in the configuration to translate coordinates into a text-only 3D representation of the molecule.

When the program starts, it first reads in the molecular geometry. It then guesses the bonds between the atoms by comparing their distances versus their Van-der-Waals-radii. Both the information on the atoms and bonds are then stored in the configuration object.

Afterwards, the main loop starts. In each loop update, the input buffer is read and if any input was given that necessitates a screen update, the screen is redrawn completely.

This screen consists of a bordered display for the molecule, as well as an info bar at the bottom of the screen that displays current viewing information and the navigation hot keys.

Molecules are drawn via an orthographic projection, with a Z-buffer^[138] determining whether objects are in front or not at a specific character coordinate. Atoms are drawn first, followed by a step-wise rasterized line in between them if they are bonded. There are three distinct ways bonds can be drawn in: Either no bonds, simple dots, or an algorithm that determines a suitable character based on the line angle.

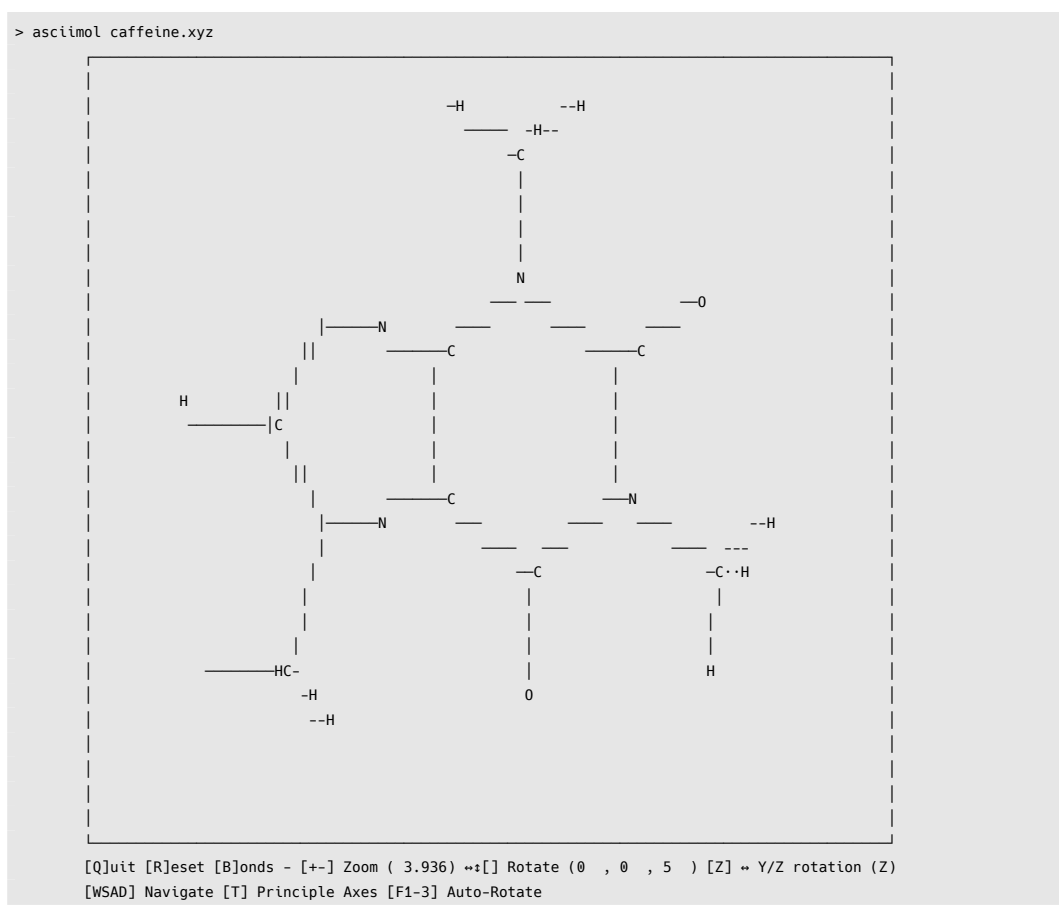
The navigation supports movement, rotation, auto-rotation and transposing the molecule into its principal axes system, as well as forward/backward jump in files containing data for more than one molecule (e.g., trajectories).

By including optional bindings to the `RDKit`^[139] and `ASE`^[140], `ASCIIMOL` can also directly open `SMILES` strings and many popular molecular data formats, making it a handy tool to, e.g., quickly check `SMILES` strings visually.

On any machine that can run `PYTHON` 3.6 or newer, everything needed to install `ASCIIMOL` is:

```
> pip3 install asciimol
```

It can then be run on any `.xyz` file to display the given molecule, giving output such as:



As can be seen, the 3D structure of the molecule is projected onto the screen using only ASCII characters. It can be freely rotated and viewed from different angles.

Additionally, when the RDKit PYTHON binding is installed, ASCIIMOL can also be used to automatically generate and view a 3D structure from any SMILES string:

```
> asciimol 'CCOCC'
```

This, for example, will then load the 3D structure of diethyl ether.

All in all, ASCIIMOL is in a fully-functional state and satisfies all the necessary and key features required, making it a quick, easy-to-use tool to view molecules in UNIX shell settings. For further development, it could certainly be improved in the performance department. Since the renderer itself is written mostly in PYTHON, a lot of speed and efficiency could be gained by rewriting computationally expensive tasks in C and using Cython bindings to incorporate them back into the main code. However, the current performance is absolutely sufficient to run this on any modern machine for moderately sized molecules without any trouble. For large molecules such as proteins, there is a tangible slowdown. However, ASCIIMOL is not specifically designed with proteins in mind anyway, and they will usually be much too large to decipher their structure in an ASCII-only setting. For that, a similar solution exists that shows only the protein backbone, called

MMTERM^[141].

9.2 A Graphical User Interface to OGOLEM: UIGOLEM

At the campusSOURCE awards 2022, a prize was offered to “[...] honor individuals or teams who make significant contributions to supporting scientifically active persons, who develop research software and thereby enable new and innovative research software solutions, for example by helping

- to make scientific software more visible as a product of research,
- to increase the quality of research software,
- to enable sustainable research software development and operation thereof,
- to support collaborations between different non-full-time software developers,
- to enable collaborations with full-time software developers, and to make research software available for open reuse.”^[142]

This inspired the conception of UIGOLEM, an easy-to-use and intuitive graphical interface to OGOLEM. Global optimization is an important asset to many research fields and the idea of this project was to enable a wider audience to use and contribute to OGOLEM.

Although the entry was not nominated in the eventual competition, the current software implementation is now publicly available on GITHUB. It is, again, written in PYTHON, for very similar reasons as ASCII-MOL. It uses the QT5 backend to produce a natural user interface on any platform that can run PYTHON.

In principle, there are many ways to approach a user interface (UI) to OGOLEM. First of all, it is certainly useful to keep the UI as a separate application, so that the link between the two remains flexible enough for e.g. an HPC setting. Thus, it would be advantageous to concentrate the UI application functionality mostly on helping the user construct valid and sane input files. This would retain maximum flexibility and portability.

One of the key problems of this approach is the fact that there is an inherent disconnect between the two applications. Thus, both have to be maintained separately from each other and each feature of OGOLEM needs to be manually implemented and fine-tuned in the UI.

This can in some ways be remedied by the fact that JAVA inherently is written in very modular code: By self-reflection, for example, one could in theory auto-generate a list of classes that implement a certain interface and expose this to a user. In practice, however, this is harder than one would think. In order to maintain portability, this would also require massive changes to the JAVA code of OGOLEM itself.

Thus, instead of trying to maximally interface a JAVA application with the inner workings of OGOLEM, a separate application was designed such that control is gained over how the functionality of OGOLEM is exposed to the user.

The eventual decision to write the program in PYTHON is based on two considerations: Firstly, writing platform-independent user interfaces in PYTHON is incredibly easy and straight-forward. Secondly, the open-source and collaboration-friendly nature of PYTHON was a very good fit for the campusSOURCE awards.

9.2.1 Functionality

Central to the functionality of the app is a library of configuration presets. This allows the user to start with a running configuration out of the box and then modify it to their exact needs. This selection presents itself to the user immediately upon initialization of the main window, which is shown in Figure 9.1. The following screenshot (Figure 9.2)

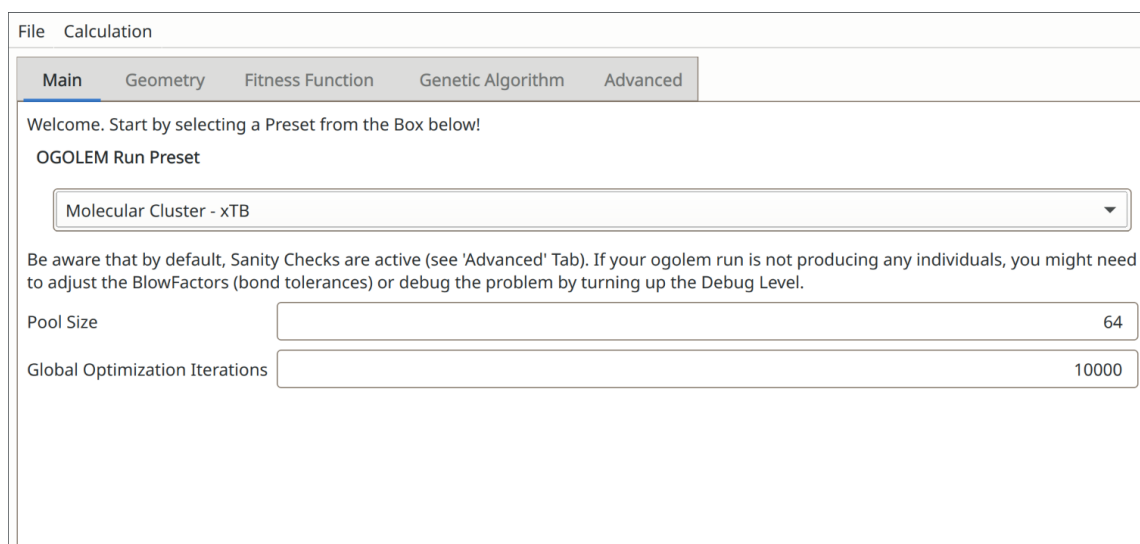


Figure 9.1: The main window (top) in UIGolem.

shows the page for customization of the geometry (adding, removing and modifying the starting atoms and molecules). It allows input of xyz style formatted molecules by hand or the loading of files of supported file formats. Next, the fitness function page (at the top of Figure 9.3) allows the user to customize the fitness function using building blocks provided to UIGolem via XML files. Since the content is loaded from XML files, new or modified building blocks can be customized and updated quickly and efficiently without modifying the codebase itself. The customization of the genetic algorithm presents itself in a similar fashion below. Lastly, there is an additional page for miscellaneous settings pertaining to debug output, verbosity customization and sanity checking.

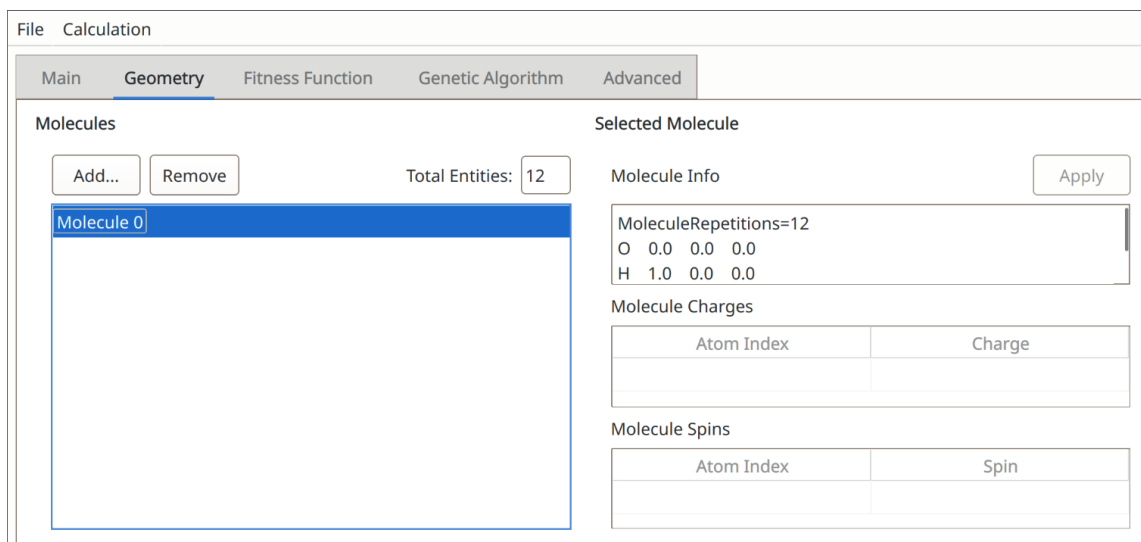


Figure 9.2: The geometry customization page in UIgolem.

When the user has finished customizing his configuration file, it can be saved and submitted to OGOLEM. A running configuration window shown in Figure 9.4 tracks the progress and output of OGOLEM and visualizes the currently best candidate in the pool using `matplotlib`.

9.2.2 Future Prospects

All in all, UIgolem is a fully functional configuration building tool for OGOLEM. As of now, it supports configurations only for cluster optimization. Support for cluster optimization on surfaces, as well as parameter optimization has yet to be implemented. In the future, it could be an interesting tool not only for research, but also for educational purposes, as it allows an easy access and overview over OGOLEM's functionality, such that production runs can be kick started for new users.

9.2. A GRAPHICAL USER INTERFACE TO OGOLEM: UIGOLEM

File
Calculation

Main
Geometry
Fitness Function
Genetic Algorithm
Advanced

Local Optimization Algorithm

chained:lbfgs:backend=xyz:untangle:sigma=1.2|mopac:pm7

Library of building Blocks

☐ Add all optional Settings (with their default values)

Local Optimizers
Generic Backends
Cartesian Backends

Evaluator	Description
1 L-BFGS	Default Built-In optimizer using both energy and gradient information for minimization. Requires a generic backend that fits the problem to be optimized (e.g. the Cartesian backend for atoms/molecules). The fitness is the minimized value returned from this Cartesian backend.
2 FIRE	Built-In optimizer using only gradient information. Requires a generic backend that fits the problem to be optimized (e.g. the Cartesian backend for atoms/molecules). The fitness is the minimized value returned from this Cartesian backend.

File
Calculation

Main
Geometry
Fitness Function
Genetic Algorithm
Advanced

Crossover Operator

sweden:

Mutation Operator

montecarlo:mode=some

Library of Operators

☐ Add all optional Settings (with their default values)

Crossovers
Mutations

Crossover	Description
	Provide a list of crossover operators. Syntax is XX%CROSSOVER YY%CROSSOVER ..., where XX is a number from 1-99. The numbers MUST add up to 100% and represent how often the following crossover operator

File
Calculation

Main
Geometry
Fitness Function
Genetic Algorithm
Advanced

Output Control

Debug Level

☐ Silent Mode

Genetic Record Buffer Size

1000

☐ Cluster Detailed Stats

Niching Printout after # Steps:

50

Serialize Genetic Records After # Steps:

10000

Serialize Geometries After # Steps:

500

☒ Serialize Pool after new Best

☐ Write every Geometry

Sanity Checks

Collision Detection

☐ Pre-Fitness

☒ Post-Fitness

Bond Detection Blow Factor

1.2

Initial Bond Detection Blow Factor

1.2

Dissociation Detection

☒ Pre-Fitness

☒ Post-Fitness

Dissociation Detection Blow Factor

3.0

Figure 9.3: The configuration pages in UIgolem.

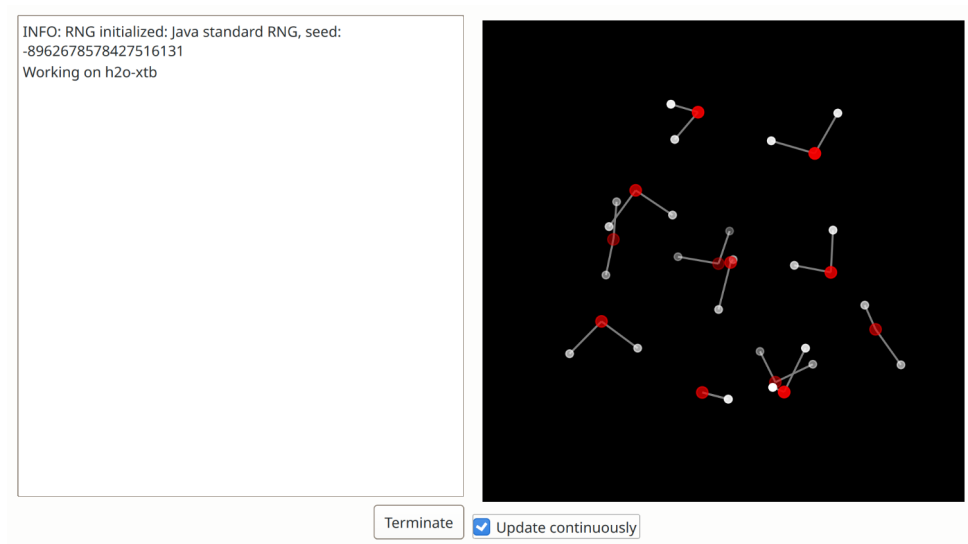


Figure 9.4: The run monitoring window of UIgolem.

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ACKNOWLEDGEMENTS

I am deeply grateful to Prof. Dr. Bernd Hartke for providing me with the opportunity to work on the GOCAT model and giving me the freedom to work on whatever outlandish idea I had during my years of PhD research. I'd also like to thank Dr. Mark Dittner for leaving me such a bountiful and interesting method to work on, even if the legacy code was, sometimes, quite a struggle!

I also owe a heartfelt thanks to Prof. Dr. Anna Hehn, who was enthusiastic about reading my Thesis and agreed to being a referee. Thank you!

Another huge thanks to my colleagues and office mates over the years: Tim Raeker, Swantje Caliebe, Jennifer Hörlöck, Florian Spenke, Julien Steffen, Mark Dittner, and Julian Müller. You guys made these years very enjoyable and I was always excited to come over into your offices to have a chat, goof around or, in the occasional case, ask for your opinion and experience.

And lastly, I thank my lovely soulmate, Jessica Wulfes. You make every day a bright and wonderful one! Without you, all of this would only be half as fun.

DECLARATION

I, Dominik M. Behrens, hereby declare that the work presented in this thesis was done by me, under the supervision of Prof. Dr. Bernd Hartke, with no other help than the referenced sources in the text. This is my first dissertation and the work has never been used in any other dissertation attempts. I have never been deprived of an academic title. The dissertation complies with the good scientific practice rules as proposed by the German Research Foundation (DFG).

Kiel, June 25, 2024

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